

A MCMC-Based Random Search Variable Selection Method for Logistic Regression with Application to Tropical Cyclone Genesis Dataset



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Abstract

This thesis investigates a stochastic model selection framework using a *Markov Chain Monte Carlo* (MCMC)-based random search procedure alongside its inverse-temperature λ variant with specific application to finding environmental variables influencing Tropical Cyclone (TC) genesis. The procedure aims to approximate the global *Bayesian Information Criterion* (BIC) minimizer across all possible submodels without necessarily enumerating all possible submodels. A factorial simulation study was designed to evaluate the random search procedure's performance under four factors: sample size, *Signal-to-Noise Ratio* (SNR), and collinearity among predictors. SNR is defined based on Wald and LRT p-values, although there is some discussion about the suitability of such definition. The random search has been able to consistently recover all ground truths, even under moderate SNR and high ground truth correlation. Metropolis-Hastings implementation of the procedure was run across multiple λ values on the TC genesis dataset, resulting in $\lambda = 0.75$ as the optimal value as it yields the BIC minimizer and near-optimal/*elite* models, which are models that differ by the minimum within 2 units. 41 *core variables*, which are variables that appear in all elite models, were also extracted; top-ranked core variables align with six canonical drivers of TC genesis proposed in previous literature. A best final model is obtained by selecting the elite model with the best cross-validation performance; benchmarking this model with LASSO demonstrated comparable predictive performance whilst remaining more parsimonious in dimension.

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1 Introduction

1.1 Tropical Cyclone (TC) and Prior Research

A *Tropical Cyclone* (TC) is a severe weather phenomenon that is known to develop over warm tropical waters. It has serious repercussions in almost all aspects of liveability (for example, societal, economic, property, environment, even life), especially when TCs are formed within populated coastal areas. Hence, given the aforementioned negative impacts of TCs, numerous researchers and related actors have taken up considerable interest to study TCs to provide skillful timely planning and measures of preparation. These endeavors are also driven by the increased abundance of knowledge on the underlying physical processes in TC formation. TC genesis studies primarily examine at least 1 of the 2 objectives below:

1. TC genesis forecasting, which forecasts whether a Tropical Cloud Center/TCC (a layreader may understand this as a set of clouds) will form into a TC on various lead times, including long-term [8], multiyear [23], seasonal [25], and short-term forecasting [10]. For example, a TC forecasting study focusing on a lead time of 12h would aim to predict whether a TC will be formed in the next 12 hours.
2. Factors of TC genesis/formation, which examines the variables or factors that affect the formation of TCs also on various lead times; for instance, a study focusing on factors affecting TC genesis on a 12h lead time would examine the factors that indicate whether a TC will be formed in the next 12 hours. The pioneering study for such goal comes from Gray [9], which identified six fundamental drivers of TC formation. Several metrics/indices have also been derived to indicate the potential formation of a TC (e.g., [14]; [27]; [4]).

However, the selected variables used for modelling and the indices are not necessarily consistent across studies and methods. To address this concern, Wijnands et al. [29] conducted a comprehensive and thorough analysis of assessing which environmental variables are useful for predicting the formation of TCs in various lead times using the Peter-Clark (PC) Algorithm applied to a global dataset.

The objective of this study aligns with the second objective – building on top of Wijnands et al. [29] to determine which environmental variables are useful for predicting TC genesis within a short lead time using a Logistic GLM.

1.2 Logistic GLM Approach and Research Goal

Contrary to the Peter-Clark algorithm used by Wijnands [29], which selects variables based on conditional independence tests within a learned joint distribution of predictors and response, we use a logistic Generalized linear Model/GLM to directly model the conditional distribution of the response (**TC.genesis**) given the predictors (environmental variables).

This way, the goal is variable selection under logistic regression, and one way to approach this goal is to identify the best submodel that optimizes a model selection criterion that accounts for both predictive accuracy and model complexity.

1.3 Research Plan

Given the above goal of finding the best submodel, one naive way to identify such model is by an exhaustive search, but quickly becomes infeasible due to the exponential growth in search space as the number of predictors increase. Alternatives include stepwise procedures (e.g. forward, backward, or both) which are computationally efficient but greedy and often suboptimal.

Therefore, this study utilizes a *Markov Chain Monte Carlo* (MCMC)-based random search procedure, which generates samples from a probability distribution of submodels induced by a model selection criterion (e.g. BIC), providing both a principled and computationally scalable approach to high-dimensional variable selection.

Before applying the random search procedure to a Tropical Cyclone Genesis dataset, we first validate the method on a simulation study, allowing a controlled evaluation of its performance.

Since the random search procedure induces a probability distribution over submodels constructed from a fit-based model selection criterion (e.g. BIC), we design this simulation study within the framework of a *factorial analysis*.

- That is, for a fixed sample size, we test the random search method in different simulated data sets corresponding to different combinations of factors/conditions that are expected to directly affect the fit of the model. These include:
 1. The effect sizes, which affects their statistical significance & signal-to-noise ratio picked up by the model, and in turn affecting the submodels being sampled in the random search procedure, and
 2. Multicollinearity between covariates.
- These allow us to better understand the behavior and robustness of this method for variable selection.
- In addition, we simulate datasets with a modest sample size (e.g. $n = 2000$) to reflect computational constraints and limited data scenarios.

Essentially, given a dataset with prior knowledge of the predictors involved in generating the response (so-called *ground truth* predictors), is the random search procedure able to recover these ground truths when:

1. The ground truth variables may not necessarily have a strong statistical significance to the response (which affects the signal-to-noise ratio that the model is able to detect), especially if the sample size is fairly modest,
2. The ground truth variables are correlated (which inflates their standard errors), and
3. Other redundant predictors not involved in generating the responses are present?

These are important questions because in real-world applications, correlated variables and ground truth predictors with low signal are not necessarily semantically meaningless and may reflect some underlying real-world

phenomena.

Ultimately, the end goal for this simulation study is to build confidence towards the result of the random search method applied to the TC genesis dataset, where its high-dimensional nature (149 variables) makes multicollinearity highly likely to be present, for instance.

2 Dataset Description and Challenges

2.1 Data Description

The dataset used for this study is compiled from Wijnands et al. [29]. It contains 30826 meteorological events, with each event representing the state of a TCC at a given time in one of the 7 global ocean basins. Of these, 2562 of them develop into a TC in the next 12 hours. Each event contains measurements of 149 environmental variables that indicate the state of the local environment in the TCCs, as well as a **TC.genesis** binary indicator (1 if the TCC develops into a TC within the next 12 hours and 0 otherwise). Several types of variables are included in this compiled dataset:

1. 10 meteorological features/physical fields/families are considered, as well as their measurements taken at nine different atmospheric pressure levels (200, 300, 400, 500, 600, 700, 850, 925, and 1000 hPa) which include: relative vorticity, potential vorticity, divergence of the wind field, relative humidity, specific humidity, air temperature, vertical velocity, geopotential and the zonal and meridional components of the wind, amounting to $10 + (10 \times 9) = 100$ variables.
2. Other variables which include: Sea Surface Temperature (SST), mean sea level pressure, 2-m dewpoint temperature, low cloud cover, medium cloud cover, high cloud cover, and total cloud cover, amounting to 8 variables.
3. Calculated variables which include (amounting to 41 variables):
 - (a) Vertical wind shear, calculated as a vector shear, using both zonal and meridional wind components. The vector shear is calculated across **all pairwise combinations** of the nine selected atmospheric levels, where each combination is treated as a variable (amounting to $\binom{9}{2} = 36$ variables).
 - (b) Coriolis parameter as a function of the Earth's rotation rate and latitude. The raw and absolute values are considered (2 variables)
 - (c) Absolute vorticity as the summation of Coriolis parameter and relative vorticity. The raw value and its value at 925 hPa is considered (2 variables)
 - (d) Equivalent potential temperature between the surface and 500 hPa (1 variable).

The goal is to identify a subset of these variables that could best explain TC genesis using a logistic regression model.

2.2 Challenges in the Data

The large and high dimensional nature of the dataset makes it difficult to apply traditional model selection methods (as outlined in Section (3.5)) due to potential challenges unique to this dataset. Some challenges are as follows:

1. **Highly correlated variables.** Some predictor variables in the TC genesis dataset are highly correlated, which could negatively affect the fitting and inference of linear models for predicting TC genesis.

Table 1: Top correlated variable pairs in the TC–genesis dataset (Pearson r). Many pairs represent the same field at adjacent pressure levels or closely related diagnostics.

Var1	Var2	r
geopotential_at_1000_hPa	mean_sea_level_pressure	0.999
geopotential_at_1000_hPa	geopotential_at_925_hPa	0.995
geopotential_at_925_hPa	geopotential_at_850_hPa	0.993
geopotential_at_925_hPa	mean_sea_level_pressure	0.993
vertical_wind_shear_between_200_hPa_and_1000_hPa	vertical_wind_shear_between_200_hPa_and_925_hPa	0.992
absolute_vorticity_averaged	potential_vorticity	0.991
wind_u_at_1000_hPa	wind_u_at_925_hPa	0.989
vertical_wind_shear_between_200_hPa_and_925_hPa	vertical_wind_shear_between_200_hPa_and_850_hPa	0.988
absolute_vorticity_at_925_hPa	absolute_vorticity_averaged	0.985
absolute_vorticity_averaged	potential_vorticity_at_600_hPa	0.984
potential_vorticity_at_500_hPa	potential_vorticity_at_400_hPa	0.984
vertical_wind_shear_between_300_hPa_and_1000_hPa	vertical_wind_shear_between_300_hPa_and_925_hPa	0.984
potential_vorticity_at_600_hPa	potential_vorticity_at_500_hPa	0.984
vertical_wind_shear_between_200_hPa_and_700_hPa	vertical_wind_shear_between_200_hPa_and_600_hPa	0.982
geopotential_at_500_hPa	geopotential	0.982
absolute_vorticity_at_925_hPa	potential_vorticity_at_850_hPa	0.981
vorticity_relative_at_1000_hPa	vorticity_relative_at_925_hPa	0.980
geopotential_at_1000_hPa	geopotential_at_850_hPa	0.980
potential_vorticity_at_700_hPa	potential_vorticity_at_600_hPa	0.979
absolute_vorticity_averaged	potential_vorticity_at_700_hPa	0.977
geopotential_at_850_hPa	mean_sea_level_pressure	0.977
vertical_wind_shear_between_200_hPa_and_600_hPa	vertical_wind_shear_between_200_hPa_and_500_hPa	0.977
vertical_wind_shear_between_300_hPa_and_925_hPa	vertical_wind_shear_between_300_hPa_and_850_hPa	0.977
absolute_vorticity_at_925_hPa	potential_vorticity_at_700_hPa	0.977
absolute_vorticity_averaged	potential_vorticity_at_500_hPa	0.976
absolute_vorticity_averaged	potential_vorticity_at_400_hPa	0.975
absolute_vorticity_averaged	Coriolis_parameter	0.975
wind_u_at_925_hPa	wind_u_at_850_hPa	0.974
absolute_vorticity_at_925_hPa	Coriolis_parameter	0.974

2. **High collinearity of variables with potentially high 'signal' to the response.** I will demonstrate an analytic and graphical approach.

- (a) Analytical approach. For each of the 149 variables, a logistic regression model is fit using that variable to the binary response. The variables are sorted based on the difference in null deviance and residual deviance, an ad-hoc measure of marginal significance to the response. The result is as follows:

Predictor	Null Deviance - Residual Deviance
vertical_wind_shear_between_200_hPa_and_700_hPa	1930.4343921
vertical_wind_shear_between_200_hPa_and_600_hPa	1885.4014736
vertical_wind_shear_between_200_hPa_and_500_hPa	1807.6463902
vertical_wind_shear_between_200_hPa_and_850_hPa	1802.8223380

Table 2: Caption

	Coef. (SE)	z	p -value
vertical_wind_shear_between_200_hPa_and_700_hPa	-0.163 (0.008)	-19.53	$< 10^{-16}$
vertical_wind_shear_between_300_hPa_and_600_hPa	-0.086 (0.014)	-6.08	1.2×10^{-9}
N	30,826		
Null deviance	17,651		
Residual deviance	15,683		
AIC	15,689		

Table 3: Logistic regression of TC genesis on two highly correlated shear measures ($r = 0.88$). Both predictors remain conditionally significant despite their high correlation.

(b) Graphical approach. Here one considers the visual distribution of TC genesis with respect to highly correlated variables representing the geopotential at different pressure levels:

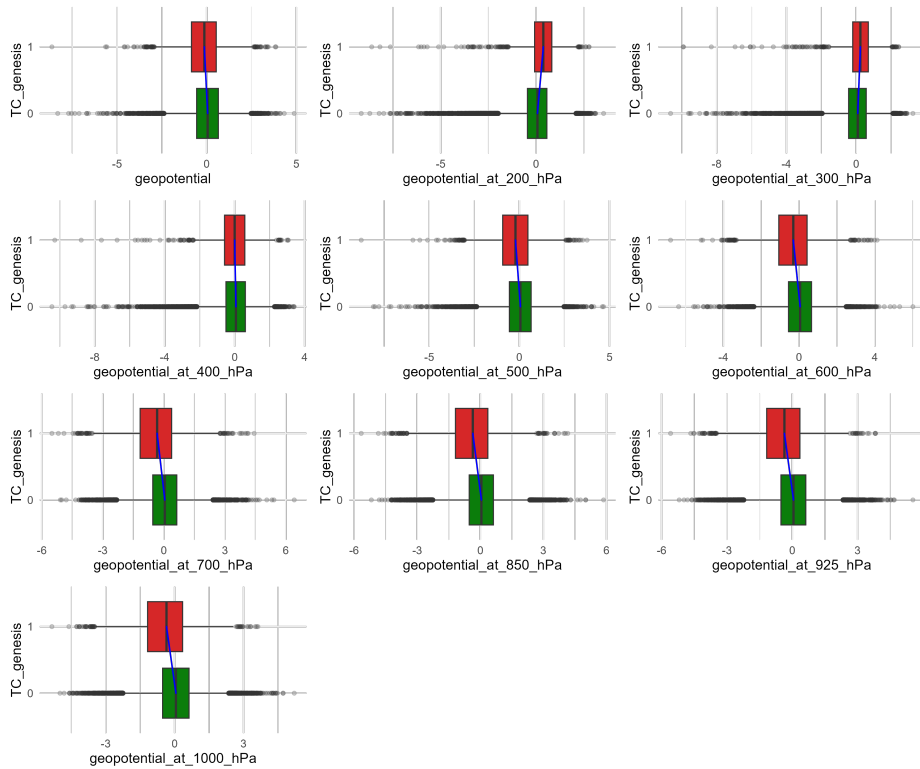


Figure 1: Distribution of TC genesis, grouped by correlated geopotential variables taken at different pressure levels. Graphically, the relationship between geopotential and TC genesis seems to differ at various pressure levels. For instance, TC genesis seems to be more prevalent at large geopotential values at small pressure levels (200-300 hPa), but not necessarily for higher pressure values.

3. Large sample size n and number of candidate predictors p .

4. **Class Imbalance?** Although the TC genesis dataset shows substantial class imbalance (only 8 percent of the events in the dataset developed into a TC), several studies have shown that severe class imbalance mainly affects the estimation of the intercept, while the slope coefficients remain statistically consistent under large sample size. For instance, King and Zeng [13] demonstrated that the maximum likelihood estimates of the slope parameters are approximately unbiased even for rare events, with the bias primarily concentrated on the intercept. Given that the TC genesis dataset contains over 30,000 observations, variable selection is therefore expected to be largely not affected by class imbalance.

3 MCMC Random Search Procedure

3.1 Preface

This section is largely adapted from the "Framework and selection criteria" section from the paper [18].

3.2 Logistic Regression Framework

Logistic Regression Model. The random search variable selection procedure assumes a logistic regression setup. That is, given a Bernoulli response variable Y and p covariates a.k.a. *explanatory variables / features / predictors / independent variables* $\mathbf{x} \equiv (1, x_1, \dots, x_p)^T$ (here an *intercept / bias* term 1 is included), one postulates the binomial model $Y \sim Bi(1, \pi(\mathbf{x}))$ for a mean function $\pi(\cdot)$ with

$$\log \frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})} = \mathbf{x}^T \beta = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p \quad (1)$$

An equivalent formulation of the model above is as follows:

$$\pi(\mathbf{x}) = \frac{\exp(\mathbf{x}^T \beta)}{1 + \exp(\mathbf{x}^T \beta)} = \sigma(\mathbf{x}^T \beta)$$

where $\sigma(\cdot)$ denotes the *logistic function* (a.k.a. *sigmoid*) and $\beta \in \mathbf{R}^{(1+p) \times 1}$ being some parameter vector to be estimated.

Alternatively, one could formulate the above in a *Generalized Linear Model* (GLM) framework, whereby we model the mean of Y as a function of the predictors $\mathbf{x}$ given by the following relationship:

$$g(\mathbf{E}(Y)) = \mathbf{x}^T \beta$$

where the *link function* $g(\cdot)$ corresponds to the function $g(t) = \log \frac{t}{1-t}$ and $\mathbf{E}(Y) = g^{-1}(\mathbf{x}^T \beta) = \pi(\mathbf{x})$.

Variable Selection Problem. Although the set of influential covariates on the response is mainly determined by domain knowledge, it is assumed that such a set exists among the p covariates. The variable selection problem, therefore, becomes identifying *the* subset of variables that is statistically significant in explaining the response.

Definition 3.1. Suppose $\tilde{\mathbf{x}} = (x_1, \dots, x_p)^T$ represents the set of all p candidate covariates available. Define α , a subvector of $(1, 2, \dots, p)$ of dimension (number of elements) $p_\alpha \leq p$ that indexes which variables are included in a submodel. Finally, define a subset of $\tilde{\mathbf{x}}$ as

$$\tilde{\mathbf{x}}_\alpha \in \mathbf{R}^{p_\alpha \times 1} \quad (2)$$

and define a submodel that corresponds to the subset $\tilde{\mathbf{x}}_\alpha$ as

$$\log \frac{\pi(\mathbf{x}_\alpha)}{1 - \pi(\mathbf{x}_\alpha)} = \mathbf{x}_\alpha^T \beta_\alpha \quad (3)$$

where $\mathbf{x}_\alpha^T = (1, \tilde{\mathbf{x}}_\alpha^T)$, an extended version of $\tilde{\mathbf{x}}_\alpha^T$ that includes an intercept term, and $\beta_\alpha \in \mathbf{R}^{(1+p_\alpha) \times 1}$, a subvector of β consisting of one intercept β_0 and p_α many β_i coefficients indexed by α .

- To illustrate using an example, suppose we have $p = 4$ candidate covariates available, denoted by $\tilde{\mathbf{x}} = (x_1, x_2, x_3, x_4)^T$ and define a submodel indexed by $\alpha = (2, 3)$. Then the model equation for the corresponding submodel with covariates $\tilde{\mathbf{x}}_\alpha = (x_2, x_3)^T$ would be:

$$\log \frac{\pi(\mathbf{x}_\alpha)}{1 - \pi(\mathbf{x}_\alpha)} = \beta_0 + \beta_2 x_2 + \beta_3 x_3$$

whereby $\mathbf{x}_\alpha^T = (1, x_2, x_3)$ and $\beta_\alpha = (\beta_0, \beta_2, \beta_3)^T$, a subvector of the full model coefficient vector $\beta = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4)^T$, which includes the intercept β_0 .

Enumerating submodels. From Definition (3.1), we can uniquely identify a submodel via α (as it can be uniquely mapped to both $\mathbf{x}_\alpha$ and β_α) as well as enumerating the set of all possible logistic regression models via the following set:

$$\mathcal{A} = \{\alpha : \text{any subvector/submodel of } (1, 2, \dots, p)\} \quad (4)$$

Here, $\mathcal{A}$ can be interpreted as the set of all logistic regression models, where each element $\alpha \in \mathcal{A}$ is a submodel having the form given in Equation (3) for the full model given in Equation (1).

Few remarks:

- Using the submodel as specified in Definition (3.1) implies that the response Y is fitted by the binomial distribution $Bi(1, \pi(\mathbf{x}_\alpha))$, but this does not necessarily mean that $Y \sim Bi(1, \pi(\mathbf{x}_\alpha))$ as the submodel may be misspecified (e.g. by omitting truly relevant variables or including irrelevant variables).
- A submodel $\alpha \in \mathcal{A}$ only proposes a dependence structure between Y and $\tilde{\mathbf{x}}_\alpha$, as β_α still needs to be estimated using Maximum Likelihood (ML) or an Iterated Weighted Least Squares Procedure (IWLS).

True Model. Suppose that the true relationship between Y and $\tilde{\mathbf{x}} = (x_1, \dots, x_p)^T$ (as defined in Definition (3.1)) is given by the following model:

$$\log \frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})} = \mathbf{x}^T \beta^{(true)} \quad (5)$$

where $\beta^{(true)} = (\beta_0^{(true)}, \beta_1^{(true)}, \dots, \beta_p^{(true)})^T \in \mathbf{R}^{(1+p) \times 1}$ is a fixed quantity with **possible zero components**. The x variables corresponding to the zero components of $\beta^{(true)}$ are deemed to have no effect i.e. not influential on the response Y .

Classifying submodels. Based on Equation (5) and the set of all models defined on (4), one can categorize the set of all models $\mathcal{A}$ into 2 disjoint categories:

$$\begin{aligned} \mathcal{A}_c &= \{\alpha : \beta_i^{(true)} = 0 \text{ for any } i \notin \alpha\}, \text{ and} \\ \mathcal{A}_w &= \{\alpha : \beta_i^{(true)} \neq 0 \text{ for some } i \notin \alpha\} \end{aligned}$$

Here, every model $\alpha \in \mathcal{A}_c$ contains all x variables that are influential on Y , thus provides a correct dependence structure between Y and $\tilde{\mathbf{x}}$. On the other hand, no model in $\mathcal{A}_w$ provides a correct dependence structure, since it misses at least one influential variable.

Each model in $\mathcal{A}_c$, while correct, may not be the most *efficient* as it may include variables that are not influential or "redundant" on explaining Y (i.e. there may exist some $i \in \alpha$ for a model $\alpha \in \mathcal{A}_c$ such that $\beta_i^{(true)} = 0$). Denote the most efficient, that is, the most simple and correct model as α^* , which has the smallest dimension (smallest number of elements) in $\mathcal{A}_c$.

As an example, consider $p = 4$ candidate variables with $\beta^{(true)} = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4)^T = (0, 0, 2, -1, 0)^T$. Then:

- $\alpha = (2, 3)$ is the simplest correct model α^* as it only includes predictors that are truly influential on Y .
- Models such as $\alpha = (1, 2, 3)$, $(0, 2, 3, 4)$, or even the full model $(0, 1, 2, 3, 4)$ are also "correctly specified" models in $\mathcal{A}_c$ simply because they include all relevant variables.
- Models such as $\alpha = (2)$, (3) , $(0, 1)$ would belong to $\mathcal{A}_w$ as they miss at least 1 truly relevant predictor (here the given examples miss 3, 2 and both respectively).

Further Narrowing the Variable Selection Problem. In this way, the problem becomes to estimate α^* based on available dataset $D = \{(\tilde{\mathbf{x}}_1, y_1), \dots, (\tilde{\mathbf{x}}_n, y_n)\}$; that is, we observe a sample of n observations of the response Y (denoted as y), as well as the candidate covariates $\tilde{\mathbf{x}}_i$. Specifically, we wish to evaluate the utility of each candidate model using some *model selection criterion* that considers both model fit and complexity, before selecting the best one as an estimate of α^* . Before proceeding, for a submodel α , denote $D_\alpha = \{(\tilde{\mathbf{x}}_{1\alpha}, y_1), \dots, (\tilde{\mathbf{x}}_{n\alpha}, y_n)\}$, where $\tilde{\mathbf{x}}_{i\alpha}$, $1 \leq i \leq n$ is defined similarly to Equation (2) in Definition (3.1), with i referring to the predictor of the i -th sample.

3.3 Model Selection Criterion

Model Selection Criterion. Several measures of the utility of a parametric model have been introduced in the literature, namely:

- *Akaike Information Criterion* (AIC): AIC estimates the relative entropy between a given fitted model and the true model underlying the data.
- *Bayesian Information Criterion* (BIC): BIC approximates the posterior distribution of the candidate models using a uniform prior among all candidate models.
- *Stochastic Complexity Criterion* (SCC): proposed and studied by Rissanen [21] and Qian and Kunsch [19], SCC measures a model's utility by the minimum code length required for encoding the data by a given fitted model in a prefix way.

These three model selection criteria have a common form. Namely, given a submodel α , its model selection criterion can be formulated as:

$$S(\alpha|D_\alpha) = \left[- \sum_{i=1}^n \left(y_i \mathbf{x}_{i\alpha}^T \hat{\beta}_\alpha - \log(1 + \exp(\mathbf{x}_{i\alpha}^T \hat{\beta}_\alpha)) \right) \right] + C(n, \alpha) \quad (6)$$

Here,

- $\mathbf{x}_{i\alpha}^T = (1, \tilde{\mathbf{x}}_{i\alpha}^T)$ corresponds to the covariate values selected by α in the i -th sample, with an intercept term preprend-ed at the beginning.
- The first term enclosed in square brackets represents the negative log-likelihood for the data D_α evaluated at the Maximum Likelihood Estimate (MLE) of β_α denoted as $\hat{\beta}_\alpha$, i.e. $\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) = \sum_{i=1}^n \left(y_i \mathbf{x}_{i\alpha}^T \hat{\beta}_\alpha - \log(1 + \exp(\mathbf{x}_{i\alpha}^T \hat{\beta}_\alpha)) \right)$. In this way, one can rewrite the criterion as:

$$S(\alpha|D_\alpha) = -\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + C(n, \alpha) \quad (7)$$

- $C(n, \alpha)$ represents a measure of model complexity which varies according to the criteria. For instance, $C(n, \alpha) = p_\alpha$ for AIC, and $C(n, \alpha) = \frac{1}{2} p_\alpha \log n$ for BIC.

Estimating the most efficient model. Denote the model that minimizes $S(\alpha|D_\alpha)$ of the form in Equation (6) as α_0 , i.e.

$$\alpha_0 = \arg \min_{\alpha} S(\alpha|D_\alpha) \quad (8)$$

The optimal model α_0 is not necessarily the same as the most simplest and correct model α^* . However, Qian and Field [17] have investigated the asymptotic properties of model selection criteria of the form given in Equation (6) in relation to the true negative log-likelihood (that is, the negative log-likelihood $-\mathcal{L}(\beta|D)$ as a function of β evaluated at $\beta = \beta^{(true)}$), which yield the following results:

- Under general conditions, for any correctly specified model $\alpha \in \mathcal{A}_c$,

$$0 \leq \mathcal{L}(\hat{\beta}_\alpha|D_\alpha) - \mathcal{L}(\beta^{(true)}|D) \in \mathcal{O}(\log \log n) \quad (9)$$

almost surely. Equivalently, there exists $c, n_0 > 0$ such that for all $n > n_0$, we have $0 \leq \mathcal{L}(\hat{\beta}_\alpha|D_\alpha) - \mathcal{L}(\beta^{(true)}|D) \leq c \cdot \log \log n$

- For any ill-specified model $\alpha \in \mathcal{A}_w$,

$$\mathcal{L}(\beta^{(true)}|D) - \mathcal{L}(\hat{\beta}_\alpha|D_\alpha) \geq \tau \cdot n \text{ (for some } \tau > 0) \quad (10)$$

almost surely.

From these results, Qian and Field [17] deduced that the optimal model under any three model selection criterion AIC, BIC and SCC is almost surely in $\mathcal{A}_c$. In particular,

1. The optimal model α_0 under AIC is almost surely in $\mathcal{A}_c$ implying that the model selected by AIC has a correct model structure. This could be verified by noticing the fact that:

$$\begin{aligned} & \min_{\alpha \in \mathcal{A}} AIC(\alpha|D) \\ &= \min_{\alpha \in \mathcal{A}} [-\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + p_\alpha] \\ &= \min\{\min_{\alpha \in \mathcal{A}_c} [-\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + p_\alpha], \min_{\alpha \in \mathcal{A}_w} [-\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + p_\alpha]\} \end{aligned}$$

provided that $\mathcal{A}_c, \mathcal{A}_w$ form a disjoint partition of the set $\mathcal{A}$. Assuming a large n and under certain conditions, Equation (9) and (10) implies that:

$$\begin{aligned} & \text{for } \alpha \in \mathcal{A}_w, -\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + p_\alpha \geq -\mathcal{L}(\beta^{(true)}|D) + p_\alpha + \tau \cdot n \\ & \text{for } \alpha \in \mathcal{A}_c, -\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + p_\alpha \geq -\mathcal{L}(\beta^{(true)}|D) + p_\alpha - c \cdot \log \log n \end{aligned} \quad (11)$$

which also implies

$$\begin{aligned} & \min_{\alpha \in \mathcal{A}} AIC(\alpha|D) \\ &= \min\{\min_{\alpha \in \mathcal{A}_c} [-\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + p_\alpha], \min_{\alpha \in \mathcal{A}_w} [-\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + p_\alpha]\} \\ &= \min\{\min_{\alpha \in \mathcal{A}_c} [-\mathcal{L}(\beta^{(true)}|D) + p_\alpha - c \cdot \log \log n], \min_{\alpha \in \mathcal{A}_w} [-\mathcal{L}(\beta^{(true)}|D) + p_\alpha + \tau \cdot n]\} \end{aligned} \quad (12)$$

as n gets arbitrarily large, $\tau \cdot n$ increases with respect to n , whereas $-c \cdot \log \log n$ decreases with respect to n . This means that the AIC for models $\alpha \in \mathcal{A}_w$ will be larger than those for $\alpha \in \mathcal{A}_c$, making models in $\mathcal{A}_c$ more likely to be selected.

2. The optimal model α_0 under BIC and SCC almost surely converges not only to a model in $\mathcal{A}_c$, but to the simplest model α^* in $\mathcal{A}_c$. To see why, say using BIC, one could use a similar argument to Equation (11) and (12) whereby assuming a large n and under certain conditions:

$$\text{for } \alpha \in \mathcal{A}_c, -\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + \frac{1}{2}p_\alpha \log n \geq -\mathcal{L}(\beta^{(true)}|D) + \frac{1}{2}p_\alpha \log n - c \cdot \log \log n \quad (13)$$

which also implies

$$\begin{aligned} & \min_{\alpha \in \mathcal{A}} BIC(\alpha|D) \\ &= \min\{\min_{\alpha \in \mathcal{A}_c} [-\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + \frac{1}{2}p_\alpha \log n], \min_{\alpha \in \mathcal{A}_w} [-\mathcal{L}(\hat{\beta}_\alpha|D_\alpha) + \frac{1}{2}p_\alpha \log n]\} \\ &= \min\{\min_{\alpha \in \mathcal{A}_c} [-\mathcal{L}(\beta^{(true)}|D) + \frac{1}{2}p_\alpha \log n - c \cdot \log \log n], \min_{\alpha \in \mathcal{A}_w} [-\mathcal{L}(\beta^{(true)}|D) + \frac{1}{2}p_\alpha \log n + \tau \cdot n]\} \end{aligned}$$

since the model complexity term $\frac{1}{2}p_\alpha \log n$ increases with p_α and grows faster than $\log \log n$, it follows that not only BIC will almost surely choose a model in $\mathcal{A}_c$, but the simplest correct model α^* in $\mathcal{A}_c$.

3.4 Usage of BIC for this project.

This project uses the BIC as a measure of model utility because:

- It provides a stronger asymptotic property than AIC (not only it almost surely selects a correctly specified model, it also selects the most efficient, i.e. the simplest correct model)
- It is easier to compute than SCC, which requires matrix inversion to compute Fisher Information.

3.5 A Short Review of Model Selection Procedures

3.5.1 Exhaustive Search

An approach to solving the minimization problem outlined in Equation (8) is to exhaustively search for all the models in $\mathcal{A}$, before finding the model that minimizes the criterion. However, this only works for very small p , say $p < 10$, as there are $\mathcal{O}(2^p)$ possible subsets/submodels of $\mathcal{A}$ to search for. For example, if $p > 20$, then one would have to search over at least 1 million models.

3.5.2 Greedy-based

Greedy-based model selection procedures attempt to solve the minimization problem in Equation (8) by evaluating a model's utility by adding or removing one variable at a time.

Well-known examples of such procedures include [7]:

- *Forward Selection*: Start with an empty model with no variables. In each iteration, a variable is included if it results in the smallest model selection criterion among all possible single-variable additions. This process is repeated until including any variable no longer produces a significant fit.
- *Backward Elimination*: Start with the full model, that is, the model including all variables. In each iteration, a variable is removed if it results in the smallest model selection criterion among all possible single-variable deletions. This process is repeated until removing any variable no longer decreases the model selection criterion.
- *(Bidirectional) Stepwise Selection*: This procedure is a combination of both Forward Selection and Backward Elimination. Each step of the algorithm considers both the addition and deletion of a variable that results in the smallest model selection criterion.

Although any model selection criterion can be used, its criticism stems from its inherent greedy-search heuristic. Desboulets [5] reviewed the drawback of such approaches as follows:

1. The procedure might get stuck on a local minimum, as each sequential iteration in any of the three variants above only considers single variable moves.
 - For instance, suppose we have $p = 3$ predictors. Starting from the null model, a bidirectional stepwise procedure might add X_1 since it yields the smallest model selection criterion, followed by adding X_2 , and eventually finds that neither adding X_3 or dropping any of X_1 , X_2 further reduces the criterion. However, the true global minimizer of the criterion might not contain X_1 – say it turns out to be X_2, X_3 , a candidate model that this stepwise procedure never considers in the second step because it would require two moves at once: dropping X_1 and adding X_3 .
2. Several studies (e.g. Whittingham et al. [28], Steyerberg et al. [24]) showed that even as the sample size grows, stepwise procedures do not output models that is close to the true model.

3.5.3 Regularization-Based Selection

There are a handful of model selection procedures that are embedded in the parameter estimation process itself via augmenting the function in the optimization problem – for instance, maximizing the log-likelihood plus some penalty term that shrinks the magnitude of the β_i coefficients. A popular method is the *Least Absolute Shrinkage and Selection Operator* (LASSO), which implicitly performs variable selection by solving the following minimization problem for parameter estimation in a logistic regression problem [31]:

$$\hat{\beta} = \arg \min_{\beta} -2 \cdot \mathcal{L}(\beta|D) + \lambda \|\beta\|_1$$

where $\mathcal{L}(\beta|D)$ is the log-likelihood of the data, and $\|\beta\|_1 = \sum_{i=1}^p |\beta_i|$ is the L1 norm of β . LASSO can be used as a variable selection method because it could shrink small parameter values to exactly zero. LASSO excels in

situations where the true value of the β_i s are *sparse* (i.e. the β vector has a small number of non-zero components), and is known to trade in a little bias in exchange with a reduction in variance, which in turn often improves prediction accuracy as per the *bias-variance tradeoff*. That said, here are some drawbacks:

- The λ parameter needs to be tuned from the data (e.g. via *cross-validation*), which incurs computational time and tends to deliver unstable solutions.
- The LASSO estimator is not guaranteed to be unique.
- When predictors are highly correlated, LASSO will usually choose only one of them and drop the rest, which contributes to the instability of LASSO estimates.

3.5.4 MCMC-based Random Search

Ultimately, greedy and pure penalty-based model selection procedures still suffer from addressing the challenges of non-convex model space and correlated predictors, respectively. In the following section, we explore a *Markov Chain Monte Carlo* (MCMC) based random search that is capable of finding the best model away from local optima.

Qian and Field [18] proposed a random search procedure that essentially performs model selection on a sample of candidate models. Suppose we have a sample of K candidate models $\mathcal{M}^{(K)} = \{\alpha^{(1)}, \dots, \alpha^{(K)}\}$ generated from some probability distribution. If $\alpha_0 \in \mathcal{M}^{(K)}$, then:

$$\arg \min_{\alpha \in \mathcal{M}^{(K)}} S(\alpha|D) = \arg \min_{\alpha \in \mathcal{A}} S(\alpha|D) = \alpha_0$$

In other words, model selection based on $\mathcal{M}^{(K)}$ will return the same result as model selection based on $\mathcal{A}$. Therefore, the high level idea of this procedure is to generate a sample of candidate models which, with the highest possible probability, contains the optimal model α_0 . Having obtained such a sample, all that remains is to find the model α with the smallest criterion value, which will often coincide with the model appearing most frequently in the sample. This model will serve as an estimate for α_0 and subsequently an estimate for α^* . I will now give the details.

3.6 A MCMC-based Random Search Procedure

Induced probability distribution on models. Define a probability distribution for a model α on $\mathcal{A}$ with probability mass defined as:

$$P(\alpha) = C \exp\{-S(\alpha|D)\}, \quad (14)$$

$$\text{where } C = \left(\sum_{\alpha \in \mathcal{A}} \exp\{-S(\alpha|D)\} \right)^{-1} \quad (15)$$

here C is the "normalizing constant". Since, by definition, $S(\alpha|D)$ achieves the global minimum at α_0 , it follows that $P(\alpha)$ achieves the highest probability mass at α_0 , i.e. α_0 is the *mode* of the probability distribution $P(\alpha)$. Here, the model selection criterion $S(\alpha|D)$ is chosen to be the BIC (denoted as $BIC(\alpha|D)$) as per the reasoning from Section (3.4).

Definition 3.2. Another representation of $P(\alpha)$. An alternative way to represent $P(\alpha)$ is through the following one-to-one transformation:

$$\begin{aligned} T: \mathcal{A} &\rightarrow \{0, 1\}^p \\ \alpha &\mapsto \mathbf{v} = (v_1, \dots, v_p)^T \end{aligned} \quad (16)$$

whereby $v_i = 1$ if $i \in \alpha$ and $v_i = 0$ if $i \notin \alpha$, i.e. only the components of $\mathbf{v}$ indexed by α are equal to one.

- For example, let $p = 4$. Then a model $\alpha = (1, 2)$ would correspond to the binary vector $\mathbf{v} = (v_1, v_2, v_3, v_4)^T = (1, 1, 0, 0)^T$.
- This transformation gives a distribution $P(\mathbf{v})$ defined on $\mathcal{V} \equiv \{0, 1\}^p$, the entire model space, and generating a sample $\mathcal{M}^{(K)} = \{\alpha^{(1)}, \dots, \alpha^{(K)}\}$ from $P(\alpha)$ is equivalent to generating a sample $\mathcal{M}^{(K)} = \{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}$ from $P(\mathbf{v})$, which is just a sequence of binary vectors of length p .

Remark 1. Consequences of Definition (3.2).

1. The following transformations are implied:

- $BIC(\alpha|D) \mapsto BIC(\mathbf{v}|D)$, indicating BIC of a model indexed by the binary vector $\mathbf{v}$.
- $P(\alpha) \mapsto P(\mathbf{v}) = C \cdot \exp\{-BIC(\mathbf{v}|D)\}$
- Here, the observed data will always be referred as D and not model specific (as written in, for example, equation (6)) as the set of predictors used for parameter estimation are implied by the model specification.
- In this way, the BIC can be rewritten as:

$$BIC(\mathbf{v}|D) = -\mathcal{L}(\hat{\beta}_{\mathbf{v}}|D) + \frac{1}{2} \cdot \text{sum}(\mathbf{v}) \cdot \log n \quad (17)$$

where $\text{sum}(\mathbf{v})$ is simply the summation of the entries in $\mathbf{v}$.

2. Further, the objective outlined in (8) can be reformulated as:

$$\alpha_0 = \arg \min_{\alpha \in \mathcal{A}} BIC(\alpha|D) \mapsto \mathbf{v}_0 = \arg \min_{\mathbf{v} \in \mathcal{V}} BIC(\mathbf{v}|D) \quad (18)$$

- The optimal model α_0 defined from (8) is mapped to the binary vector $\mathbf{v}_0$, where its components indexed by α_0 being equal to 1.
- The most efficient model α^* will also be similarly defined by a binary vector $\mathbf{v}^*$, with its components indexed by α^* being equal to 1.
- I will drop the conditioning on the observed data D on $BIC(\mathbf{v}|D)$ to $BIC(\mathbf{v})$ for brevity.

Asymptotic guarantees for i.i.d samples. Suppose K independent and identically distributed (i.i.d) models $\mathcal{M}_{iid}^{(K)} = \{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}$ can be generated from $P(\mathbf{v})$. Then the probability that $\mathbf{v}_0$, the global BIC minimizer across the entire model space, appears at least once in the sample, as $K \rightarrow \infty$ is:

$$\lim_{K \rightarrow \infty} 1 - (1 - P(\mathbf{v}_0))^K = 1 \quad (19)$$

Moreover, define the relative frequency of a model α appearing in the sample as follows:

$$f_{\mathbf{v}K} = \frac{1}{K} \sum_{i=1}^K I(\mathbf{v}^{(i)} = \mathbf{v})$$

then, under the classical *Strong Law of Large Numbers*, with probability 1,

$$\lim_{K \rightarrow \infty} f_{\mathbf{v}_0 K} = P(\mathbf{v}_0) = \max_{\mathbf{v} \in \mathcal{V}} P(\mathbf{v}) \quad (20)$$

As a consequence, with high probability, $\mathbf{v}_0$ will appear early and most frequently in the sample based on the results from (19) and (20) respectively. In other words, the model with the smallest criterion value and the one with highest relative frequency in the sample are consistent estimates of $\mathbf{v}_0$ and they both converge to $\mathbf{v}_0$ almost surely.

Infeasibility to simulate i.i.d samples. However, in practice, it is difficult to generate i.i.d samples from $P(\alpha)$ as the computation of the normalizing constant is not computationally intractable when $p \gg$. Perhaps one naive way is to utilize the following chain rule:

$$P(v_1, \dots, v_p) = P(v_1)P(v_2|v_1) \cdots P(v_p|v_1, \dots, v_{p-1})$$

here, each factor is a Bernoulli distribution (since $v_i \in \{0, 1\}$). However, it remains computationally intractable to evaluate all the factors altogether in high-dimensional settings. For example, marginalizing $P(v_1)$ alone requires summing over 2^{p-1} terms, and every conditional terms $P(v_i|v_1, \dots, v_{i-1})$ requires summing over 2^{p-i} terms, which is still intractable, as one has to sum over $\mathcal{O}(2^p)$ (i.e. exponentially many) terms.

An MCMC approach for sampling from $P(\mathbf{v})$. An alternative way to sample from $P(\mathbf{v})$ is via an MCMC approach even though computing the normalizing constant is computationally intractable as $p \gg$. The idea of MCMC in general is to generate samples from the target distribution ($P(\mathbf{v})$ in this case) that constitute a *reversible* Markov Chain. That is, it has a transition kernel that satisfies the so-called *detailed balance condition* with support covering that of $P(\mathbf{v})$. When such property is achieved, it can be shown that $P(\mathbf{v})$ is the stationary distribution of the generated Markov chain [22]. In addition, in order to use the chain to estimate mathematical expectations from the probability distribution $P(\mathbf{v})$, one needs two more requirements to make the Markov chain *ergodic*:

- *Irreducibility*: every other state can be reached from any given state in a finite number of steps, and
- *Aperiodicity*: samples in the chain do not get stuck in cycles with period greater than one.

In such case, after removing first few observations from the chain (i.e. removing *burn-in* samples), the remaining samples can be used for most purposes as an i.i.d sample to estimate expectations under $P(\mathbf{v})$ as per the *Ergodic Theorem*.

Asymptotic guarantees for an ergodic markov chain. The asymptotic results from (19) and (20) can somewhat still be achieved for non i.i.d. samples. Namely, suppose $\mathcal{M}_{(MC)}^{(K)} = \{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}$ are realisations from an ergodic Markov chain with stationary distribution $P(\mathbf{v})$. Then the following hold:

1. According to the Ergodic Theorem [6], a generalization of the Strong Law of Large Numbers,
 - $\mathbf{v}^{(K)}$ converges in distribution to $P(\mathbf{v})$ as $K \rightarrow \infty$.
 - Moreover, as $K \rightarrow \infty$, with probability 1,

$$f_{\mathbf{v}_0 K} = \frac{1}{K} \sum_{i=1}^K I(\mathbf{v}^{(i)} = \mathbf{v}_0) \rightarrow \mathbf{E}[I(\mathbf{v}^{(i)} = \mathbf{v}_0)] = P(\mathbf{v}_0) \quad (21)$$

whereby $\mathbf{E}[h(\mathbf{v})]$ represents the mathematical expectation of a transformation of $\mathbf{v}$ by a function $h(\cdot)$ taken with respect to the probability distribution $P(\mathbf{v})$. This result is an analogue of (20).

2. Since the chain is ergodic, all its regions $V \in \mathcal{V}$ are recurrent, i.e. the chain returns to any state (or model) in the region V with probability 1, which includes all regions containing the optimal model $\mathbf{v}_0$. This result is an analogue of (19).

3.6.1 Gibbs Sampler Approach.

Implementing the MCMC-Based Random Search using Gibbs Sampler. There are a number of implementations on MCMC methods, here I will present the *Gibbs Sampler*. The transition kernel used in the sampler involves sampling each component v_j sequentially via their full univariate conditional distributions $P(v_j|v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_p)$, $1 \leq j \leq p$. The univariate conditional distribution is given as follows:

$$\begin{aligned}
& P(v_j = 1 | v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_p) \\
&= \frac{P(v_1, \dots, v_{j-1}, v_j = 1, v_{j+1}, \dots, v_p)}{P(v_1, \dots, v_{j-1}, v_j = 1, v_{j+1}, \dots, v_p) + P(v_1, \dots, v_{j-1}, v_j = 0, v_{j+1}, \dots, v_p)} \\
&= \frac{P(\mathbf{v}, v_j = 1)}{P(\mathbf{v}, v_j = 1) + P(\mathbf{v}, v_j = 0)} \tag{22}
\end{aligned}$$

$$= \frac{\exp\{-BIC(\mathbf{v}, v_j = 1 | D)\}}{\exp\{-BIC(\mathbf{v}, v_j = 1 | D)\} + \exp\{-BIC(\mathbf{v}, v_j = 0 | D)\}} \tag{23}$$

which is simply a Bernoulli distribution since v_i is binary-valued. Note that the normalizing constant C is canceled out in the last step. Below is an algorithm for generating a sample of models from $P(\mathbf{v})$ using the Gibbs Sampler.

Algorithm 1 Gibbs Sampling Algorithm for Random Search

Require: Dataset D , Number of iterations K' , Burn in B

Ensure: Sample of size $K = K' - B$ from $P(\mathbf{v})$

```

1: function GIBBS_RANDOM_SEARCH( $D, K', B$ )
2:   Arbitrarily choose a starting model  $\mathbf{v}^{(0)} = (v_1^{(0)}, \dots, v_p^{(0)})^T$ 
3:   for  $i \leftarrow 1$  to  $K'$  do
4:     for  $j \leftarrow 1$  to  $p$  do
5:       Compute  $w_1 = \exp\{-BIC(v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_j = 1, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)} | D)\}$ 
6:       Compute  $w_0 = \exp\{-BIC(v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_j = 0, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)} | D)\}$ 
7:       Compute  $P(v_j = 1 | v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)}) = \frac{w_1}{w_1 + w_0}$ 
8:       Sample  $v_j^{(i)} \sim \text{Bernoulli}\left(\frac{w_1}{w_1 + w_0}\right)$ 
9:     end for
10:    Record  $\mathbf{v}^{(i)} = (v_1^{(i)}, \dots, v_p^{(i)})^T$  as a sample
11:  end for
12:  Discard first  $B$  samples as initial burn-in period
13:  return  $(\mathbf{v}^{(B+1)}, \dots, \mathbf{v}^{(K')})$  as a  $K \times p$  matrix
14: end function

```

In this setting, after an initial burn-in period, the Markov chain induced by Gibbs sampler is indeed reversible and ergodic. One reason for the ergodicity is that the full univariate conditional probabilities are strictly greater than zero.

Time complexity analysis for Algorithm (1). The running time for this algorithm is $\mathcal{O}(K'p \cdot t)$, where t is the running time needed to complete the operations in the inner loop.

- Here, computing the "weights" w_1, w_0 require fitting two logistic regression models with dimensions q and $q - 1$ respectively, where $q \leq p$ is the dimension of the model at iteration i when the j -th coordinate of $\mathbf{v}^{(i)}$ is being updated (technically q is dependent on a number of factors including i and j but I drop these dependence for brevity).
 - For a logistic regression model fitted on a sample of n data points with q predictors using the *Iteratively Weighted Least Squares* (IWLS) algorithm (first proposed by McCullagh and Nelder [15]), the running time complexity is $\mathcal{O}(nq^2 + q^3)$; in practice, IWLS converges in a constant number of iterations.
 - This is because the IWLS algorithm alternates between two steps:
 1. Calculating a $n \times 1$ working response vector and n quantities stacked in a diagonal weight matrix (which require multiplying a $n \times q$ design matrix and a $q \times 1$ parameter estimate), which costs $\mathcal{O}(nq)$.

2. Solving a weighted least squares problem from regressing the working responses on the predictor variables, which costs $\mathcal{O}(nq^2 + q^3)$, primarily due to matrix multiplication to solve the weighted normal equations.
- This means that each inner loop's running time, which is dominated by fitting two logistic regression models, is:

$$t = 2[\mathcal{O}(nq) + \mathcal{O}(nq^2 + q^3)] = \mathcal{O}(nq^2 + q^3) \quad (24)$$

here, q is assumed to be the least upper bound of the dimensions of the models being calculated in the entirety of the algorithm. In turn, the overall running time for the Gibbs sampling algorithm is:

$$\mathcal{O}(K'p \cdot [nq^2 + q^3]) \quad (25)$$

- This means that the worst-case of this algorithm is when $q = p$, to which the running time is $\mathcal{O}(K'np^3 + K'p^4) = \mathcal{O}(K'p^3(n + p)) = \mathcal{O}(K'p^3 \max(n, p))$. When $n \geq p$, $\max(n, p) = n$ and the running could be further simplified as $\mathcal{O}(K'np^3)$. In other words, if K' and n are fixed, the running time is cubic in the number of candidate variables. However, this would be a pessimistic assumption, as the empirical result of the simulation study in Section (12.4) showed that, in later iterations of the algorithm, $q \leq 24$, which is much smaller than p . This observation provides an opportunity to simplify the running time: if one has reason to believe that $q \leq c = \mathcal{O}(1)$ in almost all iterations for some value c independent of p , the running time would be:

$$\mathcal{O}(K'p[nq^2 + q^3]) = \mathcal{O}(K'p[nc^2 + c^3]) = \mathcal{O}(K'np) \quad (26)$$

i.e. for a fixed K' , n , the time complexity would be linear in the number of candidate variables.

3.6.2 Metropolis-Hastings (MH) Algorithm.

A Metropolis-Hastings (MH) algorithm can also be used to generate a sample of models from $P(\mathbf{v})$:

Algorithm 2 Metropolis Hastings Algorithm for Random Search

Require: Dataset D , Number of iterations K' , Burn in B

Ensure: Sample of size $K = K' - B$ from $P(\mathbf{v})$

```

1: function METROPOLIS_HASTINGS_RANDOM_SEARCH( $D, K', B$ )
2:   Arbitrarily choose a starting model  $\mathbf{v}^{(0)} = (v_1^{(0)}, \dots, v_p^{(0)})^T$ 
3:   for  $i \leftarrow 1$  to  $K'$  do
4:     Propose a candidate model  $\tilde{\mathbf{v}}$  from the transition kernel  $q(\mathbf{v}|\mathbf{v}^{(i-1)})$ 
5:     Generate  $u \sim \text{Uniform}(0, 1)$ .
6:     Set  $\mathbf{v}^{(i)} = \tilde{\mathbf{v}}$  if
       
$$u \leq \min\left\{\frac{P(\tilde{\mathbf{v}})q(\mathbf{v}^{(i-1)}|\tilde{\mathbf{v}})}{P(\mathbf{v}^{(i-1)})q(\tilde{\mathbf{v}}|\mathbf{v}^{(i-1)})}, 1\right\}$$

       
$$= \min\left\{\exp(-[BIC(\tilde{\mathbf{v}}) - BIC(\mathbf{v})]) \cdot \frac{q(\mathbf{v}^{(i-1)}|\tilde{\mathbf{v}})}{q(\tilde{\mathbf{v}}|\mathbf{v}^{(i-1)})}, 1\right\}$$

7:     Otherwise, set  $\mathbf{v}^{(i)} = \mathbf{v}^{(i-1)}$ .
8:   end for
9:   Discard first  $B$  samples as initial burn-in period
10:  return  $(\mathbf{v}^{(B+1)}, \dots, \mathbf{v}^{(K')})$  as a  $K \times p$  matrix
11: end function

```

Remark 2. Remarks from Algorithm (2).

- The quantity $\min\{\frac{P(\tilde{\mathbf{v}})q(\mathbf{v}^{(i-1)}|\tilde{\mathbf{v}})}{P(\mathbf{v}^{(i-1)})q(\tilde{\mathbf{v}}|\mathbf{v}^{(i-1)})}, 1\}$ is called the *acceptance probability*, as Step 6 from the algorithm is essentially equivalent to accept the proposed model $\tilde{\mathbf{v}}$ with probability $\min\{\frac{P(\tilde{\mathbf{v}})q(\mathbf{v}^{(i-1)}|\tilde{\mathbf{v}})}{P(\mathbf{v}^{(i-1)})q(\tilde{\mathbf{v}}|\mathbf{v}^{(i-1)})}, 1\}$.
- Some choices for the transition kernel q include:
 1. The discrete uniform kernel, i.e. $q(\mathbf{v}|\mathbf{v}^{(i-1)}) = \frac{1}{2^p}$.
 2. The "one-bit flip" kernel, i.e.

$$q(\mathbf{v}|\mathbf{v}^{(i-1)}) = \frac{1}{p}, \text{ if } \mathbf{v} \text{ and } \mathbf{v}^{(i-1)} \text{ differs by one component. Otherwise, } q(\mathbf{v}|\mathbf{v}^{(i-1)}) = 0.$$

In other words, $\mathbf{v}$ takes p equally likely values, with each value differing by only one component to $\mathbf{v}^{(i-1)}$.

Regardless, using either of the transition kernels lead to the probabilities of the transition kernel ratios $\frac{q(\mathbf{v}^{(i-1)}|\tilde{\mathbf{v}})}{q(\tilde{\mathbf{v}}|\mathbf{v}^{(i-1)})}$ being cancelled out, which means the acceptance probability becomes:

$$\min\{\exp(-[BIC(\tilde{\mathbf{v}}) - BIC(\mathbf{v})]), 1\}$$

3.7 Assessing MCMC Convergence/Stationarity

1. P-charts (very tentative, graphical). This graphical method tracks the stationarity of the sampled models in a chain by monitoring the stability of the marginal frequencies $\bar{v}_j$. The idea is that if the chain has reached stationarity, then estimates of the target distributions' properties, for instance, the marginal distribution, should stabilize. However this visual method is not robust to potential multi-modalities in the target distribution when $p \gg$.
2. Gelman Rubin diagnostic using $\hat{R}$ on the BIC values $\{BIC(\mathbf{v}^{(1)}), BIC(\mathbf{v}^{(2)}), \dots, BIC(\mathbf{v}^{(K)})\}$ across multiple chains. Assuming that the BIC values $\{BIC(\mathbf{v}^{(1)}), BIC(\mathbf{v}^{(2)}), \dots, BIC(\mathbf{v}^{(K)})\}$ have a one-to-one mapping to $\mathbf{v}$, then stationarity of $\{BIC(\mathbf{v}^{(1)}), BIC(\mathbf{v}^{(2)}), \dots, BIC(\mathbf{v}^{(K)})\}$ implies stationarity of $\{v_j^{(1)}, v_j^{(2)}, \dots, v_j^{(K)}\}$ for all j . However, even if this does not hold, inclusion-wise $\hat{R}$ would still not be the main primary diagnostic tool since the goal is to explore regions of models with low-BIC (or high probability mass). Vehtari et al. [26] proposed a *rank-normalized* $\hat{R}$, which computes the within-chain and between-chain variabilities of the **rank**s of the BICs instead of the raw values. Rank-normalization makes the diagnostics less sensitive to extremely high BIC values, allowing the stationarity diagnostics to reflect stationary behavior near the optimal region, rather than noise from models with high BICs.

3.8 High-dimensional issues

Random Search Power in High Dimensions. The random search procedure implemented via Algorithm (1) or (2) is highly efficient when the number of candidate variables p (in turn, the number of candidate models $\mathcal{O}(2^p)$ is relatively small, say $p \leq 20$. However, the random search procedure gradually loses its power to identify the optimal model $\mathbf{v}_0$ (as defined in (18)) as $p \gg$. To see this, assume that K i.i.d. models can be generated from $P(\mathbf{v})$. Then the probability that $\mathbf{v}_0$ appears at least once in the sample can be reformulated by its relative frequency $f_{\mathbf{v}_0 K}$ (as defined in (20)) as follows:

$$P\left(f_{\mathbf{v}_0 K} \geq \frac{1}{K}\right) = P(K \cdot f_{\mathbf{v}_0 K} \geq 1) = 1 - (1 - P(\mathbf{v}_0))^K \quad (27)$$

where $K \cdot f_{\mathbf{v}_0 K} \sim Bi(K, P(\mathbf{v}_0))$ is essentially the number of times $\mathbf{v}_0$ appears in the sample, which follows a binomial distribution since $I(\mathbf{v}^{(i)} = \mathbf{v}_0)$ are i.i.d. Bernoulli random variables by assumption. When p is extremely large, the probability $P(\mathbf{v}_0)$ could be extremely small even though it is the largest among all probabilities $\{P(\mathbf{v}) : \mathbf{v} \in \mathcal{V}\}$. An intuition for this is that, as p gets arbitrarily large, the number of terms involved in the normalizing constant $C = (\sum_{\mathbf{v} \in \mathcal{V}} \exp\{-S(\mathbf{v}|D)\})^{-1}$ increases exponentially, making the probability mass $P(\mathbf{v}_0) = C \cdot \exp\{-BIC(\mathbf{v}_0|D)\}$ extremely small. Moreover, assuming a large K and small $P(\mathbf{v}_0)$ such that $KP(\mathbf{v}_0)$ is small, we have ¹:

¹When the sample is not i.i.d but constitute a Markov chain, a similar result can be argued, as per Proposition 3.21 from Aldous and Fill [1].

$$\begin{aligned}
& P[K \cdot f_{\mathbf{v}_0 K} \geq 1] \\
&= P[Bi(K, P(\mathbf{v}_0)) \geq 1] \\
&\approx P[\text{Poisson}(\lambda) \geq 1], \lambda = KP(\mathbf{v}_0) \\
&= 1 - e^{-\lambda}
\end{aligned} \tag{28}$$

Equation (28) is simply an application of approximating the Binomial Distribution with the Poisson Distribution for rare events. Further, taking $e^{-\lambda} \approx 1 - \lambda$ around $\lambda = 0$ using a first-order Taylor Expansion, we have:

$$P[K \cdot f_{\mathbf{v}_0 K} \geq 1] \approx KP(\mathbf{v}_0) \tag{29}$$

A few implications of these results:

- From Equation (29), probability of $\mathbf{v}_0$ appearing in the sample may still be very small even when K is large or that the chain has appeared stationary. In such case, it is very likely the case that the models generated from the random search procedure are different.
- From Equation (28), an back-of-the-envelope calculation implies that, to get a 50% chance of observing the optimal model $\mathbf{v}_0$ in the sample, one would need to generate $K = \ln 2 / P(\mathbf{v}_0)$ samples. For example, if $P(\mathbf{v}_0) \approx 10^{-4}$, one would need to generate over $K \approx 6931$ models to even have a 50% chance of observing $\mathbf{v}_0$.

There is still hope: obtaining a correct model in high dimension settings. Despite the limitations of the random search procedure when $p \gg$, Qian and Field [18] showed that the model with the smallest BIC in the generated sample is still likely to be a correct model in $\mathcal{A}_c$, even for moderately large values of K . They also noted that a consequence of this is the possibility of a two-step random search: first on the entire model space, and second on the the model space from the selected variables from the first run to increase the possibility of finding smaller models with an even lower BIC.

Relative frequencies: the most important explanatory variables. If p is large and the dimension of the true model $\mathbf{v}^*$ is large, then the model that minimizes the BIC from the generated models may not be sufficient to identify the optimal model $\mathbf{v}_0$. In such case, in addition to reporting the BIC-minimizer model in the sample, one could also report the marginal relative frequencies via calculating the sample mean from the generated samples $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}$, denoted as:

$$\bar{\mathbf{v}}^{(K)} = \frac{1}{K} \sum_{k=1}^K \mathbf{v}^{(k)}$$

From this sample mean vector, we can obtain the marginal relative frequency for the j -th variable as:

$$\bar{v}_j = \frac{1}{K} \sum_{k=1}^K v_j^{(k)}, \quad 1 \leq j \leq p$$

The p relative frequencies can be used to rank the "importance" of each predictor variable's effects to the response, or could even serve as an alternative method to select the best variable set via, say, $\{X_j \mid \bar{v}_j > 0.5\}$. Note that the relative frequencies also provide an estimate of the marginal distributions of $P(\mathbf{v})$ (i.e. $\bar{v}_j$ provides an estimate of $P(v_j = 1) = \sum_{\mathbf{v}: v_j=1} P(\mathbf{v}) \propto \sum_{\mathbf{v}: v_j=1} \exp(-BIC(\mathbf{v}))$).

3.9 Tempered Target Distribution using λ

Qian et al. [20] introduced a general form of the induced probability distribution from Equation (14) and Remark (1) to control the exploration of the procedure:

$$P_\lambda(\mathbf{v}) \propto \exp(-\lambda \cdot BIC(\mathbf{v})) \quad (30)$$

where $\lambda > 0$ is an "inverse-temperature" tuning parameter. If λ is too large, the procedure getting stuck on a local minimum ("too cold"). Otherwise, if it is too small, it may slow down the procedure to navigate through the neighborhood of the optimal model $\mathbf{v}_0$ due to possibly sampling too many suboptimal models ("too hot").

3.9.1 Generalized Gibbs Sampler Algorithm.

Algorithm 3 Generalized Gibbs Sampling Algorithm for Random Search

Require: Dataset D , Number of iterations K' , Burn in B

Ensure: Sample of size $K = K' - B$ from $P(\mathbf{v})$

```

1: function GENERALIZED_GIBBS_RANDOM_SEARCH( $D, K', B$ )
2:   Arbitrarily choose a starting model  $\mathbf{v}^{(0)} = (v_1^{(0)}, \dots, v_p^{(0)})^T$ 
3:   for  $i \leftarrow 1$  to  $K'$  do
4:     for  $j \leftarrow 1$  to  $p$  do
5:       Compute  $w_1 = \exp\{-\lambda \cdot BIC(v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_j = 1, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)} | D)\}$ 
6:       Compute  $w_0 = \exp\{-\lambda \cdot BIC(v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_j = 0, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)} | D)\}$ 
7:       Compute  $P(v_j = 1 | v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)}) = \frac{w_1}{w_1 + w_0}$ 
8:       Sample  $v_j^{(i)} \sim \text{Bernoulli}\left(\frac{w_1}{w_1 + w_0}\right)$ 
9:     end for
10:    Record  $\mathbf{v}^{(i)} = (v_1^{(i)}, \dots, v_p^{(i)})^T$  as a sample
11:  end for
12:  Discard first  $B$  samples as initial burn-in period
13:  return  $(\mathbf{v}^{(B+1)}, \dots, \mathbf{v}^{(K')})$  as a  $K \times p$  matrix
14: end function

```

3.9.2 Generalized MH Algorithm.

Algorithm 4 Generalized Metropolis Hastings Algorithm for Random Search

Require: Dataset D , Number of iterations K' , Burn in B

Ensure: Sample of size $K = K' - B$ from $P(\mathbf{v})$

```

1: function GENERALIZED_METROPOLIS_HASTINGS_RANDOM_SEARCH( $D, K', B$ )
2:   Arbitrarily choose a starting model  $\mathbf{v}^{(0)} = (v_1^{(0)}, \dots, v_p^{(0)})^T$ 
3:   for  $i \leftarrow 1$  to  $K'$  do
4:     Propose a candidate model  $\tilde{\mathbf{v}}$  from the transition kernel  $q(\mathbf{v} | \mathbf{v}^{(i-1)})$ 
5:     Generate  $u \sim \text{Uniform}(0, 1)$ .
6:     Set  $\mathbf{v}^{(i)} = \tilde{\mathbf{v}}$  if

$$u \leq \min\left\{\frac{P(\tilde{\mathbf{v}})q(\mathbf{v}^{(i-1)} | \tilde{\mathbf{v}})}{P(\mathbf{v}^{(i-1)})q(\tilde{\mathbf{v}} | \mathbf{v}^{(i-1)})}, 1\right\}$$


$$= \min\left\{\exp\left(-\lambda[BIC(\tilde{\mathbf{v}}) - BIC(\mathbf{v}^{(i-1)})]\right) \cdot \frac{q(\mathbf{v}^{(i-1)} | \tilde{\mathbf{v}})}{q(\tilde{\mathbf{v}} | \mathbf{v}^{(i-1)})}, 1\right\}$$

7:     Otherwise, set  $\mathbf{v}^{(i)} = \mathbf{v}^{(i-1)}$ .
8:   end for
9:   Discard first  $B$  samples as initial burn-in period
10:  return  $(\mathbf{v}^{(B+1)}, \dots, \mathbf{v}^{(K')})$  as a  $K \times p$  matrix
11: end function

```

3.9.3 How does this change the target distribution?

Since $\lambda > 0$ by design, the mode of $P_\lambda(\mathbf{v})$ will still be $\mathbf{v}_0$, the global BIC minimizer across the entire model space. Further, the relative ordering between BICs between two models are also preserved ($BIC(\mathbf{v}_1) < BIC(\mathbf{v}_2) \implies P_\lambda(\mathbf{v}_1) > P_\lambda(\mathbf{v}_2)$).

3.9.4 Effect of λ on Gibbs

From Algorithm (3), the j -th coordinate is sampled in the i -th iteration of the procedure in the following manner:

$$\begin{aligned} v_j^{(i)} &\sim \text{Bernoulli}\left(\frac{w_1}{w_1 + w_0}\right), \\ P(v_j = 1 | v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)}) &= \frac{w_1}{w_1 + w_0} \\ w_k &= \exp\{-\lambda \cdot BIC(v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_j = k, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)} | D)\}, \quad k \in \{0, 1\}, \end{aligned}$$

Let $\Delta BIC = BIC(\mathbf{v}_{j=0}^{(i)}) - BIC(\mathbf{v}_{j=1}^{(i)})$, where $\mathbf{v}_{j=k}^{(i)} = (v_1^{(i)}, \dots, v_{j-1}^{(i)}, k, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)})^T$, $k \in \{0, 1\}$. To see the effect of λ at influencing the inclusion of variable j at the i -th iteration, notice that the Bernoulli parameter can be rewritten as the following logistic function:

$$\frac{1}{1 + \exp(-\lambda \cdot \Delta BIC)} = \sigma(\lambda \cdot \Delta BIC)$$

Suppose $\Delta BIC > 0$ i.e. the model including the j -th variable ($\mathbf{v}_{j=1}^{(i)}$) has lower BIC. This means that $\sigma(\lambda \cdot \Delta BIC) > 0.5$ since $\lambda > 0$.

To further examine the effect of ΔBIC and λ , various inclusion probability values are calculated in the following table and figure:

Table 4: Bernoulli inclusion probability $\sigma(\lambda \Delta BIC)$ for selected $\Delta BIC = BIC(\mathbf{v}_{j=0}^{(i)}) - BIC(\mathbf{v}_{j=1}^{(i)})$ (rows) and tuning λ (columns) where $\lambda = 1$ results in the usual vanilla random search procedure. Negative ΔBIC disfavors inclusion.

ΔBIC	$\lambda = 0.1$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 5$
-1.50	0.46257	0.320821	0.182426	0.000553
-1.00	0.475021	0.377541	0.268941	0.006693
-0.50	0.487503	0.437823	0.377541	0.075858
-0.30	0.492501	0.462570	0.425557	0.182426
-0.20	0.495000	0.475021	0.450166	0.268941
-0.10	0.497500	0.487503	0.475021	0.377541
-0.05	0.498750	0.493750	0.487503	0.437823
0.05	0.501250	0.506250	0.512498	0.562180
0.10	0.502500	0.512498	0.524983	0.622459
0.20	0.505000	0.524983	0.549834	0.731059
0.30	0.507500	0.537442	0.574442	0.817575
0.50	0.512498	0.562180	0.622459	0.924142
1.00	0.524983	0.622459	0.731059	0.993307
1.50	0.537429	0.679178	0.817575	0.999447

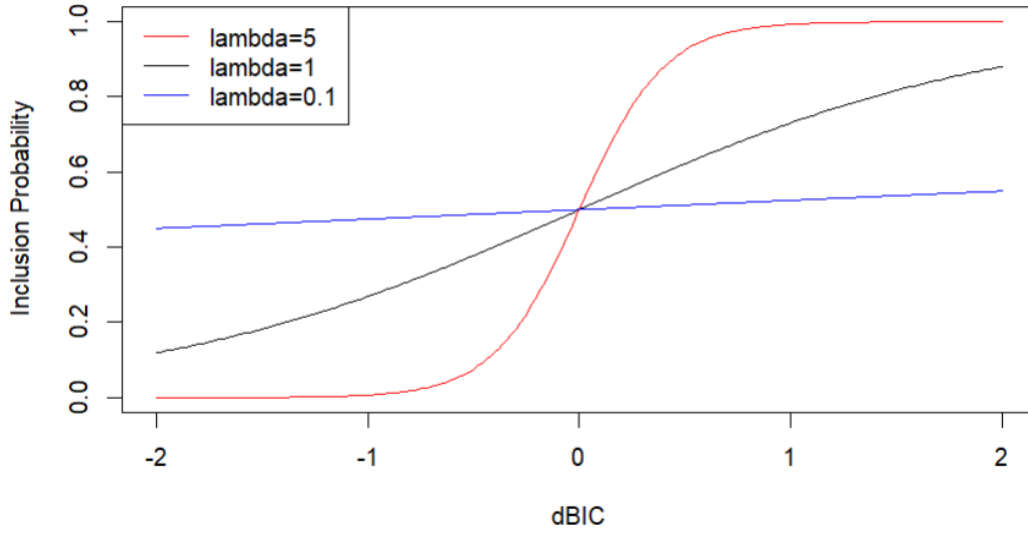


Figure 2: Bernoulli inclusion probability as a function of ΔBIC with varying λ . When λ is large, larger ΔBIC prefers the inclusion (and smaller ΔBIC discourages inclusion), whereas smaller λ discourages the inclusion by gradually flattening the inclusion probabilities, even when $\Delta BIC > 0$. In fact, as λ is closer to 0, the curve (subsequently, the Bernoulli inclusion probability), flattens to 0.5, reducing the procedure to a random walk-like behavior regardless of whether including variable j decreases the BIC.

From Table (4) and Figure (2), the behavior of the Bernoulli inclusion probability $\sigma(\lambda \cdot \Delta BIC)$ depends on $\Delta BIC = BIC(\mathbf{v}_{j=0}^{(i)}) - BIC(\mathbf{v}_{j=1}^{(i)})$:

- Case 1: $\Delta BIC > 0$.
 - Then the Bernoulli inclusion probability will always exceed 0.5.
 - The slope of the logistic curve is sharper for larger λ : for example, with $\lambda = 5$, even a small BIC reduction of 0.3 already yields at least 0.80 inclusion probability. However, for small $\lambda = 0.1$, the same BIC difference gives only 0.507 – although including variable j reduces the BIC, the decision to include it remains like a coin flip.
- Case 2: $\Delta BIC < 0$.
 - Then the Bernoulli inclusion probability will always be less than 0.5, so inclusion of variable j is not favored.
 - Larger λ pushes this probability close to 0 (for instance, if $\Delta BIC = -0.3$ then $\lambda = 5$ only gives a 0.182 inclusion probability).
 - For small λ , however, probabilities stay near 0.5, reflecting an exploratory behavior (for $\Delta BIC = -0.3$ then $\lambda = 0.1$ still gives a 0.482 inclusion probability).

3.9.5 Effect of λ on MH

From algorithm (4), assuming a transition kernel mentioned on (2), the acceptance probability of a proposed model $\tilde{\mathbf{v}}$ is:

$$\min\{\exp(\lambda \Delta^{MH} BIC), 1\}$$

where $\Delta^{MH} BIC \equiv BIC(\mathbf{v}^{(i-1)}) - BIC(\tilde{\mathbf{v}})$ such that $\Delta^{MH} BIC > 0$ favors the proposed model. Notice that when $\Delta^{MH} BIC > 0$, for any λ , the proposed model will *always* be accepted since $\exp(\lambda \Delta^{MH} BIC) > 1$.

To further examine the effect of $\Delta^{MH} BIC$ and λ , various acceptance probability values are calculated in the following table and figure:

Table 5: MH acceptance probabilities: $\min\{\exp(\lambda\Delta^{MH}BIC), 1\}$

, where $\Delta^{MH}BIC \equiv BIC(\mathbf{v}^{(i-1)}) - BIC(\tilde{\mathbf{v}})$													
λ	$\Delta^{MH}BIC=-1.5$	-1	-0.5	-0.3	-0.2	-0.1	-0.05	0.05	0.1	0.2	0.3	0.5	1
0.1	0.861	0.905	0.951	0.970	0.980	0.990	0.995	1.000	1.000	1.000	1.000	1.000	1.000
0.5	0.472	0.607	0.779	0.861	0.905	0.951	0.975	1.000	1.000	1.000	1.000	1.000	1.000
1	0.223	0.368	0.607	0.741	0.819	0.905	0.951	1.000	1.000	1.000	1.000	1.000	1.000
5	0.001	0.007	0.082	0.223	0.368	0.607	0.779	1.000	1.000	1.000	1.000	1.000	1.000

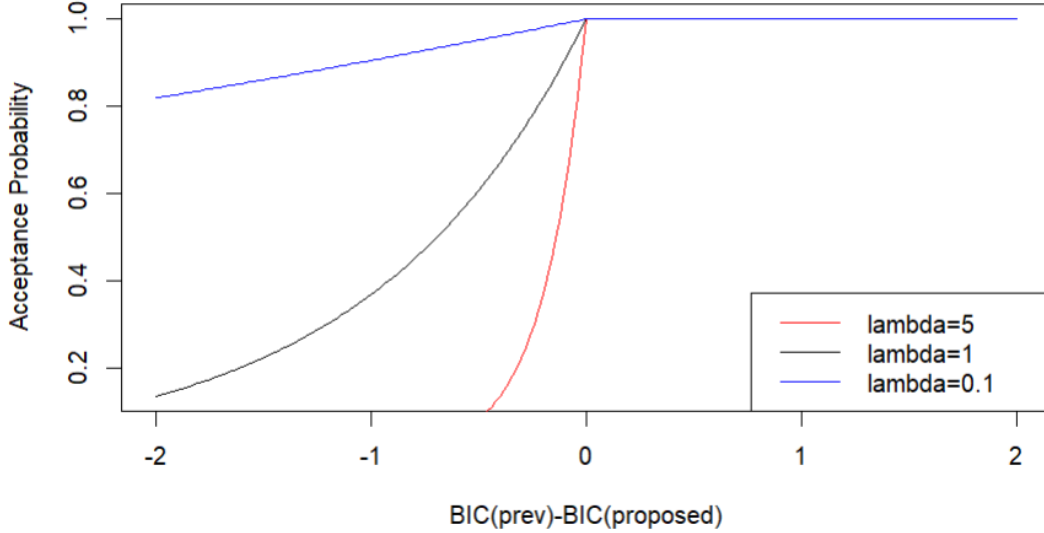


Figure 3: Acceptance probability $\min\{\exp(\lambda\Delta^{MH}BIC), 1\}$ as a function of $\Delta^{MH}BIC \equiv BIC(\mathbf{v}^{(i-1)}) - BIC(\tilde{\mathbf{v}})$. If the proposed model is better ($\Delta^{MH}BIC > 0$) it will always be accepted. Otherwise, λ controls how much one rejects this model – with larger λ being more stringent. and smaller λ still favoring the proposed model despite the higher BIC to allow for exploration.

Table (5) and Figure (3) show that positive $\Delta^{MH}BIC$ always favors the proposed model $\tilde{\mathbf{v}}$ regardless of λ . For negative $\Delta^{MH}BIC$, whoever, larger λ sharpens the proposed model’s rejection (for example, for $\Delta^{MH}BIC = -0.3$, acceptance probability drops to 0.22), while smaller λ is more explanatory (for $\Delta^{MH}BIC = -0.3$, the acceptance probability is still 0.97).

4 Simulation Study: Factorial Analysis

4.1 High-level Design

To get an understanding on how the random search procedure behaves under key challenges present in the TC genesis dataset (as outlined in (2)), a *simulation study* is conducted in a partial *factorial design* of four factors. Each scenario systematically varies four factors that reflect the four key challenges – high dimensionality, multicollinearity, varying *Signal-To-Ratio* (SNR), and sample size, whereby a synthetic dataset (X, y) is generated for each scenario. Although the datasets are generated with a known subset of “ground truth” predictors, during the simulation, the random search procedure only has access to the full dataset (X, y) without any knowledge about the underlying data generating process – just as similar to how the true model is unknown in real life settings. The procedure’s performance will then be evaluated by how well it can recover the ground truth variables. **Note that I will consider the Gibbs Sampler implementation for this simulation study.** The 4 factors under consideration are:

1. SNR of the ground truth predictor variables to the response. Three levels are considered: strong, moderate, weak
 - Unfortunately, there is no fixed definition of SNR as it is not inherently a statistical concept, but the convention is that a high SNR means that the contribution of the ground truths is strong and consistent, while a low SNR means that the true signal is drowned out by noise and sampling variation, making it statistically harder to detect (i.e. it still contributes to the data generating process, but its signal is "not obvious" from the observed dataset).
 - In the context of variable selection, it is not impossible for each ground truth variable to have differing signals, which might be influenced by collinearity. Hence, to control each variable's signal, we need a bespoke, qualitative measure of the overall SNR and variable-wise signal. This will be explained further in Definition (4.1).
2. The correlation structure between ground truth predictors. Five levels are considered: no correlation, positively moderate correlation, positively high correlation, negatively moderate correlation, and negatively high correlation.
3. The sample size n . Three levels are considered: small ($n = 200$), moderate ($n = 2000$) and large ($n = 10000$).
4. The number of candidate variables p , containing ground truth predictors and "redundant" predictors (i.e. not involved in generating the response), as per the discussion on (3.8). Two levels are considered: small ($p = 16$) and large ($p = 160$).

A few remarks on this design:

- For consistency, we do not consider the intercept in the data generating process (i.e. $\beta_0 = 0$)
- Ideally, one would consider more levels (for instance, at least 3 levels for each factor) to evaluate the procedure's *boundary performance*, i.e. at what point does the procedure's performance becomes suboptimal (for example, consistently not outputting a correct model)?
- Moreover, while it is desirable to evaluate the random search procedure under all ($3 \times 5 \times 2 \times 2 = 60$) possible combinations, a subset is chosen instead that balances computational feasibility and breadth. Therefore, the following 33 factor combinations are considered:
 1. $n = 200$, $p = 16$ across all 15 combinations of SNR and correlation structure.
 2. $n = 2000$, $p = 16$ across all 15 combinations of SNR and correlation structure.
 3. $n = 2000$, $p = 160$ across 2 combinations of SNR and correlation structure, namely moderate SNR with high correlated ground truths and moderate SNR with uncorrelated ground truths.
 4. $n = 10000$, $p = 160$ across a moderate SNR with high correlation structure.
- For each scenario/factor combination, several statistics are considered to evaluate the random search's performance:
 1. Top k variables with the largest relative frequencies. Here, k is chosen such that the top k variables include the ground truths.
 2. The "BIC Minimizer", which is the model that has the smallest BIC in the generated sample:

$$\hat{\mathbf{v}}_0 = \arg \min_{\mathbf{v} \in \{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}} BIC(\mathbf{v}|D)$$

If the BIC Minimizer is not a correct model, then the top 2-5 BIC Minimizers, namely the models with the 2nd to 5th smallest BICs, will be outputted alongside with their BIC value.

3. The BIC value of the BIC minimizer, namely $BIC(\hat{\mathbf{v}}_0|D)$.

4.2 Data generating process for each synthetic scenario.

Given a fixed sample size n and number of candidate variables p , the $n \times p$ design matrix X is comprised of n observations from a p -dimensional multinormal distribution $(x_1, \dots, x_p)^T \sim \mathcal{N}_p(\mathbf{0}, \Sigma)$ where the $p \times p$ Σ is defined as a block diagonal matrix as follows:

$$\Sigma = \begin{bmatrix} \Sigma_{gt1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \Sigma_{gt2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Sigma_r \end{bmatrix} \quad (31)$$

Here, the ground truth variables (that will be used to generate y) are set to be four variables $x_1, \dots, x_4$ – the idea is to stress-test the random search procedure under sparsity conditions whereby the number of ground truths is small compared to p . The block covariance matrices are defined as follows:

1. $\Sigma_{gt1} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$ is the covariance matrix for the first 2 ground truths $(x_1, x_2)^T$ with controlled correlation and unit variance. In other words, correlation between ground truth predictors are controlled via the ρ parameter, which could take any value from $\{-0.9, -0.5, 0, +0.5, +0.9\}$, depending on which correlation level is specified.
2. $\Sigma_{gt2} = I_2$ is the covariance matrix for the last 2 ground truths $(x_3, x_4)^T$, specified to be uncorrelated with unit variance.
3. $\Sigma_r = I_{p-4}$ is the covariance matrix for the redundant variables $(x_5, \dots, x_{p-4})^T$, specified to be uncorrelated with unit variance.
4. The 3 groups of variables are designed to be mutually uncorrelated, shown by the zero matrix $\mathbf{0}$ (with appropriate dimensions) occupying off block diagonal entries.

Given a realization of n random vectors from $\mathcal{N}_p(\mathbf{0}, \Sigma)$, the $n \times 1$ response vector is generated **only from the ground truth variables** $x_1, \dots, x_4$ – namely, $y_i \sim Bi(1, \sigma(\beta_1 x_{i1} + \dots + \beta_4 x_{i4}))$. The true coefficient values $\beta_1, \dots, \beta_4$ are generated depending on the desired level of SNR as follows:

Definition 4.1. Operational SNR of the ground truth variables. Given a design matrix X , a response y , and a fixed factor combination $(n, p, \text{SNR level, ground truth correlation } \rho)$, (observations of) the ground truth variables' SNR are defined such that:

Strong SNR $\equiv$ All $x_1, \dots, x_4$ in the data exhibit *strong signal*.

Moderate SNR $\equiv x_1, x_3$ exhibit *strong signal*, and x_2, x_4 exhibit *moderate signal*.

Weak SNR $\equiv x_1, x_3$ exhibit *strong signal*, and x_2, x_4 exhibit *weak signal*. (32)

The definition of variable-wise *signal* will be explained on Definition (4.2). Nevertheless, determining the values in $\beta_1, \dots, \beta_4$ require a trial-and-error process that could be described as follows:

1. Specify an "initial value" of $\beta_1, \dots, \beta_4$.
2. Generate a response vector y' using the current value of the β_j s.
3. Fit a logistic regression model using the dataset (X, y') .
4. If the estimated coefficients $\hat{\beta}_j$ s satisfy the criteria stated at (32) and Definition (4.2), accept y' as the synthetic response vector y for this factor combination. Otherwise, repeat step 1.

The reason x_2, x_4 is chosen to have their 'signals' altered as a proxy of the desired SNR level (and not any other pair) is due to how x_1, x_2 are already designed to be correlated, and how different 'signals' across correlated variables could be another challenge of stress-testing the random search procedure to ensure that it does not simply pick one out of the correlated pair.

Definition 4.2. Defining Ground Truth Variable Signal via Tests. Suppose we have a fixed sample size n as well as a fixed design matrix comprised of observed predictors across ground truth variables (with controlled correlation level) and redundant variables. A ground truth variable’s signal is quantified using the **marginal LRT and Wald Tests** in the true model. In particular, we label a ground-truth variable X_j as having a **strong/moderate/weak signal** by a p-value band, namely:

$$\begin{aligned} \text{A variable has strong signal} &\equiv 0 < \text{p-value} < 0.01 \\ \text{A variable has moderate signal} &\equiv 0.01 \leq \text{p-value} < 0.05 \\ \text{A variable has weak signal} &\equiv 0.05 \leq \text{p-value} < 0.1 \end{aligned} \tag{33}$$

To mitigate the HDE, the p-value band for a ground truth variable with fixed signal **must agree for both Wald and LRT tests**, and that most of the fitted probabilities from the true model should not be either close to 0 or 1. For example, if a variable X_j is considered to have a strong signal, then both Wald and LRT p-values for the test $H_0 : \beta_j = 0$ must be less than 0.01. For a detailed justification/rationale, refer to this section of the Appendix (12.1).

4.3 Discussion

4.3.1 Discussing the random search results.

For each scenario, 2 chains of 2200 models (with first 200 as burn-in samples) with different initial values were generated using the Gibbs Sampler in Algorithm (1). Chains across all 33 scenarios showed reasonable mixing (normalized $\hat{R}$ on sampled BICs across chains, P-charts of marginal frequencies’ stability), thus stationarity won’t be further discussed. Some of the P-charts are displayed in Figures (23) and (24) in the Appendix.

For detailed plots and figures, refer to this section of the Appendix (12.2). Also, for brevity, graphical plots are only presented for selected scenarios/factor combinations. For the $n = 2000$, $p = 16$ and $n = 200$, $p = 16$ case, only 2 scenarios are considered: $\text{Corr}(X1, X2) = 0.9$ (high correlation) with moderate SNR and high correlation with weak SNR.

How does the procedure perform in retrieving ground truth variables?

1. Even under moderate signal-to-noise ratio and high correlation between ground truths, the random search has been able to consistently recover all ground-truth predictors most of the time.
2. However, performance dropped slightly in the weak-signal cases, where some true variables were missed.
3. Overall, this confirms that the sampler behaves as expected — it’s robust to noise and correlation when the underlying signal can be detected.

How do the selected factors interplay with each other to influence the random search results?

1. **Sample size n sweeps off redundancy.** As n grows, redundant variables are included less frequently among the sampled models even when $p = 160$ is large (see the model path heatmaps (16) and (20) as well as the top relative frequencies (17) and (21)). The procedure is also able to find smaller models more quickly: as n increases from 2000 to 10000, the procedure is able to explore models with 5-7 fewer dimensions – 10-15 variables on average for $n = 2000$ and 5-8 variables on average for $n = 10000$ (see the trace plots (18) and (22)).
2. **Correlation on high dimensional settings.** For a fixed n and $p \gg$, correlation between ground truths could lower the importance of some ground-truths. In particular, figure (17) shows that the redundant variables X_{105} , X_{116} , X_{110} are sampled more frequently than ground truth X_4 by virtue of their relative frequencies. Nonetheless, the procedure is able to pick a correct model, albeit with one redundant variable

X_{105} (see Table (18)). I will further discuss that the high importance of X_{105} is most likely due to sampling variation (section (4.3.3)).

3. **Variable-wise signal.** Regardless of p , the strongest ground truths (X_1 and X_3 by design) remained to be the most important, (see the relative frequency bar plots (19), (17), (21)). However, it might be slightly sensitive to sample size, since the relative frequency for X_1 is slightly lower than X_3 's despite both having strong signal (bar plot (11)). This is not a concern to the TC Genesis application since the dataset has a fairly large sample size $n \approx 3 \times 10^4$.
4. **Different SNR Levels.** Lower SNR impacts inclusion of ground truths. For instance, X_2 and X_4 are less frequently included for weak SNR compared to moderate SNR (see bottom and top plots of Figure (15), and recall that X_2 and X_4 's signals are primarily used to define each SNR level, from Definition (4.1)). However, the procedure is still able to place the ground truth variables at the top even under weak SNR (see the top relative frequencies from Table (17)).
5. **Bonus: Comparison with LASSO.** Even under moderate sample size in high dimensional settings, the procedure is still able to pick simpler models containing almost all ground truths (see (18)) compared to LASSO (see (20) and (19)). Though, this difference is somewhat expected since LASSO is primarily used to minimize prediction error using a convex penalty.

4.3.2 Using relative frequencies in practice.

- Excluding the scenarios with weak SNR, defining important variables by relative frequency above 0.5 only misses at most one ground truths. For instance, the relative frequencies of X_1, X_3 , and either X_2 or X_4 are all above 0.5 (see Tables 21, 18).
- This implies that using the above 0.5 relative frequency threshold, on average, results in a near-correct model.

4.3.3 Effect of sampling variation in high-dimensional settings.

Throughout some high-dimensional scenarios, redundant variables not part of the data generating process are still included in the BIC minimizer (see inclusion of X_5 and X_{105} in the BIC minimizers from Table (18)). However, this seems to be just an artifact of sampling variation.

To demonstrate this, multiple datasets (corresponding to the $n = 2000$, $p = 160$, $Corr(X_1, X_2) = 0.9$, moderate SNR scenario) are generated and the BIC for the true model (X_1, X_2, X_3, X_4) and the "augmented model" ($X_1, X_2, X_3, X_4, X_{105}$) are compared, as shown in Figure (4). Define

$$\Delta \text{BIC} = \text{BIC}_{\text{true}} - \text{BIC}_{\text{aug}}, \text{ so that } \Delta \text{BIC} < 0 \text{ favours the true model.}$$

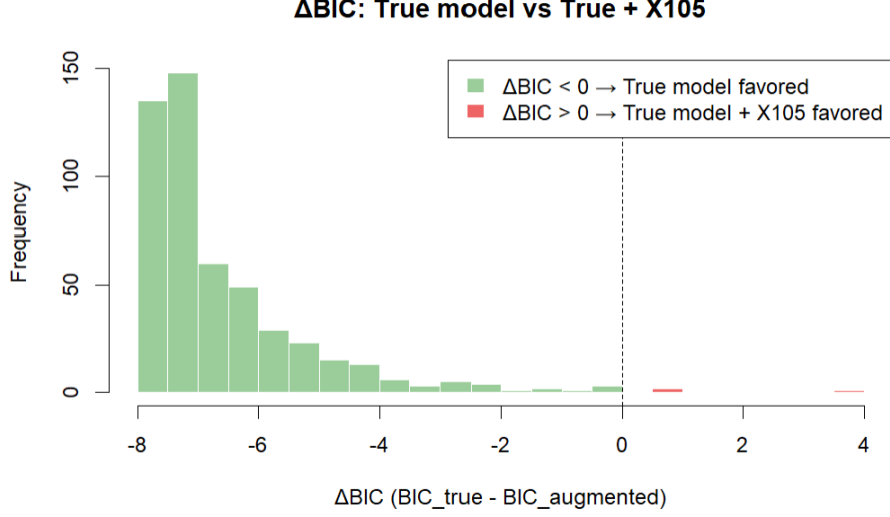


Figure 4: Histogram of $\Delta BIC = BIC_{true} - BIC_{aug}$ across 500 replicated datasets for $n = 2000$, $p = 160$, $Corr(X1, X2) = 0.9$, moderate SNR. $\text{Median}(\Delta BIC) = -7.11$ (with IQR 1.38). $\Pr(\Delta BIC > 0) = 1\%$.

Figure (4) shows that, in 99% of the 500 replicates, the true model has lower BIC than the augmented model (true model + X_{105}), with a median ΔBIC of -7.11. A few remarks of **what this result does and does not imply**:

- **Pairwise preference only.** Whenever the chain visits *both* models, it will prefer the true model on 99% of datasets under this setting.
- **No claim about the global minimum.** This two-model comparison does not estimate the probability that the true model is the *global* BIC minimizer among all 2^p models, nor does it account for finite-chain coverage or the possibility of a third model with even lower BIC.
- **Interpretation for Table (18).** Regardless, one still has confidence that the occasional BIC minima including redundant variables e.g. X_{105} are due to finite-sample variability rather than a systematic failure of the procedure.

4.3.4 Top-k BIC Minimizers?

Section (4.3.3) provides an explanation as to why redundant variables can creep in into the BIC minimizer-models, but does not necessarily tell what to do next in such situations.

Rather than reporting a single BIC minimizer (i.e. the model that minimizes the BIC among the sampled models), which may not necessarily correspond to the true model (or even a correct model) due to e.g. finite-sample variability, we therefore summarize the **Top-5 Lower BIC models** for this simulation study.

- Given two competing models, the difference in BIC (or referred to as the *Schwarz Criterion*) could be regarded as an approximation to the *Bayes Factors* of the two models (as stated by Raftery and Kass [12]), which is simply the ratio of marginal likelihoods of two model specifications.
- In particular, Bayes Factors provide a quantitative summary of the data in favor of a theory represented by a statistical model M_1 , relative to another theory (represented by another model, say M_2). Raftery and Kass [12] also provided an interpretation of differences in BIC as to how well does the data provide evidence to a model relative to another; a BIC difference of:

- 0 to 2 is regarded as "not worth more than a bare mention",
- 2 to 6 is regarded as having "positive" evidence,

- 6 to 10 regarded as having "strong" evidence,
- >10 is regarded as having "very strong" evidence.

In the first case, two models with BIC differences of no more than 2 can therefore be regarded as being statistically indistinguishable on the observed data.

- This means that under large n , if the model picked by the procedure is not correct, but the (unobservable) true model among the top 5 BIC minimizers and its BIC differs marginally, then it can serve as a proxy of the true model. Even better, domain knowledge could assist to manually identify the true model as best as possible.

Table 6: Scenarios where the BIC minimum is *not* a correct model (13 of 33 total). Columns partition cases by closeness to the true model via $\Delta\text{BIC} = \text{BIC}(\text{true}) - \text{BIC}_{\min}$.

	$\Delta\text{BIC} > 2$	$\Delta\text{BIC} \leq 2$
True model not in Top 5 BIC Minimizer (6 scenarios)	• $n=2000, p=16, \text{weak}, 0$	• $n=2000, p=16, \text{weak}, 0.5$ • $n=2000, p=16, \text{weak}, -0.5$ • $n=2000, p=16, \text{weak}, -0.5$ • $n=2000, p=160^\dagger, \text{moderate}, 0$
True model in Top 5 BIC Minimizers (8 scenarios)	• $n=2000, p=16, \text{weak}, 0.5$ • $n=2000, p=16, \text{weak}, 0.9$ • $n=2000, p=16, \text{weak}, -0.9$	• $n=200, p=16, \text{weak}, 0$ • $n=200, p=16, \text{weak}, 0.9$ • $n=200, p=16, \text{weak}, -0.9$ • $n=2000, p=16, \text{moderate}, 0.5$ • $n=10000, p=160^\dagger, \text{moderate}, 0.9$

Entries list $(n, p, \text{SNR level}, \text{corr}(X_1, X_2))$.

† High-dimensional case ($p = 160$).

" ≤ 2 " means the BIC minimum and the true model are *statistically indistinguishable* by the usual $\Delta\text{BIC} \leq 2$ heuristic.

Table (6) shows that across all scenarios/factor combinations, 13 had a BIC minimum that is not a correct model. These 13 are classified as:

1. True model not sampled at all (and not in the Top 5 BIC Minimizers). This includes 6 scenarios, but mostly:
 - Small n, p and
 - High p with fixed n
2. True model is sampled, and included in the Top 5 BIC Minimizers. This includes 8 scenarios, and out of those 8, 5 of them ($\tilde{63}\%$) have $\Delta\text{BIC} \leq 2$ with the true model, i.e. the true model is statistically indistinguishable from the reported BIC minimizer, **even in the large** ($n = 10000, p = 160$) **case**.

4.3.5 BIC's $\log n$ hurdle explains "missed" ground truths.

In the SNR definitions (see Definition (4.1) and (4.2)), X2 and X4 are intentionally set to have weaker signals than X1 and X3. For the BIC to accept a model with an additional variable, one requires $2 \cdot \Delta \log \text{lik} > \log n$ (here $\Delta \log \text{lik}$ is the difference in log-likelihood of the model with the one variable addition and the one without). Since $2 \cdot \Delta \log \text{lik} \approx z^2$ asymptotically (z^2 is the square of the Wald Statistic), we have $2 \cdot \Delta \log \text{lik} > \log n \implies z^2 > \log n$. Taking $n = 2000$ so that $\log n \approx 7.60$, "moderate" signals, (which according to Definition (4.2) is defined as the LRT test p-value between $[0.01, 0.05]$), **can be deliberately, yet correctly** excluded by the BIC. As a quick example, a variable with Wald Statistic $z = 2.1$ technically have its LRT p-value equal to $0.017 \in [0.01, 0.05]$, yet will not be included by the BIC since $z^2 \approx 4.41 < \log 2000 = 7.60$, i.e. including this variable increases the BIC.

- This is evident in the $n = 2000$, $p = 16$ scenario for $\text{Corr}(X1, X2) = 0.5$ and moderate SNR (see Table (41) for the summary table in the true model and second row of Table (16) for the BIC minimizer for that scenario), whereby only X1, X2, X3 are chosen because the z value for X4 is -2.66 , and $z^2 = (-2.66)^2 \approx 7.07 < \log 2000 = 7.6$. Hence, including X4 when X1, X2, X3 are already in the model increases the BIC and therefore not chosen, although X4 has "moderate signal" (its LRT p-value is in $[0.01, 0.05]$).

In this way, it becomes too trivial to guarantee the inclusion or exclusion of a ground truths because one could just simply set $z^2 > \log n$ to guarantee its inclusion and $z^2 < \log n$ otherwise (which correspond to unique LRT p-values). Ultimately, the SNR definitions from Definition (4.1) and (4.2) are convenient but not criterion-agnostic. Therefore, one needs a relative, bounded, and criterion-agnostic SNR measures such as McKelvey and Zavoina's pseudo R^2 , as discussed in the Future Work section at (6). Lastly, while this dependency could limit the generality of this simulation study, identifying this still clarifies the need for proper and principled SNR metrics in future works.

5 Application to Tropical Cyclone Genesis Dataset

5.1 Results: Exploratory Checks and BIC Minimum

5.1.1 Non-stationarity across chains for Gibbs Sampler

An initial random search is applied using the Gibbs Sampler with tuning parameter $\lambda = 1$ to the TC genesis dataset from Section (2). To monitor convergence/stationarity of the chains, four chains (each of size 1000, with first 200 discarded as burn-in) with the following overdispersed initial values (note the dataset contains 149 variables):

- Chain 1 starts from $\mathbf{v}^{(0)} = \mathbf{1}_{149}$ (i.e. the full model with all 149 variables – a "full" initial value).
- Chain 2 starts from $\mathbf{v}^{(0)} = \mathbf{0}_{149}$ (i.e. the intercept-only model – an "empty" initial value).
- Chain 3 starts from $\mathbf{v}^{(0)} = 149$ realizations from $\text{Bernoulli}(0.05)$ (a "sparse" initial value).
- Chain 4 starts from $\mathbf{v}^{(0)} = 149$ realizations from $\text{Bernoulli}(0.5)$ (a "random" initial value).

However, it appears that the resulting chains have not yet reached stationarity (normalized $\hat{R} \gg 1.01$ on the BIC values), which may be a problem because the resulting model that minimizes BIC in the chains may be too suboptimal.

5.1.2 Running 41,000 length MH with multiple values of λ

Since running the Gibbs Sampler on the TC genesis dataset took 9 days, in the interest of time I applied the MH algorithm (4) instead to generate 4 chains of length 41,000 each (with the same overdispersed initial values) to sample from $P_\lambda(\mathbf{v})$, for a *grid* of λ values, namely $\lambda = \{0.1, 0.14, 0.18, 0.2, 0.25, 0.5, 0.75, 1, 1.5, 1.75, 2.5, 3\}$. The BIC minimizer, i.e. the model that minimizes the BIC will be considered across all chains in all λ values.

5.1.3 Mixing behavior of chains across all λ

Here I discuss the behavior of models sampled from different target distributions $P_\lambda(\mathbf{v})$ for all λ . Visual inspections from the BIC trace plots (Figure (25)) show that for small values of $\lambda \in [0.1, 0.25]$, the sampled models' BIC values are well-mixed across all 4 chains. This also corresponds to more distinct models have been navigated (Table (56)). However, this exploratory behavior seems to be excessive; even though many models are visited, the attained minimum BIC values are still relatively high (Table (55)). Figure (25) confirms that the chains still spend most time in regions of high BIC. In other words, the sampler behaves almost like a random walk: too much exploration, but poor exploitation.

As λ increases (Figure (25)), BIC traces are smoother and the sampler has managed to explore lower BIC regions. The proportion of distinct models explored (Table (56)) subsequently decreased, indicating that the sampler has focused on specific regions of high-performing models. However, if λ is too large (e.g. $\lambda \in [1.5, 3]$), the exploitation becomes excessive: chains converge quickly but to different local minima, refusing to explore other neighborhoods with lower BIC. This is evidently shown by the proportion of distinct models explored (only $< 0.5\%$ of sampled models are distinct, see Table (56)). Across all λ , the best trade-off of exploration and exploitation thus seems to lie in $\lambda \in [0.5, 1]$. The BIC minimum occurs at $\lambda = 0.75$, specifically from the chain initialized with a sparse model, which also produces one of the lowest between-chain rank normalized $\hat{R}$ statistic as proposed by Vehtari et al. [26]. In particular, stationary checks using said $\hat{R}$ statistic across all λ (see Figure (26)) show that for $\lambda < 1$, the rank normalized $\hat{R}$ remains below the recommended cutoff $\hat{R} < 1.01$. For larger λ , $\hat{R}$ begins to rise above 1.5, consistent with the poor, overly exploitative mixing seen above. Therefore, both graphical and quantitative evidence suggest that $\lambda = 0.75$ (specifically, consider models sampled from $\lambda = 0.75$ for downstream analysis) provides a good starting point due to the following reasons:

1. the BIC traces have mixed well,
2. the BIC minimum are lowest among all runs,
3. stationarity across the 4 chains is achieved (rank-normalized $\hat{R} = 1.013$).

The following section will discuss elite/near-optimal models relative to the BIC minimum, which happens to be the models sampled from $P_{\lambda=0.75}(\mathbf{v})$ as well.

5.1.4 Near coincidence of BIC minimizer to the most frequent model!

Table 7: The top 5 most frequent models sampled in the chain with sparse initial value corresponding to $\lambda = 0.75$. The second most frequent model is exactly equal to the BIC minimizer in the chain since their *hamming distance* (which counts how many components in 2 vectors differ from each other) is exactly zero (and this minimizer is also the minimizer across all chains for every λ).

	Number of visits	Sample proportion for the mode	Hamming dist. b/w mode and global best minimizer	chain
1	659	0.02	2	3: sparse
2	550	0.01	0	3: sparse
3	535	0.01	3	3: sparse
4	525	0.01	1	3: sparse
5	525	0.01	1	3: sparse

Table (7) shows that the second most frequent model in the chain from $\lambda = 0.75$ corresponding to the sparse initial value **also coincides with its BIC minimizer** (which, again, is the BIC minimum across all chains in all λ). In fact, the top 5 most frequent models overall are fairly close to the global BIC minimizer based on their *hamming distances*. These provide more evidence that the most frequent sampled model and the BIC minimizer are optimal models.

5.2 Results: Elite Models and the Best Model

5.2.1 Identification of "Elite" Models

As discussed, the BIC minimizer (the model with the BIC minimum) is located at the chain from $\lambda = 0.75$ initialized with a sparse model. There are over 38 models whose BIC value marginally surpasses the minimum; which, according to the discussion on (4.3.4), will be deemed to be statistically indistinguishable from the best model. These models will be subsequently referred as *elite models* or near-optimal models. It turns out that near-optimal models also correspond to $\lambda = 0.75$ (Table (57)).

5.2.2 Intersection of elite sets as core variables

Across all 38 elite models ($\Delta BIC \leq 2$ from the BIC minimum) identified in Table (57), there is substantial overlap in physical variables. While each elite model differs slightly in its pressure-level configuration, they are likely to consistently reflect the factors behind tropical cyclone (TC) formation. Therefore, one might be interested to investigate the *core elite* variables (or *core variables* in short), which is defined as the set of variables present in **all** elite models. These core variables are presented on Table (59).

5.2.3 How do the core elites influence TC genesis?

Having identified the core elite sets, one might want to assess their estimated effects to predicting the probability of TC genesis. However, given the presence of elite, near optimal models, solely examining coefficient estimates from the BIC minimizer could be misleading. Burnham and Anderson [3] advocate drawing inference from the entire candidate set of models that are approximately equally supported by the data. Although Burnham and Anderson's emphasis lies in variable importance, the idea of a *multi-model* inference is still the key: by examining coefficient distributions of core variables across all 38 elite models, one could assess the stability of their estimated effects (direction and approximate magnitude). Variables whose coefficients remain consistently signed and narrowly distributed across elite models are interpreted as robust signals in the underlying (hypothetical) data generating process.

Coefficient distributions across elite models ($\beta > 0$)

Color encodes uncertainty = $0.7 \cdot \text{scaled IQR} + 0.3 \cdot (1 - \text{dip-test p-value})$

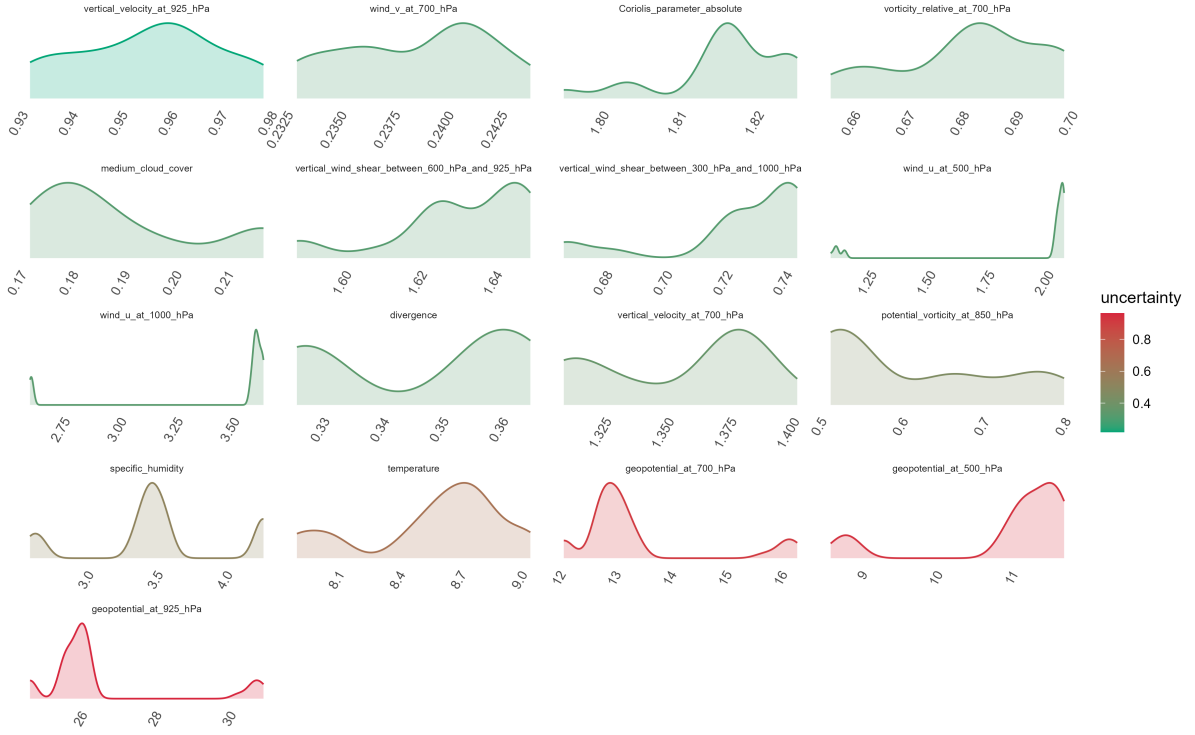


Figure 5: Estimated coefficient distributions (those with $\hat{\beta}_j > 0$ across all estimates $\hat{\beta}_j$ in the elite models) of core variables across the 38 elite models. "Uncertainty" around coefficient distributions encodes how stable the estimates are across elite models, measured by a ad-hoc weighted average of IQR and the extent of multimodality, denoted as $\text{Uncertainty} = 0.7 \cdot \text{Scaled IQR} + 0.3 \cdot (1 - \text{dip-test p-value})$, where the empirical spread (via the IQR) is more emphasized. Dip test refers to Hartigan's Dip Test of Unimodality ([1]) which tests the null hypothesis that an empirical distribution is unimodal. Most importantly, high uncertainty (red) $\neq$ bad.

Coefficient distributions across elite models ($\beta < 0$)

Color encodes uncertainty = $0.7 \cdot \text{scaled IQR} + 0.3 \cdot (1 - \text{dip-test p-value})$

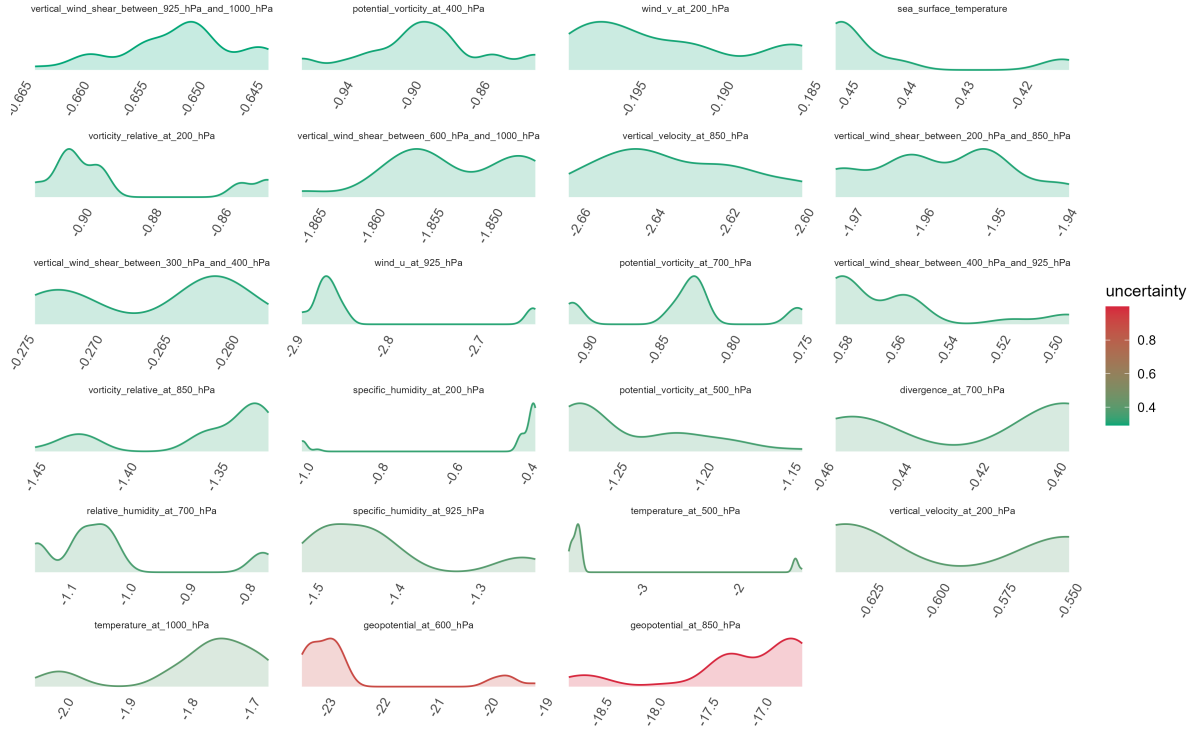


Figure 6: Estimated coefficient distributions (those with $\hat{\beta}_j < 0$ across all estimates $\hat{\beta}_j$ in the elite models) of core variables across the 38 elite models. "Uncertainty" around coefficient distributions encodes how stable the estimates are across elite models, measured by an ad-hoc weighted average of IQR and the extent of multimodality, denoted as $\text{Uncertainty} = 0.7 \cdot \text{Scaled IQR} + 0.3 \cdot (1 - \text{dip-test p-value})$, where the empirical spread (via the IQR) is more emphasized. Dip test refers to Hartigan's Dip Test of Unimodality ([11]) which tests the null hypothesis that an empirical distribution is unimodal. Most importantly, high uncertainty (red) $\neq$ bad.

Coefficient distributions across elite models (mixed: crosses zero)

Color encodes uncertainty = $0.7 \cdot \text{scaled IQR} + 0.3 \cdot (1 - \text{dip-test p-value})$

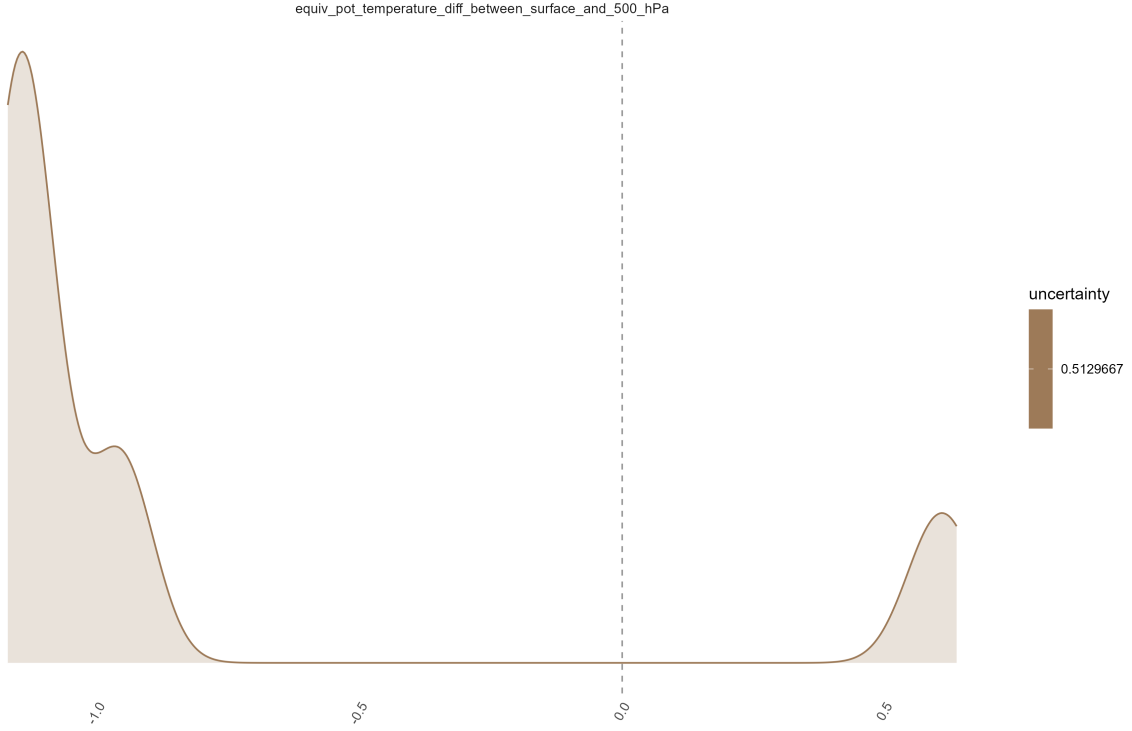


Figure 7: Estimated coefficient distributions (those with mixed signs (some $\hat{\beta}_j > 0$ and some $\hat{\beta}_j < 0$) across all estimates $\hat{\beta}_j$ in the elite models) of core variables across the 38 elite models. "Uncertainty" around coefficient distributions encodes how stable the estimates are across elite models, measured by a ad-hoc weighted average of IQR and the extent of multimodality, denoted as $\text{Uncertainty} = 0.7 \cdot \text{Scaled IQR} + 0.3 \cdot (1 - \text{dip-test p-value})$, where the empirical spread (via the IQR) is more emphasized. Dip test refers to Hartigan's Dip Test of Unimodality ([11]) which tests the null hypothesis that an empirical distribution is unimodal. Most importantly, high uncertainty (red) $\neq$ bad.

5.2.4 Model Validation: OOS Log Loss and AUC-PR

Among all the identified elite/near-optimal models, a single, representative model for final presentation is selected based on validation performance. Specifically, I seek the model that minimizes the average *out-of-sample* (OOS) deviance in a 5-fold *Cross-Validation* (CV) and maximizes the 5-fold CV OOS average *area under the precision-recall curve* (AUC-PR). The best-performing elite model (i.e., the one achieving the lowest OOS deviance and highest OOS AUC-PR) is then re-fit on the entire dataset, after which a final calibration check is conducted to ensure that predicted probabilities are not systematically biased.

- AUC-PR is considered instead of AUC-ROC due to the strong class imbalance present in the TC genesis dataset; precision-recall curves provide a more robust measure of discriminative performance than the ROC-based metrics.
- The OOS deviance (a.k.a. *log-loss*), on the other hand, is a measure of *goodness-of-fit* without being overly sensitive to absolute probability calibration, since it evaluates the relative discrepancy between predicted probabilities and observed responses, making it a suitable criterion for comparing predictive accuracy of competing models, particularly when variable selection is the goal and not probability calibration itself.

Results: Among the 38 elite/near-optimal models, the model that both minimizes AUC-PR and the OOS log-loss turns out to correspond to a single model, as summarized in the table below.

Table 8: Top 5 elite models after validation, sorted ascendingly by AUC-PR and descendingly by OOS log-loss. For more details on the top elite model, see the 25th row in Table (57).

	auc_pr	avg_ll	elite_idx
1	0.6593	0.1525	25
2	0.6585	0.1526	13
3	0.6590	0.1526	24
4	0.6584	0.1526	14
5	0.6589	0.1526	26

Clarification: Prediction or Variable Selection? Even though the primary goal for this project is not prediction but variable selection (i.e. identifying meaningful predictors of tropical cyclone genesis), I nonetheless employ out-of-sample validation as a diagnostic for model adequacy (serves as a "tie-breaker" to prefer one model over the set of 38 near-optimal models). Here I give a theoretical justification on validating near-optimal models in variable selection.

- Under the assumption that both training and test data are independently drawn from the same underlying data-generating process—assumed here to follow a binomial GLM (logistic regression with a logit link)—the empirical out-of-sample (OOS) log loss

$$\hat{L}(f) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} l(Y_i, f(\mathbf{x}_i))$$

is an unbiased estimator of the expected log loss

$$L(f) = \mathbb{E}_{(\mathbf{x}, Y)}[l(Y, f(\mathbf{x}))],$$

for any fixed regression function f , by linearity of expectation. Moreover, the expected log loss $L(f)$ is uniquely minimized if and only if the model is correctly specified and coincides with the true conditional expectation $f^*(\mathbf{x}) = \mathbb{E}[Y|\mathbf{x}]$ (see further discussion in Section 12.13), which is also equal to $P(Y = 1|\mathbf{x}) = \sigma(\mathbf{x}^T \beta^{(true)})$ if the true conditional distribution truly follows a binomial GLM with logit link. In practice, however, f must be estimated from data. Even when the model is correctly specified, the estimated function $\hat{f}$ will deviate from f^* due to sampling variability. Under standard regularity conditions and for sufficiently large samples, the maximum-likelihood estimator used to obtain $\hat{f}$ is consistent, such that $\hat{f} \xrightarrow{P} f^*$ as $n \rightarrow \infty$.

- Therefore, among a set of near-optimal models with indistinguishable BIC values, a model that maintains or improves empirical OOS log loss can be interpreted as providing a closer approximation to the true conditional structure $P(Y|\mathbf{x})$, even when the true model itself is unknown.
- Although I used 5-fold CV for validation on the same dataset (by obtaining a cross-validated version of the empirical OOS log-loss $\frac{1}{n} \sum_{k=1}^5 \left[\sum_{i=1}^{n_{\text{fold } k}} l(Y_i, \hat{f}_{-k}(\mathbf{x}_i)) \right]$ averaged across 5 folds, where in each fold $\hat{f}_{-k}$ is estimated from samples outside the fold) that is also used for running the random search procedure (no external, unseen test dataset was reserved), it still serves as a consistency check: a model that maintains predictive performance across resampled folds is likely to capture relationships that generalize to unseen data drawn from the same distribution within the observed TC genesis dataset.

5.2.5 Final model

Table (60) shows the the best elite model and its coefficient estimates. Calibration of predicted probabilities in the dataset are also examined, see Table (61).

5.3 Results: Variable Importance

Since empirical results from the simulation study showed that forming models by variables with above-0.5 marginal frequency yields a near-correct model (see Section (4.3.2)), one may consider forming additional models based on the following variable sets:

1. Variables with marginal frequency above 0.5 (there are 48 such variables), as summarized in row 1 to 48 from Table (58).
2. Core variables, as summarized in Table (59).
3. Core variables with marginal frequency above 0.5, as summarized in Table (59).

Unfortunately, none of those models produce a BIC value smaller than the BIC minimizer nor the best elite model. However, the variable set obtained from those with marginal frequency ≥ 0.5 already contains around 83% (i.e. 34 out of 41) of the core variables, signifying the influential and importance of the core variables in explaining TC genesis.

5.4 Benchmarking using Cross-Validation with LASSO

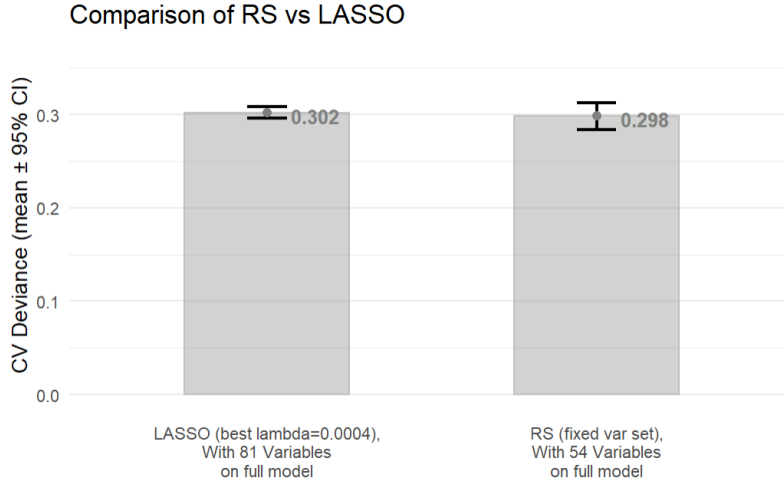


Figure 8: Cross-Validation OOS Log-Loss for Estimated Model using LASSO Versus the Variable Set from the Best Elite Model.

A LASSO logistic model was fit into the TC genesis data with the best penalty hyperparameter value that minimizes the cross-validated OOS log-loss (which yielded 0.0004 as the best penalty). Using the best elite model identified using methods described in Section (5.2.4), a cross-validation is obtained using the variable set in the best elite model. Both predictions (i.e. predictions using estimated penalized coefficients from LASSO and estimated coefficients from the best elite variable set) yield similar performance (since error bars overlap, see Figure (8)). However, when fit into the entire TC genesis dataset, LASSO outputs a much larger variable set (81 variables with non-zero coefficients), which is much larger than the variable set from the best elite model (only 54 variables).

5.5 Discussion

Summary of results. To summarize, an initial random search was run using the Gibbs Sampler with $\lambda = 1$ across 4 chains with overdispersed initial values, each of length 2000, but the generated BICs did not show evidence of stationarity (the rank-normalized $\hat{R} \gg 1.01$). Due to the computationally expensive runtime of the Gibbs Sampler, the MH algorithm was instead used to generate 4 chains, each of length 41,000 (first 1000 as burn-in), across a grid of λ values. Generated BIC values from $\lambda = 0.1, \dots, 0.75$ showed good mixing (normalized $\hat{R}$ is not significantly greater than 1.01). The BIC minimum across all chains and λ values occurred at $\lambda = 0.75$ at the chain initialized with a sparse model, and this minimum also coincides with the second most frequent model from the same sparse chain (see Section (5.1.4)). There are 38 "near-optimal" models whose BIC values are greater than the minimum but not exceeding more than 2 points, with most of them corresponding to $\lambda = 0.75$. Core variables, which are variables present across all elite models, were obtained and their coefficient estimates are examined across all elite models. The elite model that achieves optimal CV performance is considered as a representative, final model.

Three models constructed from marginal frequency cutoffs did not achieve lower BIC values. When the best elite model is validated against LASSO using cross-validation, both yield comparable performance but LASSO outputs a much larger model than the best elite model when fit in the entire TC genesis dataset. These results provide evidence that the search may have effectively converged to a local (and likely, hopefully, global) optimum in the model space, despite the high dimensionality and correlation structure of the predictors.

Consistency of results to previous literature. Most importantly, all 6 canonical *core drivers* of TC genesis postulated by Gray [9] are included in the core variables, as summarized in Table (59), namely:

1. *Low-level relative vorticity* ($\hat{p} = 1$ for relative vorticity at 200 hPa),
2. *Coriolis Parameter* ($\hat{p} = 1$),
3. *The inverse of the vertical shear of the horizontal wind between the lower and upper troposphere between 200 to 950 hPa* ($\hat{p} = 1$ for vertical wind shear between 200 to 850 hPa),
4. *Difference in equivalent potential temperature between the surface and 500 hPa* ($\hat{p} = 0.86$),
5. *Middle troposphere relative humidity from 500-700 hPa* ($\hat{p} = 0.45$ for relative humidity at 700 hPa)
6. and finally, *Sea surface temperature* ($\hat{p} = 1$)

Potential redundancy. The present study includes **all** 36 pairwise vertical wind shear combinations across the 9 selected atmospheric pressure levels ($\binom{9}{2} = 36$), which may share overlapping collinear information (and possibly redundant) processes for the same physical feature. To demonstrate this concern, individual shear variables are removed one at a time and BIC reductions are computed:

Table 9: The drop in BIC and OOS log-loss when wind shear variables are removed systematically in the best elite model. Removing wind shear measured on 200 to 850 hPa yields the largest BIC reduction, consistent to Gray [9]’s explanation of vertical wind shear from 200 to 950 hPa being a main core driver of TC genesis.

Vertical wind shear pressure levels (included in the best elite model)	BIC reduction (relative to best elite model)	OOS log-loss increase (relative to best elite model)
200 to 850 hPa	476.021	0.007
300 to 400 hPa	13.598	< 0.001
300 to 1000 hPa	26.757	0.001
400 to 925 hPa	19.863	< 0.001
600 to 925 hPa	114.701	0.002
600 to 1000 hPa	175.105	0.003
925 to 1000 hPa	223.808	0.004

Table (9) shows that when individual shear variables are removed, the BIC increase relative to the best elite model varies dramatically, from as high as ≈ 400 (for wind shear measured at 200-850 hPa) to as low as ≈ 10 (for middle layer wind shears measured at 300-400 hPa).

- This suggests that not all shear layers contribute equally, certain groupings of layers may contribute differently to influencing TC genesis, whereas others may act as redundant projections of the same underlying process.
- Hence, this empirically motivates a hierarchical ”feature-grouping” approach where multiple random search procedures are run based on how variables belonging to the same physical family are grouped together, and examine which groupings provide the most adequate model evidence, for example, which one has the lowest BIC (for more discussion see Section (6)).

Remarks on mixing challenges for the first Gibbs Sampler run. Despite the TC genesis dataset having large n , BIC values across chains in the first Gibbs Sampler run mixed very slowly. One explanation is due to the

strong collinearity of variables, particularly within physical families of predictors (e.g. geopotential across pressure levels). This is further supported by the simulation study on the $n = 10000$, $p = 160$ synthetic dataset that only manipulated the correlation of a ground truth pair and took no more than 1 day, despite the high-dimensionality ($p = 160$). Scaling was not suspected to affect the runtime significantly since the TC genesis predictors were already standardized and the synthetic data matrix were generated from standard normal distributions.

Remarks on mixing challenges for the MH runs. Statistically indistinguishable models (i.e. models with BIC difference of no more than 2 compared to the BIC minimum) only emerged after 19,000 iterations (see row 22 and 23 from Table (57)), which underscores the effect of highly collinear predictors to the procedure’s runtime, which also induced local dependence, as Figure (26) showed that even the vanilla MH algorithm $\lambda = 1$ showed poor mixing of BIC values.

Effect of "outliers"? Experimental re-runs (results not shown) showed that outliers had negligible impact on search trajectories and selected models.

- But what do we consider as an outlier? In logistic regression, outliers can be broadly categorized as either *concordant* or *discordant*. Concordant outliers are observations whose $\mathbf{x}_i$ values align with their labels y_i , while *discordant* outliers lie in the 'opposite' direction with their labels y_i . Logistic regression models are known to be robust to *concordant* outliers but less robust to *discordant* ones.
- Since the random search procedure involves repeated logistic fits, discordant outliers could, in theory, bias intermediate coefficient estimates and alter the likelihood of candidate models.
- To address this concern, *mahalanobis distances* for observations in the TC genesis dataset were used as a proxy for high *leverage* observations. Because of substantial multicollinearity among predictors, the distances were computed on a feature space transformed via *Principal Component Analysis* (PCA), retaining *Principal Components* that approximately explain 95% of the variance in the data. Observations exceeding various cutoff thresholds (e.g., the 95th percentile of the χ_p^2 distribution) were treated as potential high-leverage cases.
- Multiple re-runs of the MH-based procedure ($\lambda = 0.75$ of length 5000-10000) based on different cutoffs showed no significant changes in the selected models, both for the BIC minimizer and the set of frequently selected variables.

Spurious Correlations? The variable set from the final model still contains highly correlated variables (for example, the observed pairwise Pearson Correlation Coefficient for `geopotential_at_925_hPa` and `mean_sea_level_pressure`, with both included in the model, is over 0.99). To assess whether these inclusions are spurious, each of these variables are systematically removed and the resulting BICs and cross-validated performance metrics are recomputed.

Variable Removed	ΔBIC (compared to the final model)	ΔBIC (compared to the BIC minimizer)	ΔOOS log-loss (compared to the final model)
<code>geopotential_at_925_hPa</code>	+131.76	+130.04	+0.0015
<code>mean_sea_level_pressure</code>	+88.127	+86.39	+0.0014

Table 10: Variables from a highly correlated pair from the final model (best elite model that optimizes validation metrics) is systematically removed, and empirical OOS log-loss and BIC differences are computed.

From Table (10), removing one variable from a correlated pair increased the BIC (relative to both the final model and the BIC minimizer) by at least 80 points, providing "very strong" evidence in favor of including the variable (according to Raftery and Kass' [12] interpretation of BIC differences, see Section (4.3.4)). Thus, one has evidence of these variables contributing substantially to model fit despite their high pairwise correlation. The removals also degraded OOS log-loss, further confirming that these variables encode distinct and meaningful information to

explain TC genesis.

Given that the true model is unknown, this further provides empirical reassurance that such correlated variables improve generalization, under the assumption of a consistent data generating process between training and test sets in the Cross Validation partitions.

Dimension of the final model is still large? While the random search procedure selects a relatively large set at the 12 h lead, this should not be viewed as a weakness. The redundancy largely reflects multi-level measurements of the same physical field (e.g., vorticity, shear, humidity), consistent with findings by Wijnands et al. [29]. Indeed, their final predictive model at 12–24 h identified four dominant predictors, showing that lead time plays a key role in how redundancy collapses into a smaller set. The results can therefore be interpreted as capturing the correct drivers at the early stage, albeit across multiple correlated levels. Therefore, assuming a logistic regression setup for this dataset, there is confidence to believe that the dimension of the underlying ‘true model’ for TC genesis is large and the ‘ground truths’ with non-zero β_j s could also be highly correlated.

5.6 Answering the Research Question.

To answer the research question, I present a graphical version of Table (59). That is, I pick the top-ranked pressure levels for each family present in the 41 core variables, then summarize them into the following bar plot, containing 18 variables in total.

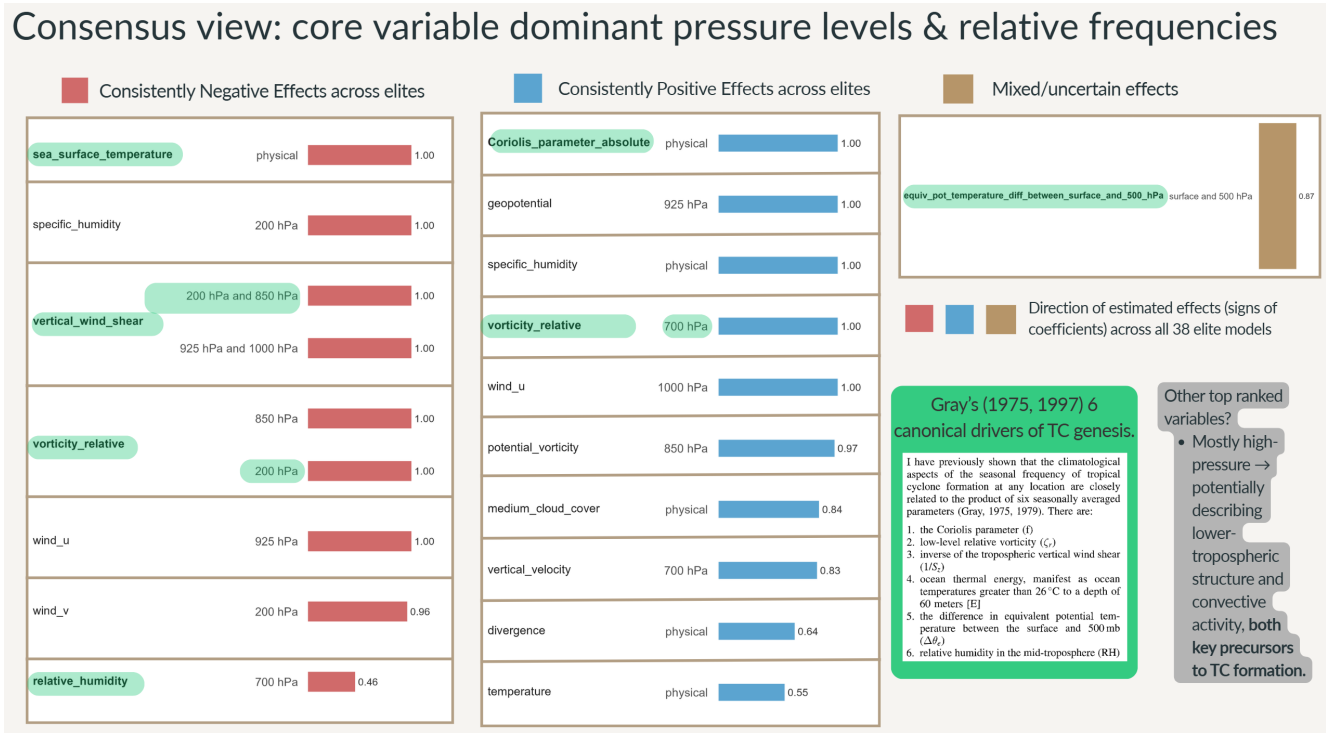


Figure 9: Bar plot of relative frequencies for the 41 core variables (family + pressure level, or ‘physical’, if the physical family itself has the highest relative frequency among its pressure-level counterparts). Bars are color coded by the signs of their estimated coefficients in the 38 elite/near-optimal models. Green variables highlight 6 canonical TC genesis drivers highlighted by Gray [9]. Other top-ranked variables mostly have high pressure levels, which may indicate lower-tropospheric structure and convective activity, both of which are key precursors to TC genesis formation.

6 Future Work

1. Since the variables in the TC genesis dataset mostly represent the same physical phenomena measured at different pressure levels, (especially wind shear variables where all pairwise combinations of pressure level measurements are included), future work might adopt a hierarchical feature-grouping framework to run the random search procedures based on different logical groupings of the variables, for example:

Stage	Detail	Example	Purpose
Coarse-grained grouping	Only one representative per physical family (e.g. <code>geopotential</code>)	Include only <code>geopotential_at_925_hPa</code>	Evaluate field effects at a broad, low resolution
Intermediate grouping	Average or cluster pressure levels into 2-3 macro layers (e.g. upper = 200-400 hPa, middle = 500-700 hPa, lower = 850-1000 hPa)	Include only <code>avg_geopotential_upper</code>	Evaluate field effects at more granular resolution
Full resolution	All pressure level combinations	All 36 wind shear pairs across all $\binom{9}{2} = 36$ combinations	Captures fine-scale pressure-level effects

Table 11: Possible groupings for each physical variable based on pressure levels, to which random search procedures will be run for each grouping.

2. Consider pre-filtering variables using distribution-agnostic methods to reduce redundancy whilst still letting the random search handling the core variable selection, especially in high dimensional, high correlation scenarios in order to speed up the MCMC procedure. For example, Wijnands et al. [29] pre-filtered variables beforehand based on their *Mutual Information* (MI) to explain the TC genesis response variable; in particular, they selected a representative pressure level for variables representing multiple pressure levels of the same physical variable.
3. In the first run of the Gibbs Sampler the chain mixed very slowly. To increase computational efficiency, one may want to consider grouping variables into *blocks* $B_1, \dots, B_K$, where each B_i contains at least one v_j such that updates are done one block at a time.
 - Some methods to build blocks include:
 - Domain-guided blocks. In the TC genesis problem, one may group by similar physical fields differing by pressure level. For example, $B_1 = \{\text{all } v_j \text{ s.t. } X_j \text{ corresponds to } \text{geopotential_on_}\ast\text{hPa}\}$.
 - Correlation-guided blocks (more data-driven). Group variables by their observed correlation $\text{Corr}(x_j, x_k)$ such that correlation of variables between blocks are minimized. Formally, consider any two distinct blocks B', B'' . For all i such that $v_i \in B'$ and all j such that $v_j \in B''$, we hope to have $\text{Corr}(x_i, x_j)$ being minimized.
 - The consequence of grouping v_j s into blocks would be to compute the conditional distribution of $P(B_i | B_{-i})$. Some methods to compute this distribution include:
 - Exact within-block Gibbs. If a block is small in size (e.g. $|B_i| \leq 10$, say) then marginalizing $P(B_{-i})$ would still be computationally manageable by enumerating all combinations of $\{0, 1\}^{|B_{-i}|}$. However, large block sizes exponentially increases the number of terms in the marginalization, likely yielding diminishing returns in efficiency.
 - When block sizes are big, one may use a *Metropolis-Within-Gibbs* update to propose a block $b \in \{0, 1\}^{|B_i|}$ with an efficient proposal.
 - However, care should be exercised to assess whether the detailed balance condition still holds for the procedure to converge to the correct stationary distribution.

4. Use a more principled approach for defining SNR. For instance, one could obtain true β values representing a certain SNR level based on a *pseudo* R^2 proposed by McKelvey and Zavoina [16], namely:

$$\text{McKelvey and Zavoina's Pseudo } R^2 = \frac{\text{Var}(\eta)}{\text{Var}(\eta) + \text{Var}(\epsilon)} = \frac{\beta^T \Sigma \beta}{\beta^T \Sigma \beta + \pi^2/3} \in [0, 1]$$

Assuming that the predictor variables has the multinormal distribution $\mathbf{x} \sim \mathcal{N}_p(\mathbf{0}, \Sigma) \rightarrow \text{Var}(\eta) = \text{Var}(\alpha + \beta^T \mathbf{x}) = \beta^T \Sigma \beta$, assuming α is the intercept. In this way, several pseudo R^2 values could be chosen as a measure of SNR (e.g. $\{0, 0.2, 0.4, 0.6, 0.8\}$), and solve for β for a prespecified value (e.g. choose a unit vector $\mathbf{b}$ such that $\beta = c\mathbf{b}, c \in \mathbb{R}$, and solve for c)

I give a possible explanation of the McKelvey and Zavoina's pseudo R^2 as follows. The binomial GLM has the form $p_i = P(y_i = 1 | \mathbf{x}_i) = \sigma(\eta_i)$, $\eta_i = \mathbf{x}_i^T \beta$, which is equivalent to proposing the following *latent-variable* model:

$$y_i^* = \eta_i + \epsilon_i, \quad y_i = \mathbf{1}[y_i^* > 0] \\ \epsilon_i \sim \text{Logistic}(0, 1), \quad \text{Var}(\epsilon_i) = \pi^2/3$$

this is because $P(y_i = 1 | \mathbf{x}_i) \equiv P(y_i^* > 0 | \mathbf{x}_i) = P(\epsilon_i > -\eta_i | \mathbf{x}_i) = F_\epsilon(\eta_i) = \sigma(\eta_i)$, since the logistic distribution is symmetric around 0. Also, F_ϵ denotes the *cummulative distribution function* (CDF) of the logistic distribution, which is precisely the sigmoid/logistic function.

In other words, specifying a logit binomial model is equivalent to specifying the above latent model with ϵ having the logistic distribution. This latent model is also akin to linear regression, where the (latent) response y_i^* can be decomposed to a deterministic component (η_i) and a random component (ϵ_i). In linear regression, SNR is defined as $\frac{\text{Var}(X\beta)}{\sigma^2}$ for the linear model $y_i = \mathbf{x}_i^T \beta + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(\mathbf{0}, \sigma^2)$, ϵ_i, ϵ_j uncorrelated for $i \neq j$. Moreover, SNR has a direct relationship with R^2 as $\text{SNR} = \frac{R^2}{1-R^2} \Leftrightarrow R^2 = \frac{\text{SNR}}{1+\text{SNR}} = \frac{\text{Var}(X\beta)}{\text{Var}(X\beta) + \text{Var}(\epsilon)}$. In this way, it is quite seamless to define a similar R^2 for the latent "linear regression" model (again, although responses y_i^* are not observable).

5. To search for the BIC minimizer global optimum, consider evaluating the random search procedure but with an adaptive, dynamic λ value (e.g. start with $\lambda \leq 1$ then move to $\lambda > 1$ under some condition).
6. The correlation levels in the current factorial design, while controlled, are conservative in scope, as we only considered *pairwise* correlations. In realistic high-dimensional settings, clusters of highly correlated variables are common (see Table (1)). A richer factorial design could introduce the scale of collinearity (e.g. defined as the number of within ground-truths or between ground-truth and redundant correlations) as a separate factor. However, such extension induces practical obstacles, since enforcing multiple high correlation often produces non-positive semi definite covariance matrices.
7. The random search procedures run in the TC genesis data are computationally intensive, as 4 chains of length 41,000 are run across a grid of λ values. While this may be feasible for research purposes, this may not be the case for practitioners who wish to apply the random search procedure. Future work could explore simulation studies that provide reliable empirical rule(s) to tune λ over a small run of the procedure of length, say, 1000-5000, before running the procedure entirely using the chosen λ .
8. Burnham and Anderson [3] advocated for *model averaging* to produce a prediction by computing a weighted average of models using *akaike weights* (i.e. an exponential, *softmax*-like transformation of the AICs). Though not directly related to the goal of variable selection, near-optimal models obtained from the random search could be used for prediction by a weighted average over relative frequencies.

7 Conclusion

This thesis introduced a MCMC-based random search procedure for high-dimensional variable selection in logistic regression. A factorial-based simulation study demonstrated the robustness of the procedure to identify influential

variables under high correlation and moderate SNR, though future work could consider a more principled measure of SNR such as a pseudo- R^2 for logistic regression. Application to TC genesis dataset produced results consistent with long-standing meteorological literature by recovering known physical drivers of TC genesis – whilst maintaining good generalization and calibration. Despite computational demands, the procedure has identified near-optimal BIC models and stable mixing under multiple runs with different (inverse temperature) λ values. Future work could focus on improving scalability via hierarchical grouping of variables, block gibbs sampling, or empirical rules from short MCMC chains to tune λ upfront. Overall, findings reaffirm that the random search-based selection method provides an interpretable pathway towards uncovering influential drivers behind complex environmental systems as well as ample opportunities for future work to further leverage the random search’s power.

8 Contribution/Significance

This study explores a stochastic variable selection procedure for high-dimensional logistic regression, showing the ability of the procedure to pick the global BIC-optimal model almost surely without having to enumerate all possible submodels. As mentioned before, several strategies could be employed to leverage the procedure’s power to the maximum. Furthermore, one possibility for extension is to apply this method beyond parametric model selection to a Monte Carlo optimization strategy. For instance, given a function $f(i) : \mathcal{D} \rightarrow \mathbb{R}$ that needs to be minimized whose domain $\mathcal{D}$ is too large to store explicitly, but $f(i)$ can be computed on demand, one could construct a stochastic search process that probabilistically converges to $\arg \min_i f(i)$ almost surely.

9 Acknowledgement

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10 Declaration

I certify that this report does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any university; and that to the best of my knowledge and belief it does not contain any material previously published or written by another person where due reference is not made in the text. The report is 12909 words in length (excluding text in images, tables, bibliographies and appendices).



October 24, 2025

11 Codebase

All the scripts and datasets used for this project can be accessed from [this public GitHub repository](#).

12 Appendix

12.1 Theoretical Justification for the 4 Chosen Factors and Definition (4.1)

The four factors (ground truth SNR, collinearity among ground truths, sample size, and candidate variables) are chosen due to how the Gibbs-sampler based random search procedure works as outlined in Algorithm (1), as well as some explanatory plots. The following describe how the selected factors are expected to influence the procedure itself.

Justification: variable selection task itself. Recall that the motivation for applying this random search procedure is to approximate the model that minimizes the BIC over all possible submodels. Furthermore, the BIC is only composed of the log-likelihood and a scaled penalty term, and the extent to which the ground truth variables produce strong signal to the response is captured in the log-likelihood.

A closer look into Algorithm (1). The marginal samples for a variable j can be denoted as $\{v_j^{(1)}, \dots, v_j^{(K')}\}$ whereby during the i -th iteration ($i = 2, \dots, K'$), $v_j^{(i)}$ is sampled as:

$$\begin{aligned} v_j^{(i)} &\sim \text{Bernoulli}\left(\frac{w_1}{w_1 + w_0}\right), \\ w_1 &= \exp\{-BIC(v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_j = 1, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)} | D)\} \\ w_0 &= \exp\{-BIC(v_1^{(i)}, \dots, v_{j-1}^{(i)}, v_j = 0, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)} | D)\} \end{aligned}$$

If variable j is truly influential in predicting the response, then throughout the iterations one would expect it to be frequently chosen by the procedure, i.e. $\frac{w_1}{w_1 + w_0}$ is consistently close to 1. If this is true, then one would expect $w_1 \gg w_0$ (equivalent to $\frac{w_1}{w_1 + w_0} \gg \frac{1}{2}$) in almost all K iterations. In other words, for "most" i in $1, \dots, K$:

$$\begin{aligned} w_1 &> w_0 \\ \implies \exp\{-BIC(\mathbf{v}_{j=1}^{(i)} | D)\} &> \exp\{-BIC(\mathbf{v}_{j=0}^{(i)} | D)\}, \\ \text{letting } \mathbf{v}_{j=1}^{(i)} &= (v_1^{(i)}, \dots, v_{j-1}^{(i)}, 1, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)})^T \\ \text{and } \mathbf{v}_{j=0}^{(i)} &= (v_1^{(i)}, \dots, v_{j-1}^{(i)}, 0, v_{j+1}^{(i-1)}, \dots, v_p^{(i-1)})^T \\ \implies BIC(\mathbf{v}_{j=1}^{(i)} | D) &< BIC(\mathbf{v}_{j=0}^{(i)} | D) \\ \implies -\mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=1}^{(i)}} | D) + \frac{1}{2} \cdot \text{sum}(\mathbf{v}_{j=1}^{(i)}) \cdot \log n &< -\mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=0}^{(i)}} | D) + \frac{1}{2} \cdot \text{sum}(\mathbf{v}_{j=0}^{(i)}) \cdot \log n \\ \implies \left[\mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=1}^{(i)}} | D) - \mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=0}^{(i)}} | D) \right] + \frac{1}{2} \cdot \log n \left(\text{sum}(\mathbf{v}_{j=0}^{(i)}) - \text{sum}(\mathbf{v}_{j=1}^{(i)}) \right) &> 0 \end{aligned}$$

where $\hat{\beta}_{\mathbf{v}} \in \mathbf{R}^{p+1}$ represent the coefficient vector for a model with variables indexed by $\mathbf{v}$ (if $v_j = 0$ then $\hat{\beta}_j = 0$ by default).

If one assumes that all the variables are continuous and numeric, in the sense that adding a variable X_j into the model only costs one parameter β_j , then the last inequality can be simplified to:

$$\left[\mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=1}^{(i)}} | D) - \mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=0}^{(i)}} | D) \right] > \frac{1}{2} \cdot \log n \quad (34)$$

Meaning that if a numeric variable X_j is truly influential, then value of $w_1 - w_0$ throughout all iterations will most likely be dominated by the difference in the log-likelihood $\left[\mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=1}^{(i)}} | D) - \mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=0}^{(i)}} | D) \right]$, possibly at least as large as $\frac{1}{2} \cdot \log n$, which acts as some sort of a "threshold" as a result of the penalization employed by BIC.

From the above heuristic, the difference in log-likelihood can then be used to control the significance/strength of association between a ground truth variable X_j and the response (which will be **used as a proxy of the ground truth variables' SNR**, explained in later sections).

- However, one does not know in advance which variables that the procedure has sampled on iteration i , i.e. the values of $\mathbf{v}_{j=1}^{(i)}$, $\mathbf{v}_{j=0}^{(i)}$ other than the j -th element could be anything.
- To work around this dilemma, one could consider the difference in log-likelihood **under the true model**. Suppose the set $G \subset (1, 2, \dots, p)$ identifies the indices for the true model such that $\beta_j^{(true)} \neq 0$, $j \in G$ and $\beta_j^{(true)} = 0$, $j \notin G$. Further, let $\mathbf{v}^*$ be the true model, i.e. $v_j^* = 1$ if $j \in G$ and $v_j^* = 0$ otherwise. Then, to control the strength of ground truth X_j to the response, one would use the following:

$$\Delta\mathcal{L}_j^* \equiv \left[\mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=1}^*} | D) - \mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=0}^*} | D) \right] \quad (35)$$

where $\hat{\beta}_{\mathbf{v}_{j=1}^*}$ is obtained by fitting a logistic model using a design matrix X that only contains the ground truths (with ground truth X_j included) and a response vector y . The idea is that when considering all candidate variables during the procedure, if a variable is "redundant" (i.e. does not contribute to the data generating process) then its estimate would be reasonably close to 0, which makes:

$$\Delta\mathcal{L}_j^* = \left[\mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=1}^*} | D) - \mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=0}^*} | D) \right] \approx \left[\mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=1}^{(i)}} | D) - \mathcal{L}(\hat{\beta}_{\mathbf{v}_{j=0}^{(i)}} | D) \right] \quad (36)$$

regardless of what other variables are present in the models $\mathbf{v}_{j=1}^{(i)}$, $\mathbf{v}_{j=0}^{(i)}$ in the i -th iteration of the procedure.

However, for a ground truth X_j , how large should be the log-likelihood difference $\Delta\mathcal{L}_j^*$ in (35) for it to be significant ("having high signal to the response") or not? One way is to control for this difference relative to the $\frac{1}{2} \cdot \log n$ "threshold" as stated in (34).

- However, one drawback to use the $\frac{1}{2} \cdot \log n$ threshold for manipulating a ground truth variable's significance is that it does not necessarily reflect a direct, calibrated measure of signal. Namely, the $\frac{1}{2} \log n$ quantity is merely a consequence of BIC's penalty term that acts as a "detection hurdle" that (asymptotically) enforces parsimony. In other words, tuning SNR based on this $\frac{1}{2} \log n$ threshold that is also used by the sampler would not provide a reliable evaluation for the simulation study.
- Instead, one could consider using the marginal p-values *Likelihood Ratio Test* (LRT) of the form $H_0 : \beta_j = 0$ for the (nested) models $\mathbf{v}^*$ such that $v_j^* = 1$ and the model with $v_j^* = 0$ by leveraging the following asymptotic distribution (larger $\Delta\mathcal{L}_j^*$ means smaller p-value):

$$2\Delta\mathcal{L}_j^* \sim \chi_1^2$$

- **However, how does this relate to signal?** The idea is that if X_j is involved in generating y , but its marginal p-value of the test $H_0 : \beta_j = 0$ in the true model is relatively large (say, close to 0.1 – more evidence *to* as opposed to *against* the null), then it is considered to have "low signal" on the response; there is insufficient evidence from the data to detect a presence of any signal from the variable.
- Using LRT p-values rather than the $\frac{1}{2} \log n$ threshold gives a more calibrated measure of signal. A back-of-the-envelope calculation for $n = 200$ and $n = 2000$ reveals the corresponding "significance threshold" γ for the LRT:

- $n = 200$. The BIC "inclusion rule" $\Delta\mathcal{L}_j^* > \frac{1}{2} \log 200$ in (34) is equivalent to the following significance threshold on the LRT: $\gamma = P(2\Delta\mathcal{L}_j^* > \log 200) = P(\chi_1^2 > \log 200) \approx 0.02$ (i.e. a p-value around 0.02 is considered as "moderately significant")
- $n = 2000$. The BIC "inclusion rule" $\Delta\mathcal{L}_j^* > \frac{1}{2} \log 2000$ in (34) is equivalent to the following significance threshold on the LRT: $\gamma = P(2\Delta\mathcal{L}_j^* > \log 2000) = P(\chi_1^2 > \log 2000) \approx 0.0058$ (i.e. a p-value around 0.0058 is *still* considered as "moderately significant")
- The takeaway is that the corresponding "significance threshold" shrinks with n . Hence to obtain a more reliable (though not perfect) measure of signal, we define variable-wise significance or SNR via **fixed LRT p-value bands** – this will be further discussed on Definition (4.1).

But, the difference in log-likelihood itself $\Delta\mathcal{L}_j^*$ does not directly reflect which aspects of the (synthetic) data does one need to tune to calibrate a ground truth variable's significance (so-called signal). One workaround to answer the above is to use the asymptotic equivalence between LRT and the *Wald Test*, namely for nested logistic models differing by one variable or parameter:

$$2\Delta\mathcal{L}_j^* \sim \chi_1^2 \text{ and } Z_j^2 \approx 2\Delta\mathcal{L}_j^* \quad (37)$$

where Z_j represents the *Wald Test Statistic*, i.e.

$$Z_j = \frac{\hat{\beta}_j^*}{se(\hat{\beta}_j^*)}$$

here:

- $\hat{\beta}_j^* \equiv [\hat{\beta}_{\mathbf{v}_{j=1}^*}]_j$ (j -th element of $\hat{\beta}_{\mathbf{v}_{j=1}^*}$), the coefficient estimate for ground truth variable j in the true model $\mathbf{v}^*$
- $se(\cdot)$ represents the standard error for the estimate using the asymptotic MLE distribution of $\hat{\beta}_j^* \sim \mathcal{N}(\beta_j^{(true)}, [I(\hat{\beta}_j^*)^{-1}]_{jj})$, where $I(\hat{\beta}_j^*) = (X^T \hat{W} X)$ denotes the Fisher Information for logistic regression and $\hat{W}$ denotes the diagonal weight matrix with $\hat{W}_{ii} = \hat{p}_i(1 - \hat{p}_i)$, $1 \leq i \leq n$.

In this way, asymptotically there are several things that indirectly influence the quantity of $\Delta\mathcal{L}_j^*$ (which is expected influence the log-likelihood differences in the procedure as per (36)):

1. $\beta_j^{(true)}$: The magnitude of the true coefficient for ground truth variable X_j .
2. The sample size n : this will affect the precision of the estimate, namely to the standard error.
3. Collinearity between ground truth variables in the true model, as the singular values of X affects the diagonal entries of the Fisher Information.
4. The fitted probabilities of the samples $\hat{p}_i$, $1 \leq i \leq n$ using the true model. This is important because of the so-called *Hauck-Donner Effect* (HDE) [30], which is a practical issue for significance of Wald Tests when the fitted probabilities in the true model are near to 0 or 1 for many observations, causing inflated standard errors and potentially undermining significance.

Reason for including sample size and collinearity as a factor. The above four points could then be directly controlled in practice to tweak a ground truth variable's signal (and consequently, the overall ground truth variables' signal) via its p-value, which is why the sample size and collinearity between ground truth variables are considered as factors for the simulation studies.

Finally, using the equivalence of the LRT to the Wald Test, accounting for the HDE, this gives birth to the definition of a ground truth variable signal, as proposed in Definition (4.1)

12.2 Simulation Study Result ($n = 200, p = 16$)

12.2.1 Random Search Results

Table 12: Summary statistics for $n = 200, p = 16$ **strong** signal-to-noise across 5 levels of $Corr(X1, X2)$

Correlation	Highest relative frequencies	Empirical mode	BIC model	Min. BIC
0	X1=1.0000	X1, X2, X3, X4, X8, X15	X1, X2, X3, X4, X8	90.10103
	X2=1.0000			
	X3=1.0000			
	X4=1.0000			
+0.5	X1=1.0000	X1, X2, X3, X4	X1, X2, X3, X4	78.65617
	X2=1.0000			
	X3=1.0000			
	X4=1.0000			
+0.9	X3=1.0000	X3, X4	X1, X2, X3, X4	112.4208
	X4=1.0000			
	X2=0.7715			
	X1=0.7680			
−0.5	X1=1.0000	X1, X2, X3, X4, X15, X16	X1, X2, X3, X4	90.58299
	X2=1.0000			
	X3=1.0000			
	X4=1.0000			
−0.9	X3=1.0000	X1, X2, X3, X4	X1, X2, X3, X4	63.69947
	X4=0.9955			
	X2=0.9900			
	X1=0.9665			

Table 13: Summary statistics for $n = 200$, $p = 16$, **moderate** signal-to-noise across 5 levels of $Corr(X1, X2)$

Correlation	Highest relative frequencies (top 6)	Empirical mode	BIC model	Min. BIC
0	X1=1.0000 X3=1.0000 X2=0.9085 X4=0.8880	X1, X2, X3, X4, X7	X1, X2, X3, X4	111.0479
+0.5	X1=1.0000 X3=0.9990 X2=0.9105 X4=0.8025	X1, X2, X3	X1, X2, X3, X4, X8	102.4557
+0.9	X3=1.0000 X1=0.7955 X4=0.7690 X2=0.6950	X1, X2, X3, X4, X6	X1, X2, X3, X4	122.9075
-0.5	X3=1.0000 X1=0.9865 X4=0.9085 X2=0.8630	X1, X2, X3, X4, X14	X1, X2, X3, X4, X14	117.6152
-0.9	X3=1.0000 X1=0.9980 X4=0.8565 X2=0.7835	X1, X2, X3, X4	X1, X2, X3, X4	67.84139

Table 14: Summary statistics for $n = 200$, $p = 16$, **weak** signal-to-noise across 5 levels of $Corr(X1, X2)$

Correlation	Highest relative frequencies (top 6)	Empirical mode	BIC model	Min.BIC
0	X1=1.0000	X1, X3	X1, X3, X4	107.06
	X3=1.0000			Top 2-5 BIC Minimizers:
	X4=0.5615			BIC for (X1,X3): 107.26
	X14=0.4030			BIC for (X1,X3,X4,X14): 107.58
	X2=0.3745			BIC for (X1,X2,X3,X4): 107.63 BIC for (X1,X3,X14): 107.74
+0.5	X1=1.0000	X1, X2, X3, X8, X14	X1, X3, X8, X14	105.09
	X3=1.0000			BIC for True Model (X1,X2,X3,X4): 106.77
	X14=0.6450			Top 2-5 BIC Minimizers:
	X8=0.5810			BIC for (X1,X2,X3,X8,X14): 105.26
	X2=0.4700			BIC for (X1,X3,X14): 105.58
+0.9	X3=1.0000	X1, X3	X1, X3	121.76
	X1=0.8460			Top 2-5 BIC Minimizers:
	X4=0.5080			BIC for (X1,X3,X4): 122.0069
	X2=0.3675			BIC for (X1,X2,X3): 122.39
				BIC for (X1,X2,X3,X4): 122.43 BIC for (X1,X3,X8): 122.57
-0.5	X3=1.0000	X1, X3	X1, X3	118.86
	X1=0.9880			BIC for True Model (X1,X2,X3,X4): 119.54
	X2=0.4880			Top 2-5 BIC Minimizers:
	X6=0.4495			BIC for (X1,X2,X3): 119.01
	X14=0.4430			BIC for (X1,X2,X3,X14): 119.03
-0.9	X4=0.3545	X1, X2, X3, X4, X6, X9	X1, X3, X6	72.86
	X3=1.0000			Top 2-5 BIC Minimizers:
	X1=0.9960			BIC for (X1,X2,X3,X4,X6): 72.87
	X6=0.5595			BIC for (X1,X3,X4,X6): 73.02
	X4=0.5350			BIC for (X1,X2,X3,X6): 73.11
	X2=0.5305			BIC for (X1,X2,X3,X4): 73.34

12.2.2 Plots

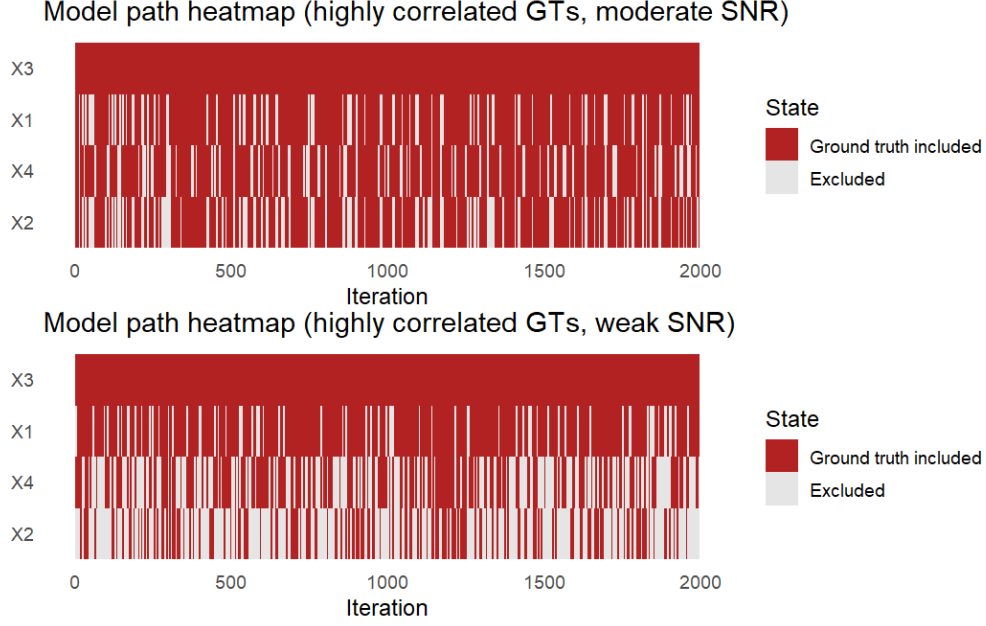


Figure 10: Heatmaps for the $n = 200$, $p = 16$, highly correlated, with both moderate & weak SNR scenarios, showing the inclusion of each variable in the sampled models. The variables are sorted from top to bottom by their relative frequencies in decreasing order.

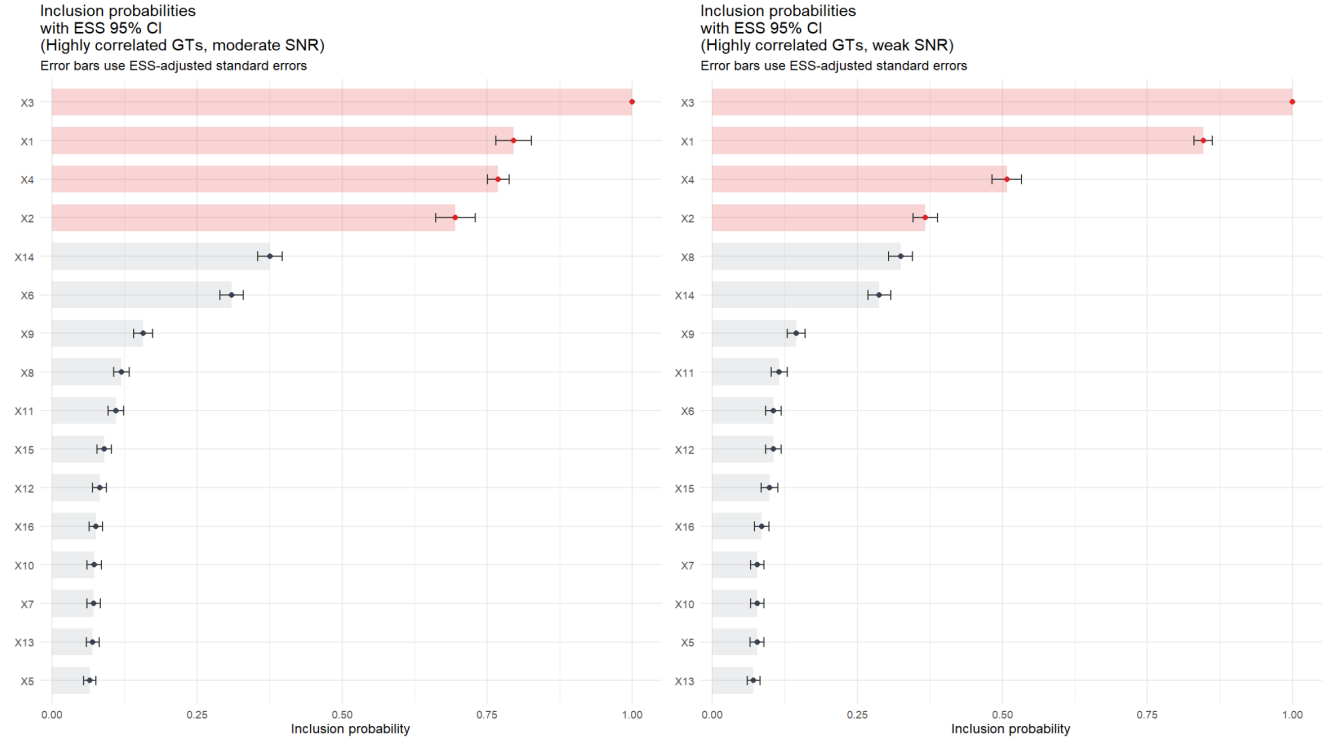


Figure 11: Bar charts of relative frequencies $\bar{v}_j = K^{-1} \sum_{k=1}^K v_j^{(k)}$ for each variable j in the $n = 200$, $p = 16$ with high correlation and moderate & weak SNR scenario, sorted from top to bottom by relative frequencies. A back-of-the-envelope 95% normal approximation confidence interval is added for each relative frequency with width $1.96 \cdot \sqrt{\frac{\bar{v}_j(1-\bar{v}_j)}{ESS(j)}}$, where $ESS(j)$ represents the effective sample size calculated from the chain $\{v_j^{(1)}, \dots, v_j^{(K)}\}$.

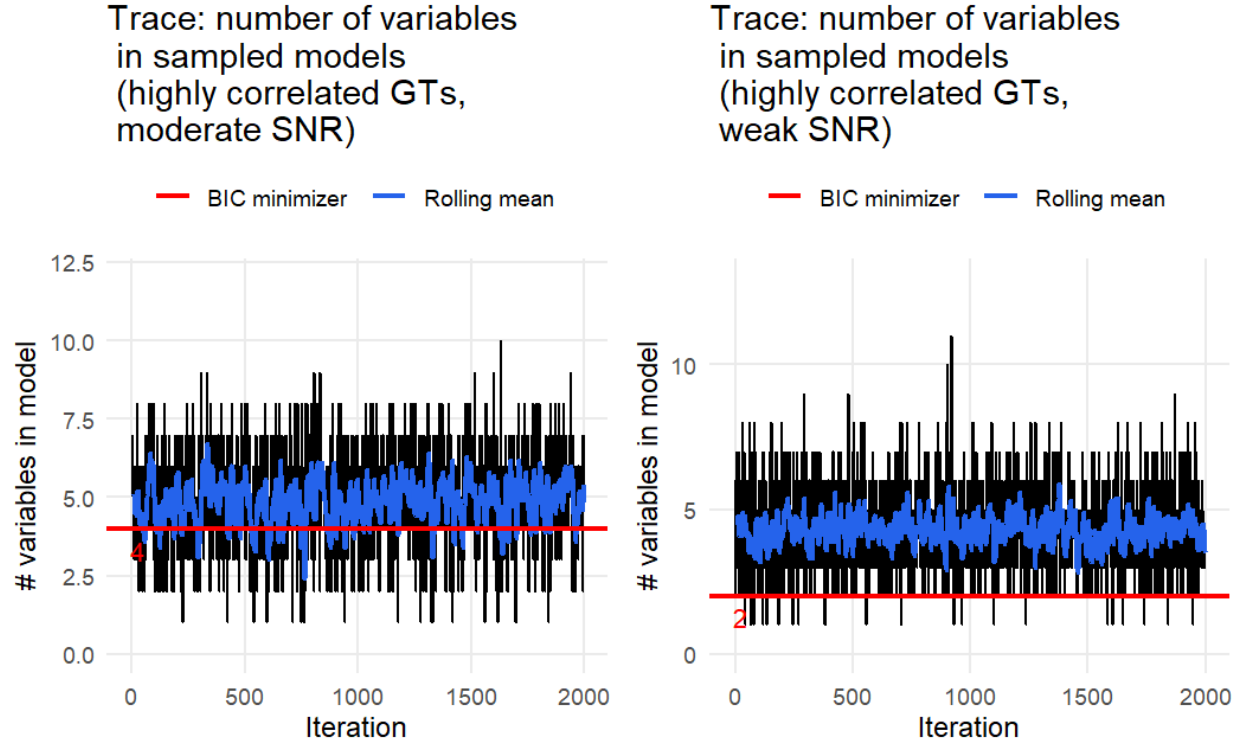


Figure 12: A trace plot of the number of variables in each sampled model (i.e. dimensions of $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}$) for the $n = 200$, $p = 16$, highly correlated, with both moderate and weak SNR displayed side-by-side. A 50-iteration moving average is calculated and plotted as a blue curve, and a red horizontal line is attached showing the dimension of the BIC minimizer; i.e. the model that minimises the BIC in the sample.

12.3 Simulation Study Result ($n = 2000, p = 16$)

12.3.1 Random Search Results

Table 15: Summary statistics for $n = 2000, p = 16$, **strong** signal-to-noise across 5 levels of $Corr(X1, X2)$

Correlation	Highest relative frequencies	Empirical mode	BIC model	Min. BIC
0	v1=1	X1, X2, X3, X4	X1, X2, X3, X4	379.3128
	v2=1			
	v3=1			
	v4=1			
+0.5	v1=1	X1, X2, X3, X4	X1, X2, X3, X4	322.3009
	v2=1			
	v3=1			
	v4=1			
+0.9	v1=1	X1, X2, X3, X4	X1, X2, X3, X4	320.8129
	v2=1			
	v3=1			
	v4=1			
-0.5	v1=1	X1, X2, X3, X4	X1, X2, X3, X4	388.7256
	v2=1			
	v3=1			
	v4=1			
-0.9	v1=1	X1, X2, X3, X4	X1, X2, X3, X4	457.2138
	v2=1			
	v3=1			
	v4=1			

Table 16: Summary statistics for $n = 2000$, $p = 16$, **moderate** signal-to-noise across 5 levels of $Corr(X1, X2)$

Correlation	Highest relative frequencies	Empirical mode	BIC model	Min. BIC
0	v1=1 v3=1 v4=0.66 v2=0.51	X1, X2, X3, X4	X1, X2, X3, X4	562.7601
+0.5	v1=1 v3=1 v2=0.85 v4=0.47	X1, X2, X3	X1, X2, X3	572.22 Top 2-5 BIC Minimizers: BIC for (X1,X2,X3,X4): 572.45 BIC for (X1,X3): 573.85 BIC for (X1,X3,X4): 574.09 BIC for (X1,X2,X3,X16): 574.89
+0.9	v1=1 v3=1 v2=0.66 v4=0.52	X1, X2, X3, X4	X1, X2, X3, X4	515.2396
-0.5	v1=1 v3=1 v4=0.7 v3=0.69	X1, X2, X3, X4	X1, X2, X3, X4	561.6066
-0.9	v1=1 v3=1 v2=0.76 v4=0.59	X1, X2, X3, X4	X1, X2, X3, X4	594.5822

Table 17: Summary statistics for $n = 2000$, $p = 16$, **weak** signal-to-noise across 5 levels of $Corr(X1, X2)$. Red fonts signify that the true model is not sampled, and blue fonts signify the true model being in the top-5 BIC minimizers.

Correlation	Highest relative frequencies	Empirical mode	BIC model	Min. BIC
0				555.95
	v1=1			BIC for True Model (X1,X2,X3,X4): 558.07
	v3=1			Top 2-5 BIC Minimizers:
	v4=0.33	X1, X3	X1, X3	BIC for (X1,X3,X4): 556.56
	v2=0.19			BIC for (X1,X2,X3): 557.45 BIC for (X1,X3,X14): 557.82 BIC for (X1,X3,X8): 558.01
+0.5				568.73
	v1=1			Top 2-5 BIC Minimizers:
	v3=1			BIC for (X1,X2,X3): 569.85
	v2=0.25	X1, X3	X1, X3	BIC for (X1,X3,X4): 570.34
	v4=0.18			BIC for (X1,X2,X3,X4): 571.47 BIC for (X1,X3,X16): 571.823
+0.9				509.39
	v1=1			Top 2-5 BIC Minimizers:
	v3=1			BIC for (X1,X3,X4): 510.54
	v4=0.27	X1, X3	X1, X3	BIC for (X1,X2,X3): 510.91
	v2=0.17			BIC for (X1,X3,X8): 511.32 BIC for (X1,X2,X3,X4): 511.89
-0.5				551.07
	v1=1			BIC for True Model (X1,X2,X3,X4): 552.93
	v3=1			Top 2-5 BIC Minimizers:
	v4=0.3	X1, X3	X1, X3	BIC for (X1,X3,X16): 551.41
	v2=0.26			BIC for (X1,X3,X4): 551.82 BIC for (X1,X2,X3): 551.98 BIC for (X1,X2,X3,X16): 552.22
-0.9				587.86
	v1=1			Top 2-5 BIC Minimizers:
	v3=1			BIC for (X1,X3,X4): 588.46
	v4=0.32	X1, X3	X1, X3	BIC for (X1,X2,X3): 589.66
	v2=0.16			BIC for (X1,X2,X3,X4): 590.22 BIC for (X1,X3,X8): 590.42

12.3.2 Plots

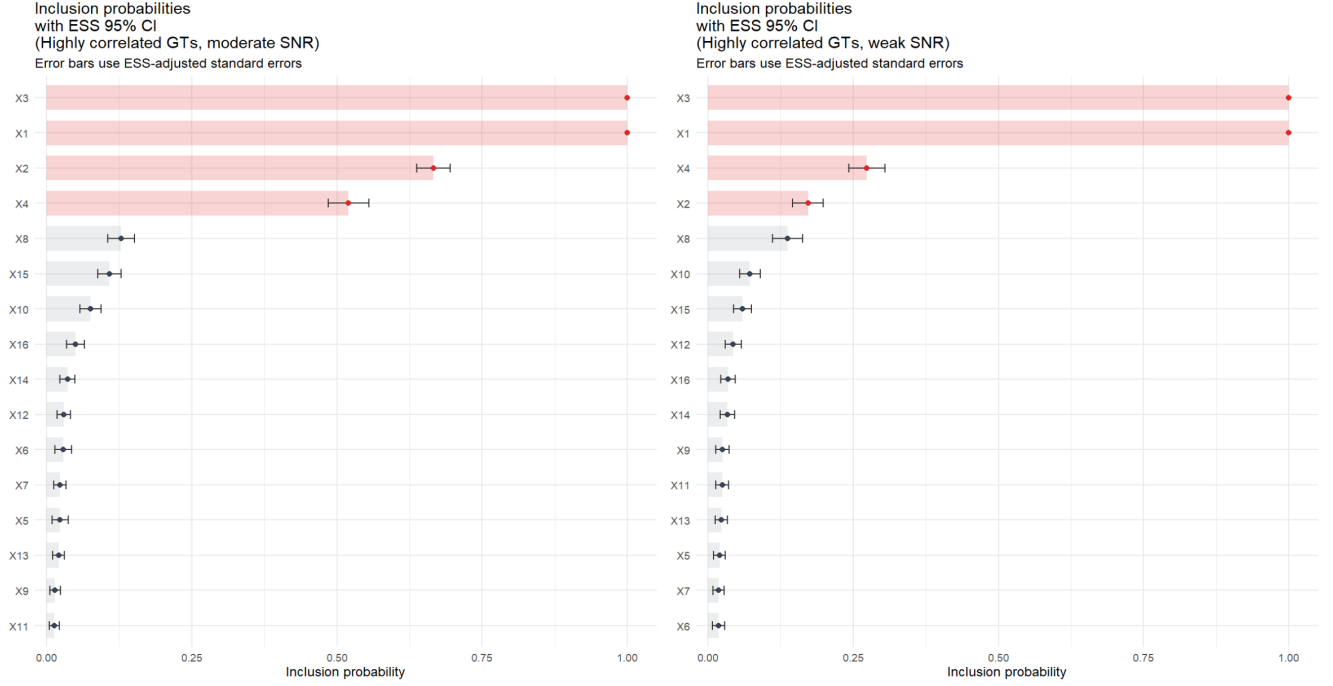


Figure 13: Bar charts of relative frequencies $\bar{v}_j = K^{-1} \sum_{k=1}^K v_j^{(k)}$ for each variable j in the $n = 2000$, $p = 16$ with high correlation and moderate & weak SNR scenario, sorted from top to bottom by relative frequencies. A back-of-the-envelope 95% normal approximation confidence interval is added for each relative frequency with width $1.96 \cdot \sqrt{\frac{\bar{v}_j(1-\bar{v}_j)}{ESS(j)}}$, where $ESS(j)$ represents the effective sample size calculated from the chain $\{v_j^{(1)}, \dots, v_j^{(K)}\}$.

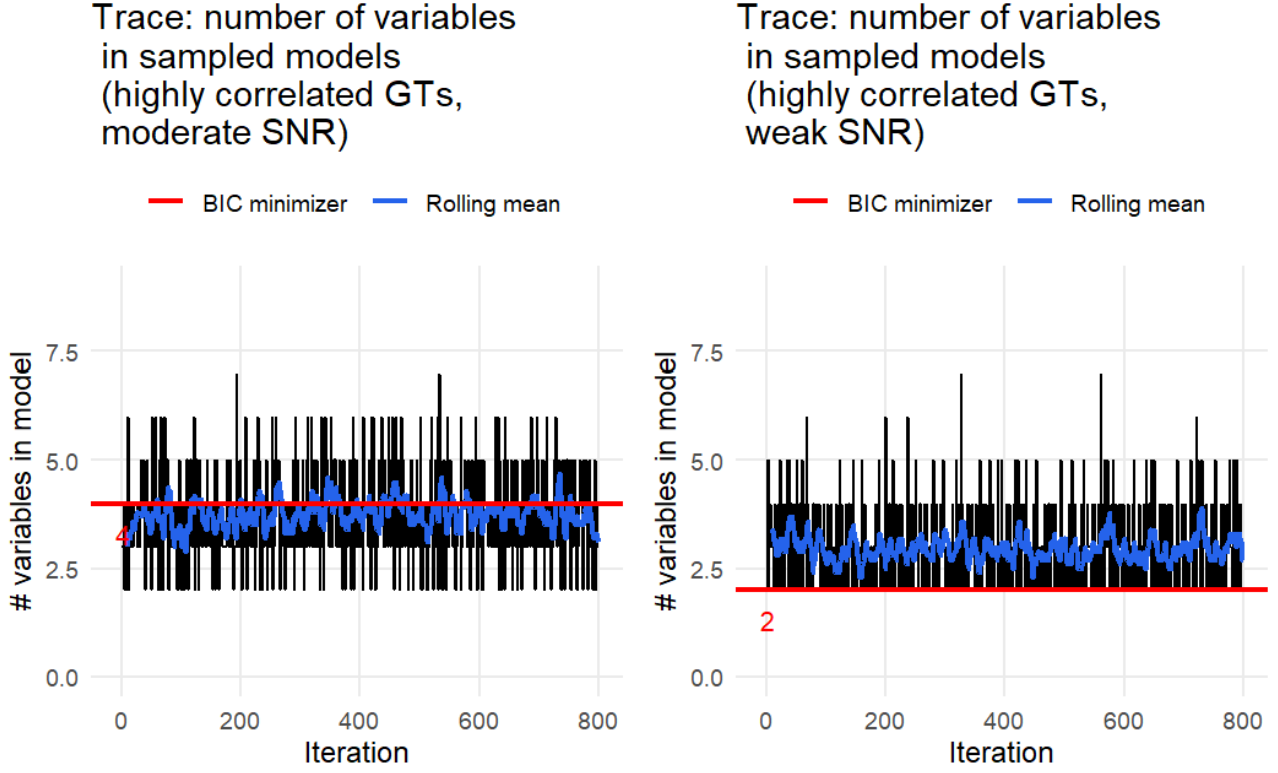


Figure 14: A trace plot of the number of variables in each sampled model (i.e. dimensions of $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}$) for the $n = 2000$, $p = 16$, highly correlated, with both moderate and weak SNR displayed side-by-side. A 50-iteration moving average is calculated and plotted as a blue curve, and a red horizontal line is attached showing the dimension of the BIC minimizer; i.e. the model that minimises the BIC in the sample.

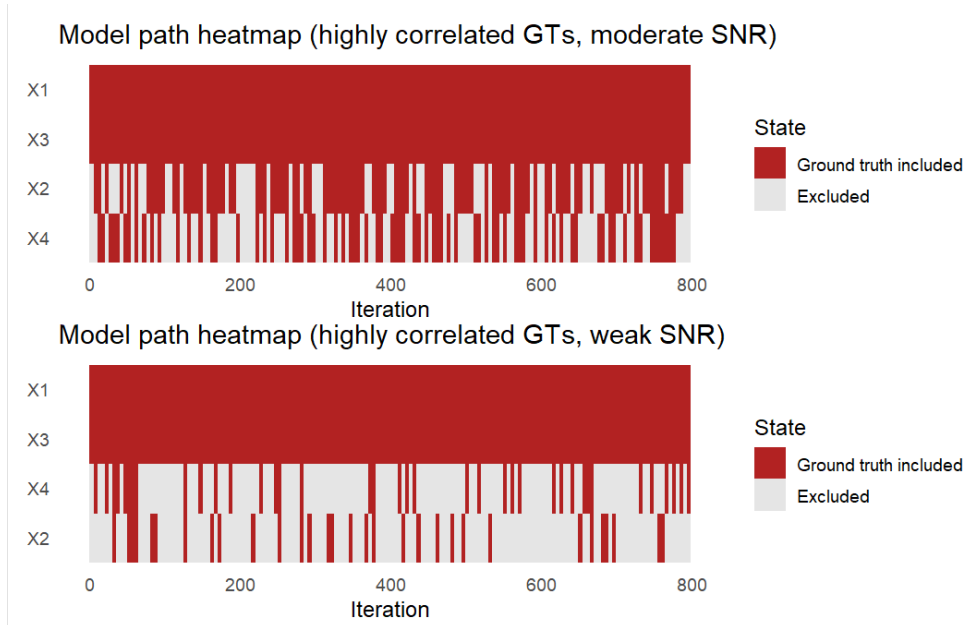


Figure 15: Heatmaps for the $n = 2000$, $p = 16$, highly correlated, with both moderate & weak SNR scenarios, showing the inclusion of each variable in the sampled models. The variables are sorted from top to bottom by their relative frequencies in decreasing order.

12.4 Simulation Study Result ($n = 2000, p = 160$)

12.4.1 Random Search Results

Table 18: Summary statistics $n = 2000, p = 160$, for **moderate** signal-to-noise across 5 levels of $Corr(X1, X2)$

Correlation	Highest relative frequencies	Empirical mode	BIC model	Min. BIC
0	v1=1	X1, X2, X3, X5, X94	X1, X2, X3, X5	539.51
	v3=1			BIC for True Model (X1,X2,X3,X4): 540.37
	v5=0.65			Top 2-5 BIC Minimizers:
	v2=0.52			BIC for (X1,X3, X5): 539.56
	v4=0.45			BIC for (X1,X2,X3): 540.11
0.9				BIC for (X1,X3,X4,X5): 540.16
				BIC for (X1,X2,X3,X5,X94): 540.73
	v1=1	X1, X3, X4, X20, X35, X47, X49, X71, X79, X102, X105, X110, X113, X116, X119, X126, X129	X1, X2, X3, X4, X105	598.31
	v3=1			BIC for True Model (X1,X2,X3,X4): 598.67
	v105=0.66			Top 2-5 BIC Minimizers:
	v116=0.65			BIC for (X1,X2,X3,X4,X110): 598.79
	v110=0.62			BIC for (X1,X3,X105,X116,X126): 599.74
	v4=0.58			BIC for (X1,X2,X3,X4,X105,X113,X157): 599.98
	v157=0.5			BIC for (X1,X3,X4,X105,X113,X116): 600.03
	v2=0.47			

12.4.2 Plots

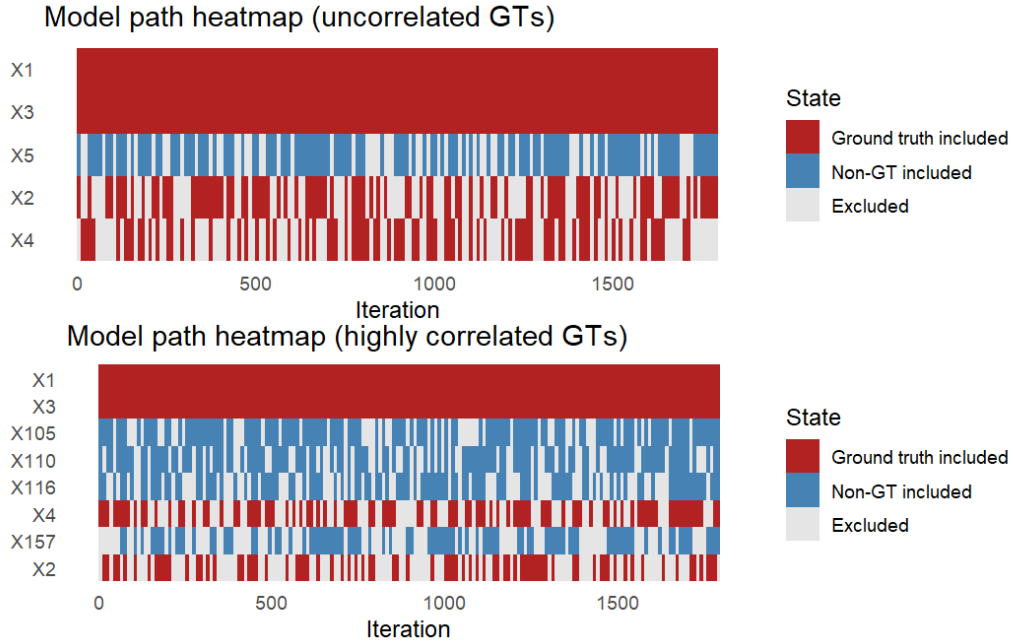


Figure 16: Heatmaps for the $n = 2000, p = 160$, highly correlated, with both moderate & weak SNR scenarios, showing the inclusion of each variable in the sampled models. The variables are sorted from top to bottom by their relative frequencies in decreasing order.

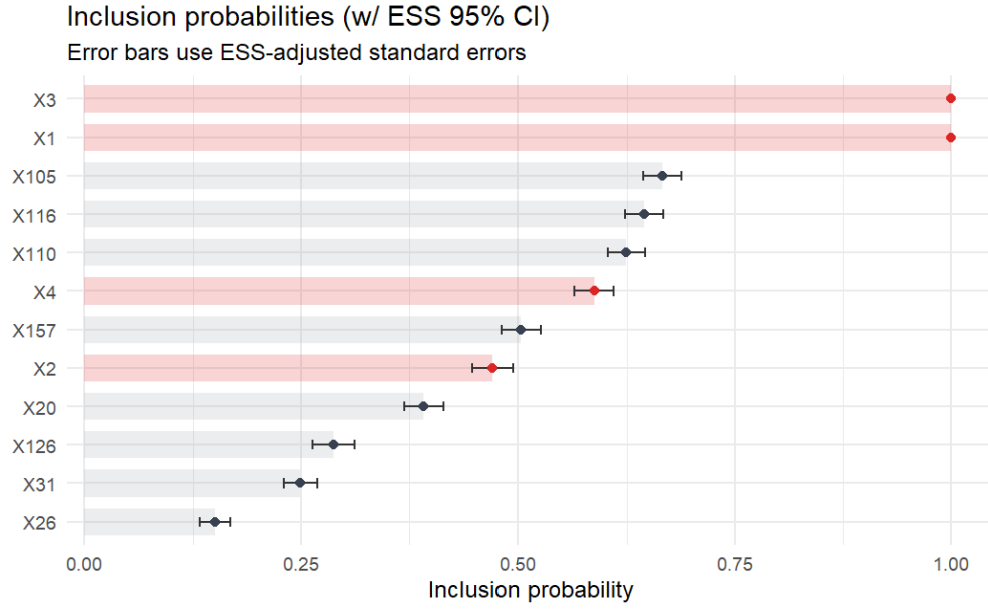


Figure 17: A bar chart of relative frequencies $\bar{v}_j = K^{-1} \sum_{k=1}^K v_j^{(k)}$ for each variable j in the $n = 2000$, $p = 160$ with moderate SNR and high correlation scenario, sorted from top to bottom by relative frequencies. A back-of-the-envelope 95% normal approximation confidence interval is added for each relative frequency with width $1.96 \cdot \sqrt{\frac{\bar{v}_j(1-\bar{v}_j)}{ESS(j)}}$, where $ESS(j)$ represents the effective sample size calculated from the chain $\{v_j^{(1)}, \dots, v_j^{(K)}\}$.

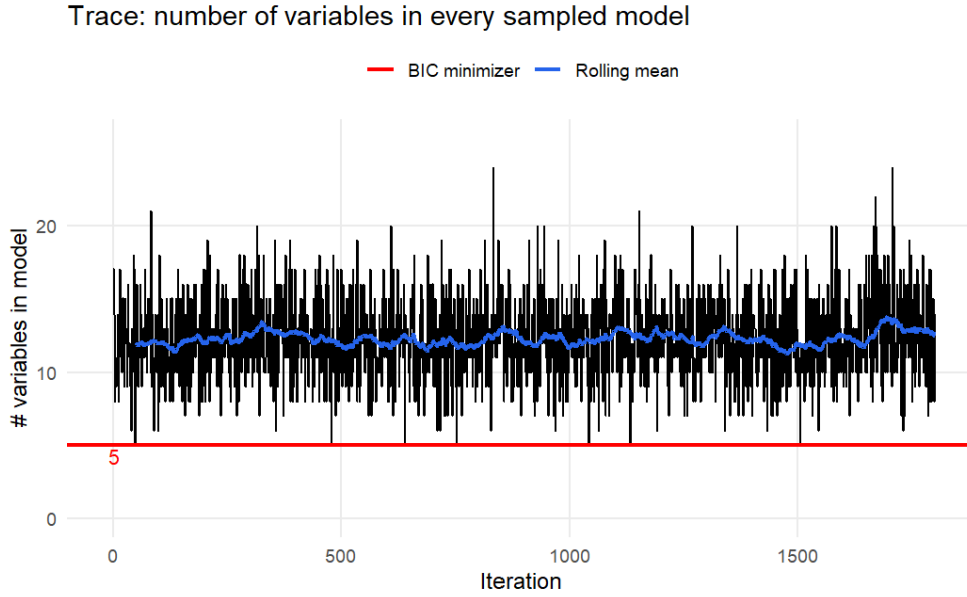


Figure 18: A trace plot of the number of variables in each sampled model (i.e. dimensions of $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}$) for the $n = 2000$, $p = 160$, moderate SNR, and high correlation scenario. A 50-iteration moving average is calculated and plotted as a blue curve, and a red horizontal line is attached showing the dimension of the BIC minimizer; i.e. the model that minimises the BIC in the sample.

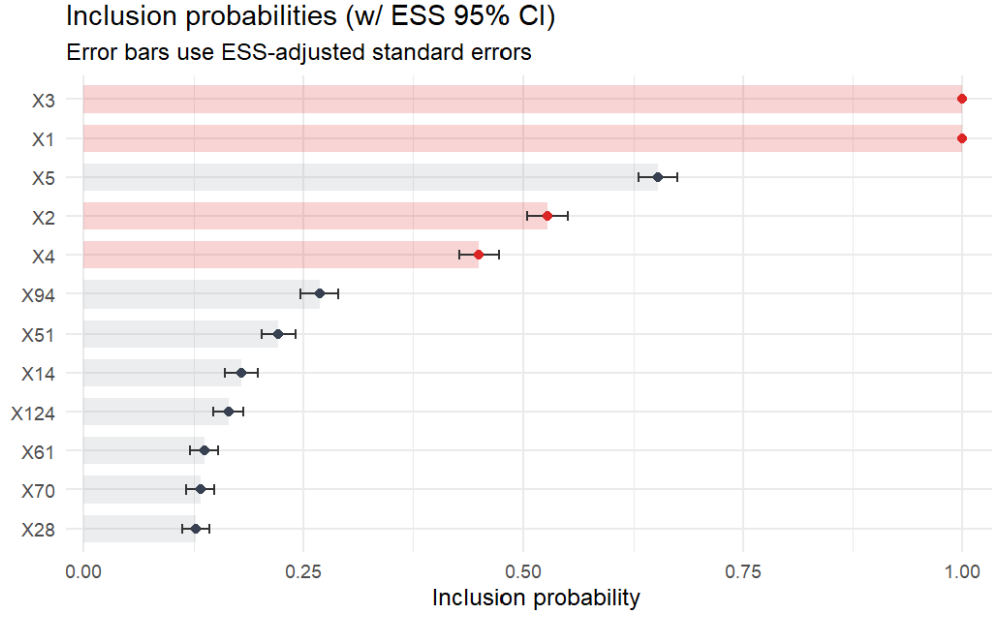


Figure 19: A bar chart of relative frequencies $\bar{v}_j = K^{-1} \sum_{k=1}^K v_j^{(k)}$ for each variable j in the $n = 2000$, $p = 160$ with moderate SNR and uncorrelated scenario, sorted from top to bottom by relative frequencies. A back-of-the-envelope 95% normal approximation confidence interval is added for each relative frequency with width $1.96 \cdot \sqrt{\frac{\bar{v}_j(1-\bar{v}_j)}{ESS(j)}}$, where $ESS(j)$ represents the effective sample size calculated from the chain $\{v_j^{(1)}, \dots, v_j^{(K)}\}$.

12.4.3 Baseline Comparison with LASSO

Table 19: LASSO (corr=0)

X1	X3	X4	X5	X44	X61	X121	X142
0.2529	-0.2437	-0.0039	-0.0041	-0.001	-0.0043	-0.0027	0.001

Table 20: LASSO (corr=0.9)

X1	X2	X3	X105	X110	X116	X157
0.2009	0.0493	-0.2282	0.0039	0.0036	2e-04	-0.0023

12.5 Simulation Study Result ($n = 10000$, $p = 160$)

12.5.1 Random Search Results

Table 21: Summary statistics for $n = 10000$, $p = 160$, **moderate** signal-to-noise across 5 levels of $Corr(X1, X2)$

Correlation	Highest relative frequencies	Empirical mode	BIC model	Min. BIC
0.9	v3=1			6766.59
	v1=0.99			Top 2-5 BIC Minimizers:
	v4=0.548	X1,X3,X4	X1,X3,X4	BIC for (X1,X3): 6766.76
	v2=0.269			BIC for (X1,X2,X3,X4): 6767.61
	v42=0.21			BIC for (X1,X2,X3): 6767.80
				BIC for (X1,X3,X4,X42): 6767.89

12.5.2 Plots

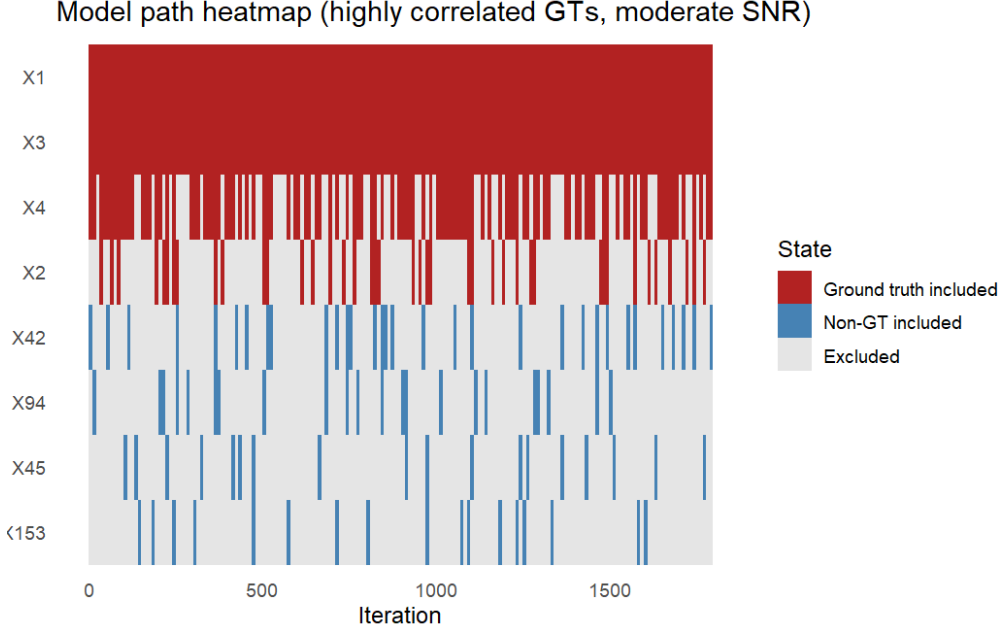


Figure 20: A heatmap for the $n = 10000$, $p = 160$ and moderate SNR scenario, showing the inclusion of each variable in the sampled models. The variables are sorted from top to bottom by their relative frequencies in decreasing order.

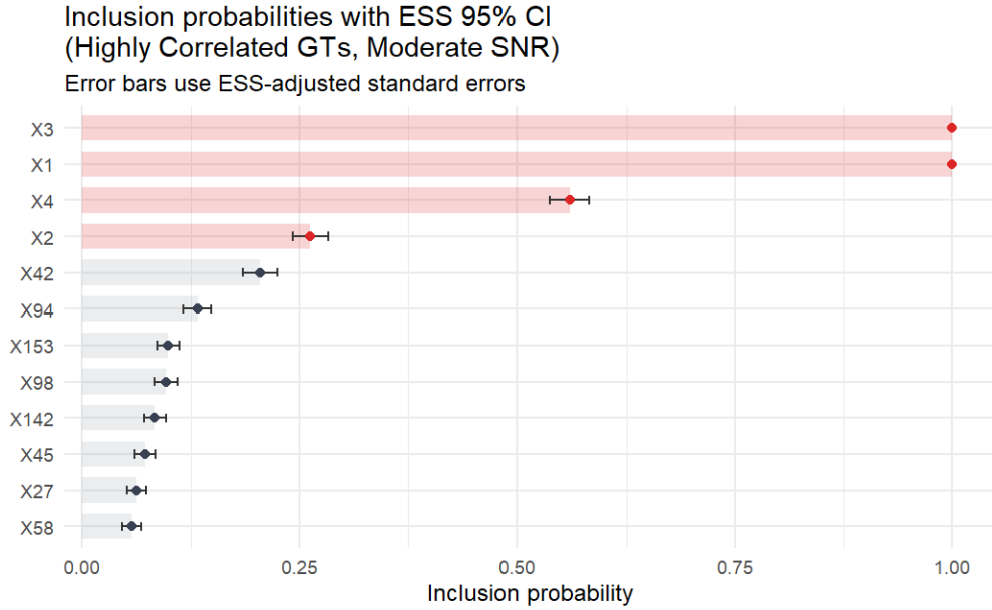


Figure 21: A bar chart of relative frequencies $\bar{v}_j = K^{-1} \sum_{k=1}^K v_j^{(k)}$ for each variable j in the $n = 10000$, $p = 160$, moderate SNR scenario, sorted from top to bottom by relative frequencies. A back-of-the-envelope 95% normal approximation confidence interval is added for each relative frequency with width $1.96 \cdot \sqrt{\frac{\bar{v}_j(1-\bar{v}_j)}{ESS(j)}}$, where $ESS(j)$ represents the effective sample size calculated from the chain $\{v_j^{(1)}, \dots, v_j^{(K)}\}$.

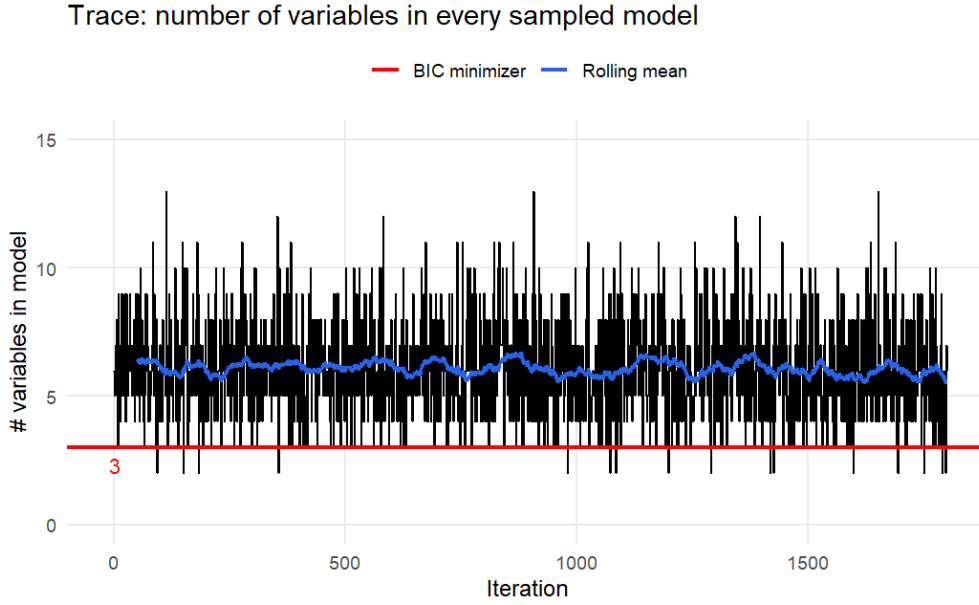


Figure 22: A trace plot of the number of variables in each sampled model (i.e. dimensions of $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(K)}\}$) for the $n = 10000$, $p = 160$, moderate SNR scenario. A 50-iteration moving average is calculated and plotted as a blue curve, and a red horizontal line is attached showing the dimension of the BIC minimizer; i.e. the model that minimises the BIC in the sample.

12.6 True coefficients used to generate synthetic data in simulation study.

12.6.1 $n = 200$, $p = 16$

	Estimate	Std. Error	z value	$\Pr(> z)$
(Intercept)	-0.168406082	0.198530052	-0.848264934	0.396290442
X1	1.229252818	0.260846026	4.712561039	0.000002446
X2	1.404351801	0.267703000	5.245932256	0.000000155
X3	-1.469011959	0.257656624	-5.701432921	0.000000012
X4	-1.028836105	0.220623855	-4.663304002	0.000003112

Table 22: $\text{Corr}(X1, X2) = 0$, SNR=strong

	Estimate	Std. Error	z value	$\Pr(> z)$
(Intercept)	-0.124842355	0.173855336	-0.718081815	0.472706842
X1	0.905155599	0.203808676	4.441202495	0.000008946
X2	0.565975821	0.187214551	3.023140132	0.002501664
X3	-1.177939794	0.209517018	-5.622167610	0.000000019
X4	-0.515276379	0.174470670	-2.953369636	0.003143253

Table 23: $\text{Corr}(X1, X2) = 0$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.099905735	0.177459820	0.562976651	0.573450784
X1	1.002303782	0.210769703	4.755445244	0.000001980
X2	0.366182726	0.183808444	1.992197520	0.046349390
X3	-1.383355076	0.225348139	-6.138746389	0.000000001
X4	-0.407763185	0.171365842	-2.379489290	0.017336647

Table 24: $Corr(X1, X2) = 0$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.008365180	0.214894616	-0.038926896	0.968948673
X1	1.789895840	0.339815425	5.267258948	0.000000138
X2	1.272216609	0.305996549	4.157617510	0.000032158
X3	-1.300590367	0.268430217	-4.845171239	0.000001265
X4	-1.602141320	0.300429060	-5.332844029	0.000000097

Table 25: $Corr(X1, X2) = 0.5$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.014930960	0.181721579	0.082163936	0.934516351
X1	1.348492108	0.260823484	5.170133030	0.000000234
X2	0.659903556	0.222984865	2.959409627	0.003082291
X3	-0.813851294	0.200460940	-4.059899625	0.000049094
X4	-0.522922127	0.192400832	-2.717878723	0.006570192

Table 26: $Corr(X1, X2) = 0.5$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.002401951	0.178069242	0.013488860	0.989237773
X1	1.296347071	0.250622759	5.172503396	0.000000231
X2	0.468645755	0.209713617	2.234693968	0.025437447
X3	-0.896456475	0.201592221	-4.446880301	0.000008713
X4	-0.364348649	0.184360999	-1.976278341	0.048123261

Table 27: $Corr(X1, X2) = 0.5$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.078514212	0.171409914	-0.458049421	0.646916936
X1	1.422047975	0.421416754	3.374445754	0.000739645
X2	-1.455625979	0.428231534	-3.399156446	0.000675940
X3	1.033084687	0.208070003	4.965082299	0.000000687
X4	-1.165407368	0.215715008	-5.402532611	0.000000066

Table 28: $Corr(X1, X2) = 0.9$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.172738468	0.162292636	-1.064364176	0.287163756
X1	1.316152411	0.389207458	3.381621769	0.000720593
X2	-1.094893594	0.384376977	-2.848489005	0.004392737
X3	0.978003486	0.186778647	5.236163234	0.000000164
X4	-0.455967134	0.173196945	-2.632651135	0.008472132

Table 29: $Corr(X1, X2) = 0.9$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.095701237	0.162765962	-0.587968366	0.556553526
X1	1.171669705	0.387731360	3.021859527	0.002512271
X2	-0.790396275	0.379535358	-2.082536603	0.037293483
X3	1.024074266	0.187929407	5.449249716	0.000000051
X4	-0.382092193	0.171005631	-2.234383687	0.025457838

Table 30: $Corr(X1, X2) = 0.9$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.162580183	0.199721122	0.814036002	0.415624326
X1	1.585061082	0.288233714	5.499221664	0.000000038
X2	1.755823119	0.294542348	5.961190751	0.000000003
X3	-1.449034529	0.251417384	-5.763461960	0.000000008
X4	-1.107806761	0.243184773	-4.555411707	0.000005228

Table 31: $Corr(X1, X2) = -0.5$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.154571668	0.167207332	0.924431165	0.355261875
X1	0.820396820	0.207718958	3.949551968	0.000078298
X2	0.568098146	0.198350613	2.864110866	0.004181816
X3	-1.037896246	0.192544089	-5.390434227	0.000000070
X4	-0.553831325	0.178753797	-3.098291252	0.001946400

Table 32: $Corr(X1, X2) = -0.5$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.091308122	0.165763712	0.550832995	0.581748163
X1	0.796699895	0.205460014	3.877639638	0.000105475
X2	0.424631404	0.193267795	2.197114130	0.028012296
X3	-1.076346668	0.192041866	-5.604750103	0.000000021
X4	-0.349806736	0.172356887	-2.029548934	0.042402410

Table 33: $Corr(X1, X2) = -0.5$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.503979969	0.262600530	-1.919188697	0.054960457
X1	2.194398811	0.657218025	3.338920611	0.000841046
X2	-2.333959778	0.689866022	-3.383207324	0.000716445
X3	1.509113311	0.299551607	5.037907579	0.000000471
X4	-1.041668724	0.278812151	-3.736095153	0.000186900

Table 34: $Corr(X1, X2) = -0.9$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.512871199	0.251194709	-2.041727718	0.041178548
X1	2.417987629	0.646162251	3.742075038	0.000182507
X2	-1.662515937	0.625219500	-2.659091629	0.007835165
X3	1.436192682	0.279931975	5.130506026	0.000000289
X4	-0.752142714	0.255842862	-2.939862026	0.003283584

Table 35: $Corr(X1, X2) = -0.9$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.382055760	0.232873050	-1.640618182	0.100876698
X1	2.124011421	0.595190313	3.568625654	0.000358859
X2	-1.288800393	0.569445429	-2.263255314	0.023619955
X3	1.458149531	0.269597286	5.408620958	0.000000064
X4	-0.569890679	0.239761774	-2.376903833	0.017458639

Table 36: $Corr(X1, X2) = -0.9$, SNR=weak

12.6.2 $n = 2000$, $p = 16$

Table 37: $Corr(X1, X2) = 0$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.1306	0.0948	-1.38	0.1684
X1	2.9592	0.1849	16.00	0.0000
X2	3.4150	0.2098	16.27	0.0000
X3	-2.9289	0.1848	-15.85	0.0000
X4	-4.4251	0.2579	-17.15	0.0000

Table 38: $Corr(X1, X2) = 0$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.0679	0.0767	0.89	0.3757
X1	2.9624	0.1503	19.72	0.0000
X2	0.2081	0.0754	2.76	0.0058
X3	-2.9993	0.1505	-19.92	0.0000
X4	-0.2340	0.0789	-2.97	0.0030

Table 39: $Corr(X1, X2) = 0$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.0339	0.0770	0.44	0.6592
X1	2.9747	0.1510	19.70	0.0000
X2	0.1610	0.0755	2.13	0.0330
X3	-3.0387	0.1525	-19.93	0.0000
X4	-0.1979	0.0790	-2.51	0.0122

Table 40: $Corr(X1, X2) = 0.5$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.0636	0.1034	-0.61	0.5387
X1	3.2558	0.2192	14.86	0.0000
X2	3.5095	0.2384	14.72	0.0000
X3	-3.2362	0.2113	-15.32	0.0000
X4	-4.7092	0.2902	-16.23	0.0000

Table 41: $Corr(X1, X2) = 0.5$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.0463	0.0757	-0.61	0.5406
X1	2.8031	0.1505	18.63	0.0000
X2	0.2893	0.0885	3.27	0.0011
X3	-2.9763	0.1522	-19.55	0.0000
X4	-0.1976	0.0743	-2.66	0.0079

Table 42: $Corr(X1, X2) = 0.5$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.0068	0.0758	-0.09	0.9289
X1	2.8818	0.1533	18.79	0.0000
X2	0.2027	0.0880	2.30	0.0212
X3	-2.9702	0.1523	-19.50	0.0000
X4	-0.1548	0.0743	-2.08	0.0371

Table 43: $Corr(X1, X2) = 0.9$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.0312	0.1039	-0.30	0.7641
X1	2.5744	0.2863	8.99	0.0000
X2	3.7987	0.3152	12.05	0.0000
X3	-2.9992	0.1994	-15.04	0.0000
X4	-4.3824	0.2715	-16.14	0.0000

Table 44: $Corr(X1, X2) = 0.9$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.0349	0.0804	0.43	0.6638
X1	3.0322	0.2318	13.08	0.0000
X2	0.5539	0.1855	2.99	0.0028
X3	-3.1021	0.1601	-19.38	0.0000
X4	-0.2160	0.0778	-2.78	0.0055

Table 45: $Corr(X1, X2) = 0.9$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.0161	0.0806	0.20	0.8415
X1	3.1771	0.2366	13.43	0.0000
X2	0.4077	0.1853	2.20	0.0277
X3	-3.1494	0.1627	-19.35	0.0000
X4	-0.1836	0.0779	-2.36	0.0185

Table 46: $Corr(X1, X2) = -0.5$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.1431	0.0939	1.52	0.1277
X1	3.1252	0.1936	16.14	0.0000
X2	3.6617	0.2147	17.06	0.0000
X3	-3.0273	0.1859	-16.28	0.0000
X4	-4.4785	0.2508	-17.86	0.0000

Table 47: $Corr(X1, X2) = -0.5$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.0519	0.0768	-0.68	0.4990
X1	3.0900	0.1602	19.28	0.0000
X2	0.2643	0.0882	3.00	0.0027
X3	-3.0958	0.1563	-19.80	0.0000
X4	-0.2333	0.0770	-3.03	0.0024

Table 48: $Corr(X1, X2) = -0.5$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.0706	0.0774	-0.91	0.3622
X1	3.1393	0.1634	19.21	0.0000
X2	0.2049	0.0886	2.31	0.0208
X3	-3.1370	0.1591	-19.72	0.0000
X4	-0.1836	0.0773	-2.37	0.0176

Table 49: $Corr(X1, X2) = -0.9$, SNR=strong

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.1144	0.0858	1.33	0.1825
X1	2.9133	0.2377	12.26	0.0000
X2	3.5037	0.2538	13.81	0.0000
X3	-2.8247	0.1613	-17.51	0.0000
X4	-4.4391	0.2304	-19.27	0.0000

Table 50: $Corr(X1, X2) = -0.9$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.0845	0.0743	1.14	0.2558
X1	2.9271	0.2145	13.65	0.0000
X2	0.5485	0.1738	3.16	0.0016
X3	-3.1565	0.1561	-20.22	0.0000
X4	-0.2123	0.0725	-2.93	0.0034

Table 51: $Corr(X1, X2) = -0.9$, SNR=weak

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.0566	0.0746	0.76	0.4475
X1	2.8380	0.2130	13.32	0.0000
X2	0.3503	0.1738	2.02	0.0438
X3	-3.1195	0.1549	-20.14	0.0000
X4	-0.1834	0.0724	-2.53	0.0114

12.6.3 $n = 2000$, $p = 160$

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.068707883	0.078351608	-0.876917332	0.380531542
X1	3.164182593	0.159396328	19.851038140	0.000000000
X2	0.223377562	0.078377672	2.850015274	0.004371713
X3	-3.122430539	0.158858731	-19.655391459	0.000000000
X4	-0.214611376	0.081127674	-2.645353516	0.008160561

Table 52: $Corr(X1, X2) = 0$, SNR=moderate

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.086821527	0.073947705	1.174093586	0.240357542
X1	2.625446391	0.206623431	12.706431099	0.000000000
X2	0.440823402	0.165700685	2.660359572	0.007805727
X3	-2.739883276	0.137781731	-19.885679051	0.000000000
X4	-0.214938746	0.073722038	-2.915529101	0.003550860

Table 53: $Corr(X1, X2) = 0.9$, SNR=moderate

12.6.4 $n = 10000$, $p = 16$

	Estimate	Std. Error	z value	$\Pr(> z)$
(Intercept)	0.005830767	0.020368241	0.286267566	0.774673198
X1	0.212275757	0.046831000	4.532804248	0.000005821
X2	0.125083860	0.046794668	2.673036575	0.007516805
X3	-0.203675421	0.020572193	-9.900520586	0.000000000
X4	-0.063084720	0.020404321	-3.091733320	0.001989915

Table 54: $\text{Corr}(X1, X2) = 0.9$, SNR=moderate

12.7 P-charts

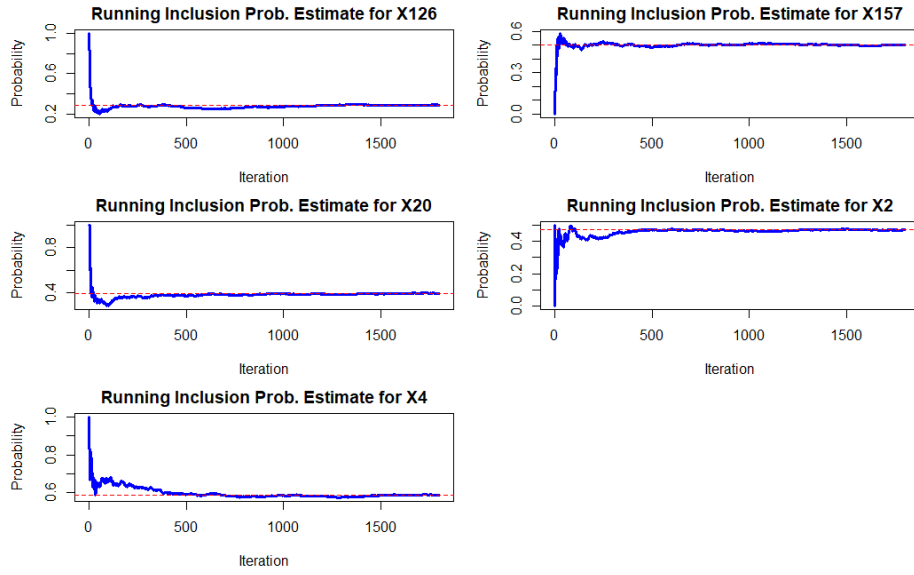


Figure 23: P-chart for the 'slowest converging' variable in the $n=2000$, $p=160$, high correlation scenario. Slowest convergence is defined as having the largest Monte Carlo Standard Error.

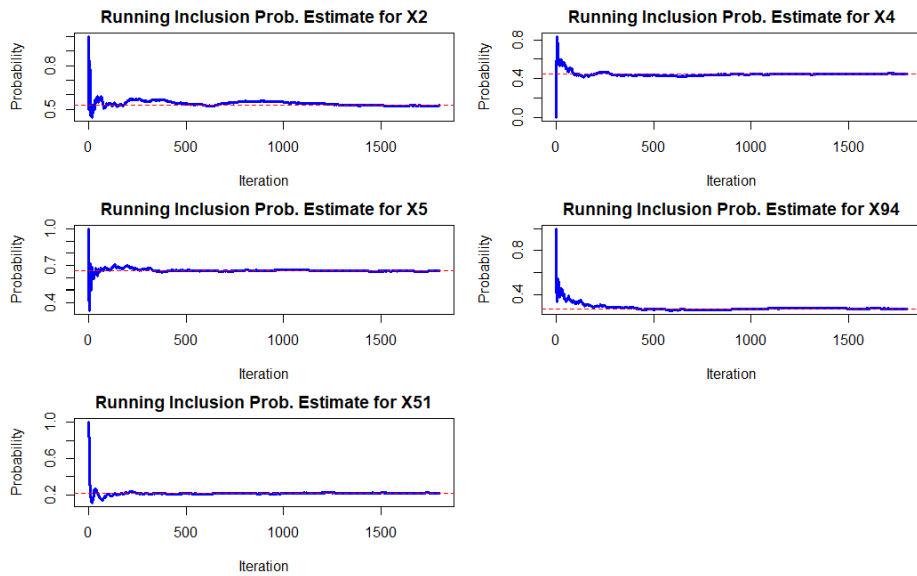


Figure 24: P-chart for the 'slowest converging' variable in the $n=2000$, $p=160$, no correlation scenario. Slowest convergence is defined as having the largest Monte Carlo Standard Error.

12.8 Mixing behavior of chains across all λ

Table 55: BIC Minimum values for each of the 4 chains as well as the minimum across all 4 chains sampled from $P_\lambda(\mathbf{v})$ for all prespecified values of λ . `min_bic` encodes the BIC minimum for a specific chain and λ value, `global_min_bic` encodes the BIC minimum across all chains for a specific λ , `min_bic.diff_to_global` encodes how different a chain-specific BIC minimum to the BIC minimum across all chains (again, for a specific λ). Across all chains from every λ , the minimum comes from $\lambda = 0.75$ from the chain with the sparse initial value.

#	lambda	chain	min_bic	global_min_bic	min_bic.diff_to_global
1	0.75	3: sparse	4795.27	4795.27	0.00
2	0.75	2: empty	4798.56	4795.27	3.29
3	0.75	1: full	4798.62	4795.27	3.35
4	0.75	4: random	4812.86	4795.27	17.59
5	0.50	4: random	4795.94	4795.94	0.00
6	0.50	2: empty	4796.96	4795.94	1.02
7	0.50	3: sparse	4799.47	4795.94	3.53
8	0.50	1: full	4802.42	4795.94	6.48
9	1.00	2: empty	4804.78	4804.78	0.00
10	1.00	3: sparse	4809.66	4804.78	4.88
11	1.00	4: random	4817.60	4804.78	12.82
12	1.00	1: full	4820.95	4804.78	16.16
13	0.25	3: sparse	4806.78	4806.78	0.00
14	0.25	2: empty	4811.53	4806.78	4.75
15	0.25	1: full	4812.70	4806.78	5.92
16	0.25	4: random	4819.81	4806.78	13.02
17	1.50	3: sparse	4814.02	4814.02	0.00
18	1.50	4: random	4823.92	4814.02	9.90
19	1.50	2: empty	4834.58	4814.02	20.56
20	1.50	1: full	4838.73	4814.02	24.70
21	1.75	4: random	4814.73	4814.73	0.00
22	1.75	2: empty	4834.58	4814.73	19.85
23	1.75	1: full	4838.73	4814.73	23.99
24	1.75	3: sparse	4857.62	4814.73	42.89
25	2.50	4: random	4818.92	4818.92	0.00
26	2.50	2: empty	4835.28	4818.92	16.36
27	2.50	1: full	4844.17	4818.92	25.25
28	2.50	3: sparse	4855.12	4818.92	36.20
29	0.20	4: random	4819.71	4819.71	0.00
30	0.20	1: full	4825.90	4819.71	6.19
31	0.20	2: empty	4829.13	4819.71	9.42
32	0.20	3: sparse	4835.34	4819.71	15.64
33	3.00	3: sparse	4825.94	4825.94	0.00
34	3.00	4: random	4835.29	4825.94	9.35
35	3.00	1: full	4844.17	4825.94	18.22
36	3.00	2: empty	4857.74	4825.94	31.79
37	0.18	1: full	4827.92	4827.92	0.00
38	0.18	2: empty	4833.82	4827.92	5.90
39	0.18	4: random	4837.65	4827.92	9.72

Continued on next page

Table 55 (continued)

#	lambda	chain	min_bic	global_min_bic	min_bic_diff_to_global
40	0.18	3: sparse	4838.91	4827.92	10.99
41	0.14	1: full	4850.66	4850.66	0.00
42	0.14	3: sparse	4858.13	4850.66	7.48
43	0.14	4: random	4859.19	4850.66	8.53
44	0.14	2: empty	4860.28	4850.66	9.62
45	0.10	1: full	4867.10	4867.10	0.00
46	0.10	3: sparse	4877.60	4867.10	10.50
47	0.10	4: random	4881.09	4867.10	13.98
48	0.10	2: empty	4884.14	4867.10	17.03

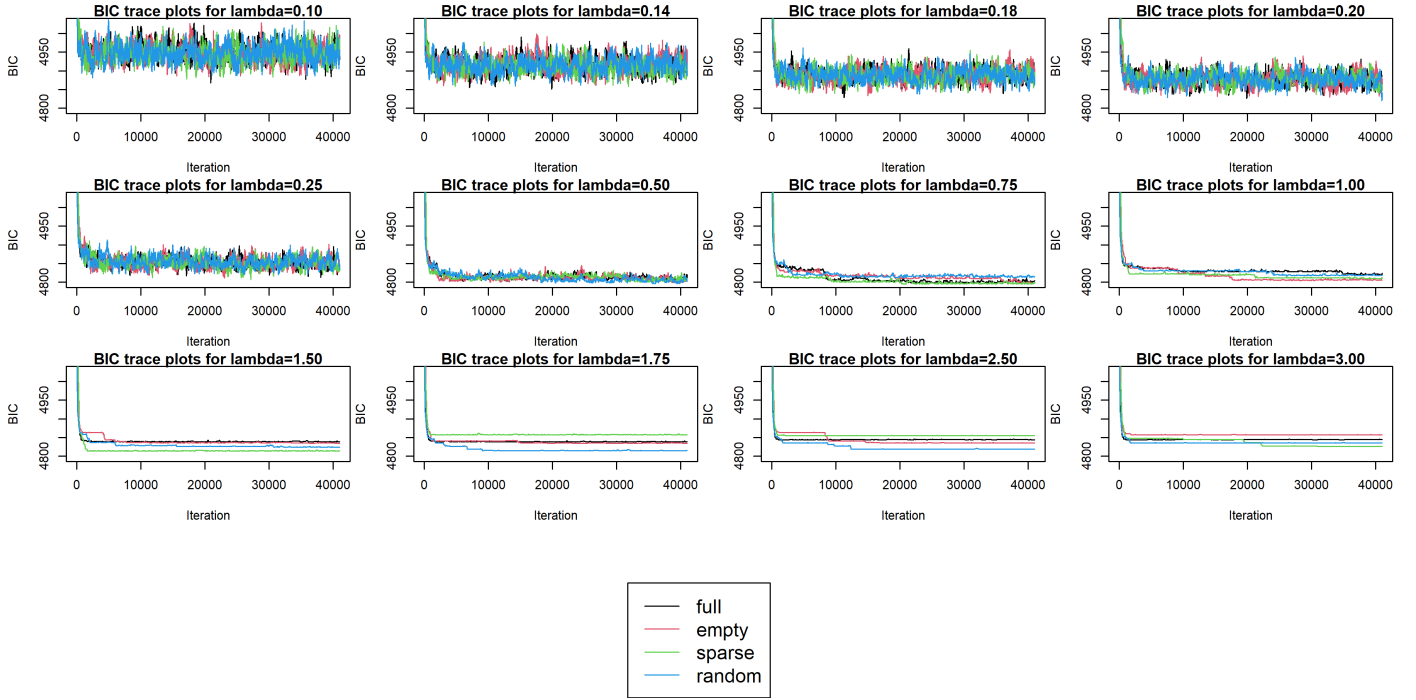


Figure 25: BIC trace plots across all chains and all λ values. All plots' y-axis scale is ordered from the BIC minimum across all samples from all λ to a predefined maximum.

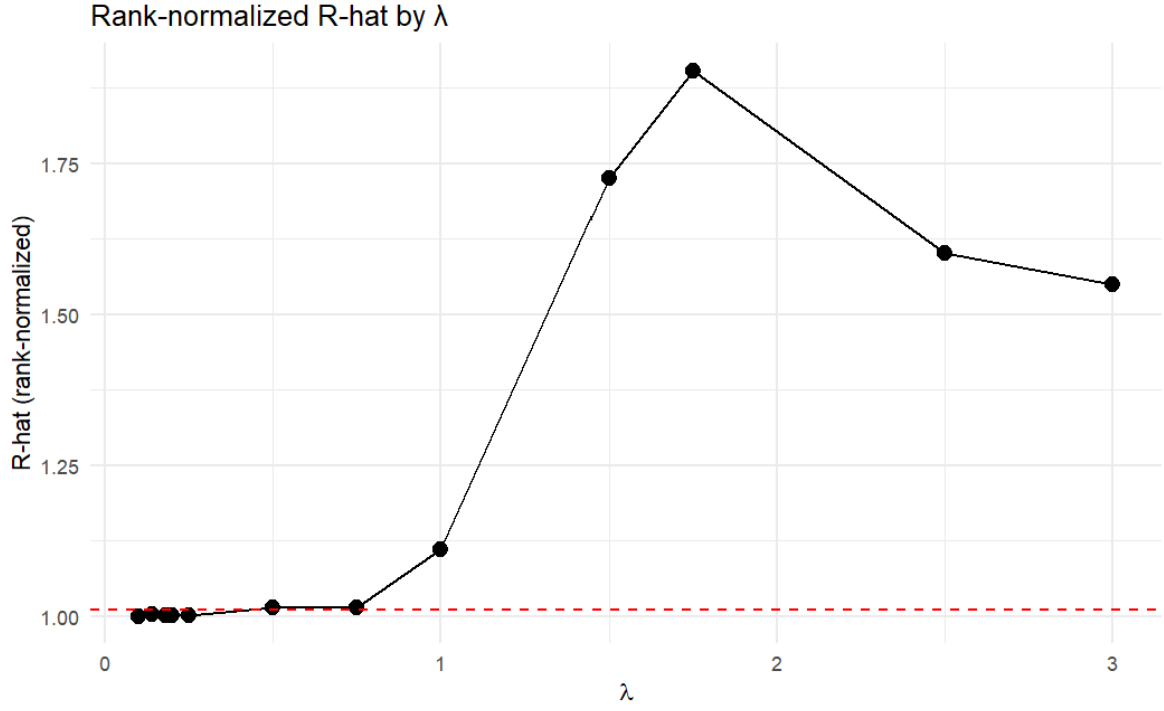


Figure 26: Rank-normalized $\hat{R}$ that computes within and between variability of ranks of BICs across 4 overdispersed chains for all λ values. Dashed horizontal line represents $\hat{R} = 1.01$, the proposed threshold of stationarity from Vehtari et al. [26]. For $\lambda = 0.75$, $\hat{R} = 1.013$ which is very close to the proposed cutoff.

Table 56: Proportion of distinct models obtained across 4 chains for each λ . Each cell represents Proportion of distinct model (average model dimension sampled)

	lambda	1: full	2: empty	3: sparse	4: random
1	0.10	0.549 (74)	0.546 (73)	0.550 (74)	0.551 (74)
2	0.14	0.443 (70)	0.437 (70)	0.432 (69)	0.441 (70)
3	0.18	0.347 (67)	0.346 (66)	0.345 (67)	0.348 (66)
4	0.20	0.308 (65)	0.304 (65)	0.308 (64)	0.311 (64)
5	0.25	0.222 (61)	0.225 (59)	0.228 (61)	0.232 (61)
6	0.50	0.061 (57)	0.060 (51)	0.056 (56)	0.066 (52)
7	0.75	0.023 (52)	0.016 (50)	0.018 (55)	0.021 (55)
8	1.00	0.010 (57)	0.011 (48)	0.007 (54)	0.010 (57)
9	1.50	0.004 (62)	0.004 (50)	0.003 (54)	0.006 (56)
10	1.75	0.002 (61)	0.002 (52)	0.000 (51)	0.003 (56)
11	2.50	0.002 (61)	0.001 (51)	0.000 (50)	0.003 (56)
12	3.00	0.002 (61)	0.000 (51)	0.003 (55)	0.001 (59)

12.9 38 Elite/Near-Optimal Models

Table 57: BICs of *elite* ('near optimal') models; that is, all sampled models $\mathbf{v}$ such that $0 \leq BIC(\mathbf{v}) - BIC(\mathbf{v}') \leq 2$ where $\mathbf{v}'$ is the model that minimises the BIC across all sampled models in all λ values. Almost all elite models (the top 16, for instance) come from a chain sampled from $P_{\lambda=0.75}(\mathbf{v})$, and they are sampled half to three-quarters of the way in the run (of length 41,000).

	lambda	chain	iter	BIC	Dimension
1	0.75	3	21518	4795.27	52.00
2	0.75	3	19238	4795.27	52.00
3	0.75	3	19695	4795.27	52.00
4	0.75	3	19442	4795.27	53.00
5	0.75	3	19280	4795.47	53.00
6	0.75	3	21485	4795.47	53.00
7	0.75	3	19463	4795.47	53.00
8	0.75	3	19459	4795.47	54.00
9	0.75	3	21259	4795.62	53.00
10	0.75	3	21305	4795.62	54.00
11	0.75	3	21358	4795.62	53.00
12	0.75	3	26447	4795.62	53.00
13	0.75	3	22042	4795.63	54.00
14	0.75	3	21344	4795.63	54.00
15	0.75	3	22103	4795.63	54.00
16	0.75	3	21329	4795.63	55.00
17	0.50	4	38482	4795.94	52.00
18	0.50	4	34608	4795.97	50.00
19	0.50	4	38456	4795.97	51.00
20	0.75	3	21619	4796.04	53.00
21	0.75	3	19825	4796.04	53.00
22	0.75	3	19393	4796.04	54.00
23	0.75	3	19395	4796.04	53.00
24	0.75	3	29445	4796.14	54.00
25	0.75	3	30364	4796.14	54.00
26	0.75	3	27962	4796.14	54.00
27	0.75	3	26976	4796.14	55.00
28	0.75	3	28915	4796.21	53.00
29	0.75	3	21159	4796.21	53.00
30	0.75	3	23496	4796.21	54.00
31	0.75	3	26224	4796.21	53.00
32	0.50	4	38479	4796.22	51.00
33	0.50	4	38477	4796.22	52.00
34	0.50	4	34885	4796.22	51.00
35	0.75	3	23128	4796.23	53.00
36	0.75	3	28269	4796.23	53.00
37	0.75	3	22702	4796.23	53.00
38	0.75	3	27565	4797.23	54.00

12.10 Relative Frequencies

12.10.1 For all variables

Table 58: Marginal probabilities (and ESS) from $P_{\lambda=0.75}(\mathbf{v})$. Probabilities are sorted in descending order, and ESS are summed across all 4 chains.

#	variable	$\hat{p}$	ESS	SE	lower	upper
1	specific_humidity	1.00	0.00			
2	sea_surface_temperature	1.00	0.00			
3	geopotential_at_925_hPa	1.00	0.00			
4	specific_humidity_at_200_hPa	1.00	0.00			
5	vorticity_relative_at_200_hPa	1.00	0.00			
6	vorticity_relative_at_700_hPa	1.00	0.00			
7	vorticity_relative_at_850_hPa	1.00	0.00			
8	wind_u_at_925_hPa	1.00	0.00			
9	wind_u_at_1000_hPa	1.00	0.00			
10	vertical_wind_shear_between_200_hPa_and_850_hPa	1.00	0.00			
11	vertical_wind_shear_between_925_hPa_and_1000_hPa	1.00	0.00			
12	Coriolis_parameter_absolute	1.00	0.00			
13	geopotential_at_500_hPa	1.00	1521.85	0.00	1.00	1.00
14	geopotential_at_600_hPa	0.99	52.07	0.02	0.96	1.00
15	potential_vorticity_at_850_hPa	0.97	283.42	0.01	0.95	0.99
16	wind_v_at_200_hPa	0.96	56.18	0.03	0.91	1.00
17	wind_u_at_500_hPa	0.95	11.03	0.06	0.83	1.00
18	potential_vorticity_at_700_hPa	0.93	387.80	0.01	0.90	0.95
19	potential_vorticity_at_500_hPa	0.91	117.51	0.03	0.86	0.96
20	potential_vorticity_at_400_hPa	0.88	226.86	0.02	0.83	0.92
21	equiv_pot_temperature_diff_between_surface_and_500_hPa	0.87	45.12	0.05	0.77	0.97
22	medium_cloud_cover	0.84	95.32	0.04	0.77	0.92
23	vertical_velocity_at_700_hPa	0.83	13.90	0.10	0.63	1.00
24	vertical_wind_shear_between_600_hPa_and_1000_hPa	0.78	8.94	0.14	0.51	1.00
25	geopotential_at_700_hPa	0.75	14.27	0.11	0.53	0.98
26	geopotential_at_850_hPa	0.75	8.65	0.15	0.47	1.00
27	wind_u_at_200_hPa	0.75	2.83	0.26	0.25	1.00
28	absolute_vorticity_averaged	0.74	632.61	0.02	0.70	0.77
29	temperature_at_200_hPa	0.73	7.82	0.16	0.42	1.00
30	wind_u_at_600_hPa	0.70	6.71	0.18	0.36	1.00
31	specific_humidity_at_925_hPa	0.70	5.91	0.19	0.33	1.00
32	relative_humidity_at_1000_hPa	0.69	19.71	0.10	0.49	0.90
33	Coriolis_parameter	0.69	468.13	0.02	0.65	0.74
34	vertical_wind_shear_between_600_hPa_and_925_hPa	0.68	7.14	0.17	0.33	1.00
35	vorticity_relative	0.67	467.81	0.02	0.62	0.71
36	relative_humidity_at_200_hPa	0.65	123.83	0.04	0.57	0.73
37	divergence	0.64	4.62	0.22	0.20	1.00
38	vertical_velocity_at_200_hPa	0.64	4.78	0.22	0.21	1.00
39	vertical_velocity	0.61	3.78	0.25	0.12	1.00
40	vertical_wind_shear_between_300_hPa_and_400_hPa	0.61	115.96	0.05	0.52	0.70
41	surface_dewpoint_temperature	0.59	5.72	0.21	0.19	1.00
42	wind_v_at_700_hPa	0.59	6.91	0.19	0.22	0.96
43	temperature	0.55	6.47	0.20	0.17	0.94
44	relative_humidity_at_600_hPa	0.54	2.43	0.32	0.00	1.00
45	vertical_wind_shear_between_200_hPa_and_1000_hPa	0.54	25.46	0.10	0.35	0.73

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Table 58 (continued)

#	variable	$\hat{p}$	ESS	SE	lower	upper
46	temperature_at_500_hPa	0.51	5.50	0.21	0.10	0.93
47	divergence_at_925_hPa	0.51	9.73	0.16	0.19	0.82
48	geopotential_at_200_hPa	0.51	6.51	0.20	0.12	0.89
49	vertical_velocity_at_850_hPa	0.49	3.50	0.27	0.00	1.00
50	divergence_at_1000_hPa	0.49	18.59	0.12	0.26	0.71
51	vertical_wind_shear_between_300_hPa_and_1000_hPa	0.47	19.20	0.11	0.25	0.70
52	surface_air_temperature	0.47	54.73	0.07	0.34	0.60
53	divergence_at_600_hPa	0.47	130.80	0.04	0.38	0.55
54	relative_humidity_at_700_hPa	0.46	6.70	0.19	0.08	0.83
55	potential_vorticity	0.39	212.96	0.03	0.32	0.45
56	geopotential_at_300_hPa	0.39	6.67	0.19	0.02	0.76
57	vertical_velocity_at_925_hPa	0.37	8.76	0.16	0.05	0.69
58	vertical_velocity_at_400_hPa	0.37	4.59	0.23	0.00	0.81
59	wind_v_at_400_hPa	0.37	7.32	0.18	0.02	0.72
60	divergence_at_200_hPa	0.36	4.61	0.22	0.00	0.80
61	vertical_wind_shear_between_400_hPa_and_925_hPa	0.36	22.17	0.10	0.16	0.56
62	specific_humidity_at_850_hPa	0.34	60.20	0.06	0.22	0.46
63	specific_humidity_at_1000_hPa	0.33	41.71	0.07	0.19	0.48
64	geopotential	0.33	13.03	0.13	0.08	0.59
65	wind_v	0.33	5.54	0.20	0.00	0.72
66	wind_v_at_925_hPa	0.33	3.33	0.26	0.00	0.83
67	temperature_at_1000_hPa	0.31	9.90	0.15	0.03	0.60
68	temperature_at_925_hPa	0.29	6.03	0.19	0.00	0.66
69	specific_humidity_at_600_hPa	0.28	25.08	0.09	0.11	0.46
70	geopotential_at_1000_hPa	0.28	43.58	0.07	0.15	0.41
71	mean_sea_level_pressure	0.26	24.97	0.09	0.09	0.43
72	wind_u_at_850_hPa	0.26	14.89	0.11	0.04	0.48
73	temperature_at_850_hPa	0.26	5.05	0.19	0.00	0.64
74	temperature_at_400_hPa	0.26	12.66	0.12	0.01	0.50
75	vertical_wind_shear_between_700_hPa_and_850_hPa	0.25	204.80	0.03	0.20	0.31
76	temperature_at_300_hPa	0.25	4.88	0.20	0.00	0.64
77	wind_u	0.25	10.76	0.13	0.00	0.51
78	wind_u_at_300_hPa	0.25	8.12	0.15	0.00	0.55
79	divergence_at_500_hPa	0.25	5.98	0.18	0.00	0.60
80	relative_humidity_at_500_hPa	0.25	5.50	0.18	0.00	0.61
81	specific_humidity_at_400_hPa	0.25	2.17	0.29	0.00	0.83
82	vertical_wind_shear_between_500_hPa_and_1000_hPa	0.23	17.44	0.10	0.03	0.43
83	specific_humidity_at_700_hPa	0.23	3.03	0.24	0.00	0.70
84	vertical_wind_shear_between_200_hPa_and_500_hPa	0.23	164.36	0.03	0.16	0.29
85	vertical_wind_shear_between_500_hPa_and_925_hPa	0.22	6.38	0.16	0.00	0.54
86	vertical_wind_shear_between_850_hPa_and_1000_hPa	0.22	21.55	0.09	0.04	0.39
87	divergence_at_400_hPa	0.22	136.24	0.04	0.15	0.28
88	divergence_at_700_hPa	0.21	16.01	0.10	0.01	0.41
89	divergence_at_850_hPa	0.20	46.68	0.06	0.08	0.31
90	potential_vorticity_at_1000_hPa	0.15	412.12	0.02	0.12	0.18
91	vertical_velocity_at_1000_hPa	0.13	21.57	0.07	0.00	0.28
92	vorticity_relative_at_1000_hPa	0.12	25.35	0.06	0.00	0.25

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Table 58 (continued)

#	variable	$\hat{p}$	ESS	SE	lower	upper
93	relative_humidity_at_400_hPa	0.11	113.12	0.03	0.06	0.17
94	vertical_wind_shear_between_700_hPa_and_925_hPa	0.11	10.34	0.10	0.00	0.30
95	vertical_wind_shear_between_700_hPa_and_1000_hPa	0.11	10.60	0.10	0.00	0.29
96	vertical_wind_shear_between_400_hPa_and_1000_hPa	0.10	43.78	0.04	0.01	0.18
97	wind_v_at_600_hPa	0.09	41.02	0.05	0.00	0.18
98	specific_humidity_at_500_hPa	0.09	23.73	0.06	0.00	0.21
99	vorticity_relative_at_400_hPa	0.08	21.95	0.06	0.00	0.19
100	vertical_wind_shear_between_600_hPa_and_850_hPa	0.08	64.44	0.03	0.01	0.14
101	low_cloud_cover	0.06	462.49	0.01	0.04	0.09
102	potential_vorticity_at_200_hPa	0.05	43.66	0.03	0.00	0.12
103	relative_humidity	0.05	32.83	0.04	0.00	0.12
104	wind_u_at_400_hPa	0.05	32.90	0.04	0.00	0.12
105	wind_u_at_700_hPa	0.05	21.99	0.04	0.00	0.13
106	vertical_wind_shear_between_200_hPa_and_400_hPa	0.05	283.69	0.01	0.02	0.07
107	relative_humidity_at_300_hPa	0.04	29.06	0.04	0.00	0.12
108	relative_humidity_at_925_hPa	0.04	11.86	0.06	0.00	0.16
109	vorticity_relative_at_300_hPa	0.04	39.99	0.03	0.00	0.10
110	relative_humidity_at_850_hPa	0.04	533.04	0.01	0.02	0.05
111	vertical_wind_shear_between_200_hPa_and_925_hPa	0.04	491.87	0.01	0.02	0.05
112	vertical_wind_shear_between_400_hPa_and_700_hPa	0.03	149.62	0.01	0.01	0.06
113	vertical_wind_shear_between_300_hPa_and_500_hPa	0.03	47.37	0.03	0.00	0.08
114	potential_vorticity_at_925_hPa	0.03	549.71	0.01	0.02	0.05
115	vertical_wind_shear_between_300_hPa_and_850_hPa	0.03	67.29	0.02	0.00	0.06
116	temperature_at_700_hPa	0.02	730.10	0.00	0.01	0.02
117	vertical_velocity_at_600_hPa	0.01	608.40	0.00	0.00	0.02
118	temperature_at_600_hPa	0.01	277.23	0.01	0.00	0.02
119	vertical_velocity_at_500_hPa	0.01	265.06	0.01	0.00	0.02
120	vertical_wind_shear_between_400_hPa_and_600_hPa	0.01	163.37	0.01	0.00	0.03
121	vertical_wind_shear_between_200_hPa_and_300_hPa	0.01	383.09	0.00	0.00	0.02
122	vertical_wind_shear_between_200_hPa_and_600_hPa	0.01	488.22	0.00	0.00	0.02
123	potential_vorticity_at_300_hPa	0.01	266.44	0.01	0.00	0.02
124	high_cloud_cover	0.01	218.81	0.01	0.00	0.02
125	geopotential_at_400_hPa	0.01	439.36	0.00	0.00	0.01
126	vertical_wind_shear_between_500_hPa_and_850_hPa	0.01	271.76	0.00	0.00	0.02
127	vorticity_relative_at_600_hPa	0.01	172.31	0.01	0.00	0.02
128	vertical_wind_shear_between_300_hPa_and_600_hPa	0.01	256.97	0.00	0.00	0.02
129	vertical_wind_shear_between_400_hPa_and_500_hPa	0.01	897.15	0.00	0.00	0.01
130	total_cloud_cover	0.00	364.58	0.00	0.00	0.01
131	absolute_vorticity_at_925_hPa	0.00	379.04	0.00	0.00	0.01
132	vertical_wind_shear_between_300_hPa_and_925_hPa	0.00	908.27	0.00	0.00	0.01
133	potential_vorticity_at_600_hPa	0.00	414.98	0.00	0.00	0.01
134	specific_humidity_at_300_hPa	0.00	636.60	0.00	0.00	0.01
135	vorticity_relative_at_500_hPa	0.00	840.74	0.00	0.00	0.01
136	vertical_wind_shear_between_300_hPa_and_700_hPa	0.00	850.71	0.00	0.00	0.01
137	vertical_wind_shear_between_200_hPa_and_700_hPa	0.00	942.10	0.00	0.00	0.01
138	vertical_velocity_at_300_hPa	0.00	218.19	0.00	0.00	0.01
139	vertical_wind_shear_between_400_hPa_and_850_hPa	0.00	567.15	0.00	0.00	0.00

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Table 58 (continued)

#	variable	$\hat{p}$	ESS	SE	lower	upper
140	divergence_at_300_hPa	0.00	3501.81	0.00	0.00	0.00
141	wind_v_at_1000_hPa	0.00	4292.22	0.00	0.00	0.00
142	vorticity_relative_at_925_hPa	0.00	0.00			
143	wind_v_at_300_hPa	0.00	0.00			
144	wind_v_at_500_hPa	0.00	0.00			
145	wind_v_at_850_hPa	0.00	0.00			
146	vertical_wind_shear_between_500_hPa_and_600_hPa	0.00	0.00			
147	vertical_wind_shear_between_500_hPa_and_700_hPa	0.00	0.00			
148	vertical_wind_shear_between_600_hPa_and_700_hPa	0.00	0.00			
149	vertical_wind_shear_between_850_hPa_and_925_hPa	0.00	0.00			

12.10.2 For core variables

Table 59: Core variables, that is, variables that are present in all 38 elite models (i.e. models with greater BIC than the best model of no more than 2), grouped by pressure levels. Relative frequencies are also displayed, and sorted descendingly by pressure levels for each physical family. If several pressure levels are chosen for a physical variable, the pressure level with the highest marginal frequency is shown with boldface. Overall there are 41 core variables present in all 38 elite models.

# Physical variable	Relative Frequencies (physical + on selected pressure levels, if also present)
Coriolis parameter (absolute)	physical ($\hat{p} = 1$)
geopotential	925 hPa ($\hat{p} = 1$) 500 hPa ($\hat{p} = 0.99$) 600 hPa ($\hat{p} = 0.98$) 700 hPa ($\hat{p} = 0.75$) 850 hPa ($\hat{p} = 0.75$)
sea surface temperature	physical ($\hat{p} = 1$)
specific humidity	physical ($\hat{p} = 1$) 200 hPa ($\hat{p} = 1$) 925 hPa ($\hat{p} = 0.7$)
relative vorticity	200 hPa ($\hat{p} = 1$) 700 hPa ($\hat{p} = 1$) 850 hPa ($\hat{p} = 1$)
vertical wind shear	200–850 hPa ($\hat{p} = 1$) 925–1000 hPa ($\hat{p} = 1$) 600–1000 hPa ($\hat{p} = 0.78$) 600–925 hPa ($\hat{p} = 0.67$) 300–400 hPa ($\hat{p} = 0.61$) 300–1000 hPa ($\hat{p} = 0.47$) 400–925 hPa ($\hat{p} = 0.36$)
wind u	925 hPa ($\hat{p} = 1$) 1000 hPa ($\hat{p} = 1$) 500 hPa ($\hat{p} = 0.95$)

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Table 59 (continued)

# Physical variable	Relative Frequencies (physical + on selected pressure levels, if also present)
potential vorticity	850 hPa ($\hat{p} = 0.97$) 700 hPa ($\hat{p} = 0.92$) 500 hPa ($\hat{p} = 0.90$) 450 hPa ($\hat{p} = 0.87$)
wind v	200 hPa ($\hat{p} = 0.95$) 700 hPa ($\hat{p} = 0.58$)
equivalent potential temperature diff (surface–500 hPa)	physical ($\hat{p} = 0.86$)
medium cloud cover	physical ($\hat{p} = 0.84$)
vertical velocity	700 hPa ($\hat{p} = 0.82$) 200 hPa ($\hat{p} = 0.63$) 850 hPa ($\hat{p} = 0.49$) 925 hPa ($\hat{p} = 0.37$)
divergence	physical ($\hat{p} = 0.63$) 700 hPa ($\hat{p} = 0.2$)
temperature	physical ($\hat{p} = 0.55$) 500 hPa ($\hat{p} = 0.51$) 1000 hPa ($\hat{p} = 0.31$)
relative humidity	700 hPa ($\hat{p} = 0.45$)

12.11 Final model coefficients

Table 60: Coefficient estimates and standard errors for best model (i.e. chosen elite/near-optimal model with the highest AUC-PR and log-loss) on the **standardized** design matrix, where the model has the form $P(\text{TC_genesis}=1|\text{event } j) = \sigma(\hat{\beta}_0 + \sum_j \hat{\beta}_j x_j)$. $\hat{\beta}_0 = -4.77$ with $se(\hat{\beta}_0) = 0.067$. Variables are sorted by absolute values of coefficients.

#	variable x_j	$\hat{\beta}_j$ ($se(\hat{\beta}_j)$)
1	geopotential_at_925_hPa	25.92 (2.23)
2	geopotential_at_600_hPa	-22.73 (1.79)
3	geopotential_at_850_hPa	-16.79 (1.84)
4	geopotential_at_700_hPa	12.85 (1.39)
5	geopotential_at_500_hPa	10.94 (1.15)
6	temperature	9.02 (0.92)
7	absolute_vorticity_averaged	8.69 (0.84)
8	mean_sea_level_pressure	-7.99 (0.83)
9	Coriolis_parameter	-5.75 (0.69)
10	wind_u	-5.38 (0.44)
11	temperature_at_500_hPa	-3.62 (0.32)
12	wind_u_at_1000_hPa	3.58 (0.31)
13	specific_humidity	3.47 (0.38)
14	geopotential_at_200_hPa	-3.34 (0.50)
15	wind_u_at_925_hPa	-2.85 (0.46)
16	vertical_velocity_at_850_hPa	-2.63 (0.25)
17	specific_humidity_at_600_hPa	-2.43 (0.42)

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Table 60 (continued)

#	variable x_j	$\hat{\beta}_j$ ($se(\hat{\beta}_j)$)
18	relative_humidity_at_600_hPa	2.32 (0.39)
19	wind_u_at_500_hPa	2.04 (0.18)
20	vertical_wind_shear_between_200_hPa_and_850_hPa	-1.95 (0.09)
21	wind_u_at_300_hPa	1.86 (0.22)
22	vertical_wind_shear_between_600_hPa_and_1000_hPa	-1.85 (0.14)
23	temperature_at_1000_hPa	-1.83 (0.20)
24	Coriolis_parameter_absolute	1.82 (0.05)
25	wind_u_at_850_hPa	1.71 (0.33)
26	surface_dewpoint_temperature	1.66 (0.25)
27	relative_humidity_at_500_hPa	-1.63 (0.18)
28	vertical_wind_shear_between_600_hPa_and_925_hPa	1.63 (0.15)
29	specific_humidity_at_925_hPa	-1.44 (0.16)
30	vorticity_relative_at_850_hPa	-1.43 (0.12)
31	vertical_velocity_at_700_hPa	1.32 (0.13)
32	potential_vorticity_at_500_hPa	-1.22 (0.23)
33	equiv_pot_temperature_diff_between_surface_and_500_hPa	-1.12 (0.21)
34	relative_humidity_at_700_hPa	-1.04 (0.13)
35	vertical_velocity_at_925_hPa	0.95 (0.14)
36	potential_vorticity	-0.87 (0.29)
37	vorticity_relative_at_200_hPa	-0.86 (0.09)
38	potential_vorticity_at_400_hPa	-0.85 (0.22)
39	potential_vorticity_at_700_hPa	-0.82 (0.23)
40	potential_vorticity_at_850_hPa	0.77 (0.16)
41	vertical_wind_shear_between_300_hPa_and_1000_hPa	0.73 (0.12)
42	vorticity_relative_at_700_hPa	0.68 (0.12)
43	vertical_wind_shear_between_925_hPa_and_1000_hPa	-0.65 (0.04)
44	vertical_wind_shear_between_400_hPa_and_925_hPa	-0.56 (0.11)
45	vertical_velocity_at_200_hPa	-0.55 (0.05)
46	divergence_at_700_hPa	-0.45 (0.07)
47	sea_surface_temperature	-0.45 (0.06)
48	specific_humidity_at_200_hPa	-0.39 (0.06)
49	divergence	0.33 (0.05)
50	vertical_wind_shear_between_300_hPa_and_400_hPa	-0.27 (0.06)
51	wind_v_at_700_hPa	0.24 (0.03)
52	wind_v_at_200_hPa	-0.19 (0.04)
53	medium_cloud_cover	0.17 (0.04)
54	divergence_at_400_hPa	-0.13 (0.04)

12.12 Calibration plots for the best model

Table 61: Data Summary for Calibration Plots. The idea is to bin the predicted probabilities into 10 equal frequency bins i.e. *deciles*. For each bin, we average the predicted probabilities in that bin (**predRate**) and count the proportions of observed $y_i = 1$ in that bin (**obsRate**). A well calibrated probability model should have these two quantities match as close as possible.

	decile	obsRate	obsRate_SE	obsNo	predRate	obsRate_UCL	obsRate_LCL
1	1	0.000973	0.000562	3083	0.000059	0.002074	-0.000128
2	2	0.000973	0.000562	3083	0.000459	0.002074	-0.000128
3	3	0.001622	0.000725	3083	0.001442	0.003042	0.000201
4	4	0.003244	0.001024	3083	0.003314	0.005251	0.001236
5	5	0.004217	0.001167	3083	0.006830	0.006504	0.001929
6	6	0.013299	0.002063	3083	0.013249	0.017343	0.009254
7	7	0.023037	0.002703	3082	0.026421	0.028334	0.017740
8	8	0.050616	0.003949	3082	0.057228	0.058357	0.042876
9	9	0.148605	0.006408	3082	0.149069	0.161165	0.136045
10	10	0.584685	0.008878	3082	0.573200	0.602086	0.567285

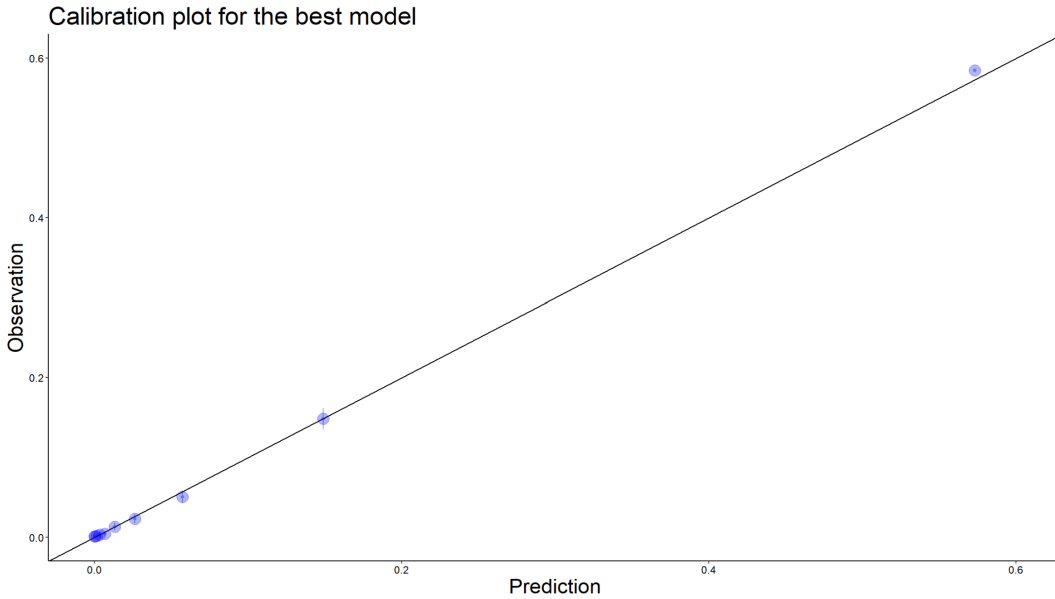


Figure 27: Calibration plot for predicted probabilities from the best model and observed TC genesis events (complementary to Table (61)). Despite the class imbalance, the best model's predicted probabilities align closely with observed event frequencies across the entire predicted probability range.

12.13 Unique minimizer of the expected log loss

Referencing Proposition 2.30 from Bernardo and Smith [2], for preferences described by the logarithmic scoring rule, the expected loss incurred by reporting a predictive distribution $q = \{q_j, j \in J\}$ when the true beliefs are $p = \{p_j, j \in J\}$ is given by

$$\delta\{q|p\} = A \sum_{j \in J} p_j \log \frac{p_j}{q_j}, \quad A > 0.$$

This expression is proportional to the Kullback–Leibler divergence $D_{\text{KL}}(p||q)$, which satisfies $\delta\{q|p\} \geq 0$ with equality if and only if $q = p$. Hence, the expected logarithmic loss (where I used the empirical OOS log-loss as an estimate) is uniquely minimized when the predictive distribution matches the true data-generating distribution,

$q = p$. In the context of logistic regression, this corresponds to $p(x) = P(Y = 1 \mid X = x)$, the true conditional probability.

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